

CS610 MID-Term BY Attiq Kundi

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AND Many MoRe

Question No: 1 Marks: 1) - Please choose one

The number of connections needed for N computer in direct point to point communication is equal to:

► $(N^2 - N)/2$

(Page 23)

► $N(N-1)$

► N^2

► None of the given

Question No: 2(Marks: 1) - Please choose one

When an application----- data, it makes a copy of the data available to all other computers on the network.

► **Broadcasting**

Click here for detail

► Multicasting

► Unicasting

► None of the given

Question No: 3

(Marks: 1) - Please choose one

Ethernet uses a ----- bit static addressing scheme in which each device is assigned a unique

address by the manufacturer.

▶ 64

48 (Computer Networks and Internets, page 109)

▶ 32

▶ 8Question No: 4

(Marks: 1) - Please choose one

A system with redundant bridges might have a problem with
in the system.

▶ Loop

Click here for detail

rep

▶ Filters

▶ Spanning Trees

▶ All given choices

Question No: 5

(Marks: 1) - Please choose one

The product of delay and throughput measures the

of data that can be present on the network.

▶ Area

▶ Volume

(Page 80) rep

- ▶ Length
- ▶ None of the given

Question No: 6

(Marks: 1) - Please choose one

Connectionless service, Message-Oriented protocol, best effort delivery service, arbitrary interaction and operating system independent are the characteristics of

- ▶ TCP
- ▶ UDP (Page 110)
- ▶ IP
- ▶ None of the given

Question No: 7

(Marks: 1) - Please choose one

Connection-oriented service, Point-to-point, Complete reliability, Full-duplex communication, Stream interface, Reliable connection startup and Graceful connection shutdown are the services provided by

- ▶ None of the given
- ▶ TCP (Page 123) rep
- ▶ UDP
- ▶ IP

Question No: 8

(Marks: 1) - Please choose one

The process of using a routing table to select a next hop for a given datagram is called

- ▶ Encapsulation
- ▶ Reassembling
- ▶ **Routing or forwarding**

(Computer Networks and Internets, page 265)

- ▶ None of the given

Question No: 9(Marks: 1) - Please choose one

uses distance vector approach to define routing

- ▶ BGP
- ▶ OSPF
- ▶ **RIP (Computer Networks and Internets, page 332) rep**
- ▶ None of the given

Question No: 10

(Marks: 1) - Please choose one

A multicast routing scheme in which the protocol software builds a delivery tree from a central point is called

- ▶ Distance Vector Multicast Routing Protocol (DVMRP)
- ▶ **Core Based Trees (CBT)**

(Page 114)

- ▶ Protocol Independent Multicast_ *Sparse Mode (PIM-SM)*
- ▶ Protocol Independent Multicast _ *Dense Mode (PIM-DM)*

Question No: 11

(Marks: 1) - Please choose one

One repeater

, two repeaters

the maximum cable length limitation.

► doubles, cancel

► doubles, triple

(Page 49) rep

► square roots, cube roots

► and, triple

Question No: 12

(Marks: 1) - Please choose one

Whenever it handles a packet, IP software needs to separate the destination address into a and

.

► postfix, Infix

► non of these

► Infix, prefix

► prefix, suffix

(Page 87)

Question No: 13

(Marks: 1) - Please choose one

Although message exchange can be used to bind addresses, sending a request for each binding is hopelessly inefficient.

► True

(Page 99) rep

► False

Question No: 14

(Marks: 1) - Please choose one

ARP is almost always used to bind a

-bit IP address to a

-bit Ethernet address.

► 32, 48

(Page 98)

► 24, 32

► 32, 64

► 32, 128

Question No: 15

(Marks: 1) - Please choose one

End-to-end delivery service is connection oriented.

► True

► False

(Page 101)

Question No: 16

(Marks: 1) - Please choose one

A single networking technology is best for all needs.

► True

► False

(Page 81) rep

Question No: 17

(Marks: 1) - Please choose one

In the 1970s large organizations began to acquire multiple networks. Each network in the organization formed island. Employees needed to choose a computer appropriate for each task. So they needed multiple screens, keyboards and computers.

► False

► True

(Page 81)

Question No: 18

(Marks: 1) - Please choose one

Which method of Address Resolution Protocol is useful with any hardware?

Were "T" stands for Table lookup, "C" for Closed-form Computation and "D" for Data Exchange?

► T

(Page 97)

▶ C

▶ D

▶ C, D

Question No: 19

(Marks: 1) - Please choose one

In which method of Address Resolution Protocol the protocol address is determined by hardware address?

Were "T" stands for Table lookup, "C" for Closed-form Computation and "D" for Data Exchange?

▶ T

▶ D

▶ C

(Page 97) rep

▶ T, C

Question No: 20

(Marks: 1) - Please choose one

We use the term

_ to refer to a measure of the path that routing software use when choosing a route.

► routing path

► routing metric

► **routing**

(Computer Networks and Internets, page330)

► switching

Question No: 1 (M - 1) .

In broadcast topology there are two categories_____.

1. SATELLITE\ RADIO
2. RING TOPOLOGY
3. **Both a & b**
4. None of the above

Question No: 2 (M - 1) .

The termrefers to the general concept of a small block of data

- a. **Packet**
- b. Frame
- c. Data

- d. None of the given

Question No: 3 (M - 1) :

The term ----- is used to denote the definition of a packet used with a specific type of network.

- 5. Packet

- 6. Frame**

- 7. Data

- 8. None of the given

Question No: 4 (M - 1) :

9. **CRC can detect more errors than a simple checksum.....**

10. **True**

11. **False**

Question No: 5 (M - 1) :

The Ethernet standard specifies that frames are sent using the -----Manchester

Question No: 6 (M - 1) :

Most LANs that employ ring topology use an access mechanism known as-----

- a. CSMA/CD
- b. CSMA/CA
- c. **TOKEN PASSING**
- d. None of the given

Question No: 7 (M - 1) :

An -----method, the network hardware designers specify how type

information is included in the frame and the value use to identify various frame types.

12. **Explicit frame type**

13. Ideal frame type

14. Implicit frame type

15. None of the given

Question No: 8 (M - 1) .

IEEE LLC/SNAP header is -----, which is used to specify the type of data.

16. **8 octets**

17. 8 bytes

18. 8 bits

19. None of the given

Question No: 9 (M - 1) .

Formally named _____ informally known as the thick wire Ethernet or Thick net.

20. 10 Base 2

21. 10 Base 5

22. 10 Base T

23. None of the given vuzs.net

Question No: 10 (M - 1) .

Most NICs contain _____ circuitry that allows the NIC to operate independent of the CPU.

Question No: 11 (M - 1) .

A bridges function in the _____ layers(s).

24. Physical(MAC)

25. Data Link

26. Network

27. Physical (MAC) and Datalink

Question No: 12 (M - 1) .

A Bridge forwards or filters a frame by comparing the information in its address table to the frame's_____

28. Layer 2 source address

29. Source node's physical address

30. Layer 2 destination address

31. Layer 3 destination address

Question No: 13 (M - 1) .

The next hop to which a packet is sent depends only on

- a. **Packet's destination**
- b. Packet's original source
- c. Path the packet has taken
- d. Non of the given

Question No: 14 (M - 1) .

_____ is used for compressed audio and video where the data rate depends on the level of compression that can be achieved.

32. Constant Bit Rate (CBR) service

33. Variable Bit Rate (VBR) service

34. Available Bit Rate (ABR) service

35. None of the given

Question No: 15 (M - 1) .

ATM assigns each VC a _____ identifier that is divided two parts to produce a hierarchy.

36. 21-bit

37. 22-bit

38. 23-bit

39. **24 bit**

Question No: 16 (M - 1) .

ATM was designed for _____ and use a _____

Question No: 17 (M - 1) .

The product of delay and throughput measures the _____ of data that can be present on the network.

40. Area

41. Volume

42. Length

43. None of the given

Question No: 18 (M - 1) .

A typical port on an ATM switch operates at _____ or higher.

44. OC-2 speed (155Mbps)

45. OC-3 speed (100Mbps)

46. **OC-3 speed (155Mbps)**

47. OC-3 speed (155Gbps)

Question No: 19 (M - 1) .

A- interface ---- provides mechanism that automatically assigns a physical address to a station when the station first boots.

Question No: 20 (M - 1) .

By size there are.....classifications of networks.

► 2

► 3

► 4

► 1

Question No: 1 (M - 1) :

No error detection scheme is perfect because transmission errors can affect the additional information as well as the data.

48.True

49.False

No error detection scheme is perfect because transmission errors can affect the additional information as well as the data. A cyclic redundancy check, which is slightly more difficult to compute, can detect more errors than a Parity or Check sum mechanism.

Question No: 2 (M - 1) :

----- Program sends a message to a remote computer and reports whether the computer responds.

► Ping

- 50.Traceroute
- 51.ICMP
- 52.Non of the given

Question No: 3 (M - 1) :

----- was especially concerned about the lack of high powered computers.

53.ARP

- 54.IEEE
- 55.EIA
- 56.Non of the given

The US Department of Defense Advanced Research Projects Agency (ARPA) was especially concerned about the lack of high-powered computers. Many of the ARPA research projects needed access to the latest equipment

Question No: 4 (M - 1) :

The term ----- is used to denote the definition of a packet used with a specific type of network.

- 57.Packet

58.Frame

- 59.Data
- 60.None of the given

To help distinguish between the general idea of packet transmission and the specific definition of packets for a given hardware technology, we use the term frame to denote the definition of a packet used with a specific type of network. Thus one might refer the maximum amount of data a frame can hold or the order in which bits of a frame are sent across a network.

Question No: 5 (M - 1) :

Computer networks are often called ----- because they use packet technology.

- 61.Ethernet
- 62.Switch networks
- 63.Packet networks
- 64.None of the given

Question No: 6 (M - 1) :

----- have advantages arisen from the size and ease of computation.

65.CRC

66.Parity

[67.Checksums](#)

68.None of given

Question No: 7 (M - 1) :

Most LANs that employ ring topology use an access mechanism known as-----

69.CSMA/CD

70.CSMA/CA

[71.TOKEN PASSING](#)

72.None of the given

Most LANs that employ ring technology use an access mechanism known as *token passing*. A token ring operates as a single, shared medium. When a computer wants to send data, it must wait until it obtains the token, when it is in control of the ring

Question No: 8 (M - 1) :

IEEE LLC/SNAP header is -----, which is used to specify the type of data.

73.8 octets

74.8 bytes

75.8 bits

76.None of the given

Question No: 9 (M - 1) :

Formally named _____ informally known as the twisted pair Ethernet or TP Ethernet.

77.10 Base 2

78.10 Base 5

79.10 Base T

80.None of the given

twisted pair ethernet

- Formally called 10 Base –T
- Also twisted pair Ethernet or simply TP Ethernet

Question No: 10 (M - 1) :

An interface for twisted pair Ethernet must have an _____ connector , and must generate signals according to the _____ specification.

81.RJ-45, 10 Base T

82.RJ-45, 10 Base 5

83.BNC, 10 Base 2

84.BNC, 10 Base T

an interface for twisted pair Ethernet must have an RJ-45 connector, and must generate signals according to the 10BaseT specification. (P# 683 E-books)

Question No: 11 (M - 1) :

A bridges function in the _____ layers(s).

85.Physical (MAC)

86.Data link

► Network

87.Physical (MAC) and Data link

Question No: 12 (M - 1) :

A Bridge can _____

88.Filter a frame

89.Forward a frame

90.Extend a LAN

► Do all the above

A bridge is a hardware device also used to connect two LAN segments to extend a LAN. It listens to all traffic and recognizes frame format. It also forwards only correct complete frames and discards the collided and error frames.

The most important task a bridge performs is frame filtering.

Question No: 13 (M - 1) :

A Bridge forwards or filters a frame by comparing the information in its address table to the frame's _____

91.Layer 2 source address

92.Source node's physical address

93.Layer 2 destination address

94.Layer 3 destination address

Question No: 14 (M - 1) :

_____ computes shortest paths in a graph by using weights on edges as a measure of distance.

95.Greedy algorithm

96.Distance vector algorithm

97.Dijkstra's algorithm

98.Non of the given

Dijkstras algorithm computes shortest paths in a graph by using weights on edge as a measure of distance. A path with the fewest number of edges may not be the path with least weight. (P# 173 e-books)

Question No: 15 (M - 1) :

_____ is used for audio and video, since these have predefined maximum data rates

99.Constant Bit Rate (CBR) service

100. Variable Bit Rate (VBR) service

101. Available Bit Rate (ABR) service

102. None of the given
CONSTANT BIT RATE (CBR):

It is used for audio and video, since these have predefined maximum data rates.

Question No: 16 (M - 1) :

Unlike Frame Relay and ATM, SMDS (Switched multi-megabit Data service) offers_____ .

103. Connectionless service paradigm

104. Connection oriented service paradigm

105. Both Connectionless and Connection-oriented service paradigm

106. None of the given

Switched Multi megabit Data Service (SMDS) is also a Telco service. It is a connection less service.

FRAME RELAY:

It is used for Telco service for delivering blocks of data. It is connection based
Service ATM uses the concept of connection-oriented networking

Question No: 17 (M - 1) :

A network with throughput T and delay D has a total of _____ bit in transit at any time.

107. T / D

108. T x D

109. T + D

110. None of the given

A network with throughput T and delay D has a total of T x D bits in transit at any time. (e-books)

Question No: 18 (M - 1) :

ATM is designed to work on_____.

111. Twisted Pair

112. Coaxial

113. Radio Frequency

114. Fiber

ATM is designed to work on fiber (but can be used with twisted pair). (P# 72)

Question No: 19 (M - 1) :

Computers attached to an ethernet use ----- in which a computer waits for the ether to be idle before transmitting a frame.

115. CSMA/CD

116. CSMA/CA

117. TOKEN PASSING

118. None of the given

All computers attached to the Ethernet use CSMA/CD to co-ordinate their activities. A computer wishing to transmit checks for electrical activity on the cable, informally called a *carrier*. If there is no carrier, the computer can transmit. If a carrier is present, the computer waits for the sender to finish before proceeding

Question No: 20 (M - 1) :

FDDI can transmits data at a rate of -----

119. 100 million bits per second
120. 10 million bits per second
121. 1000 million bits per second
122. None of the given

Question No: 1 (M - 1) .

In Point-to-Point topology there are two topologies.

123. Star and Tree
124. Tree and Ring
125. Star and Ring

126. Star and Tree

127. None of the given

In *Point-to-Point* topology there are two topologies.

128. STAR topology

129. TREE topology

Question No: 2 (M - 1) .

----- Program sends a message to a remote computer and reports whether the computer responds.

► Ping

- Traceroute
- ICMP

► Non of the given

Ping program tests reach ability. It sends datagram from B to A, that echoes back to B. it uses ICMP echo request and echo reply messages.

Question No: 3 (M - 1) .

----- has no way to determine the cause of the problem.

130. ICMP

131. Ping

132. Trace route

133. Non of the given

Ping has no way to determine the cause of the problem. Tracing A Route Network administrators use another tool, trace route.

Question No: 4 (M - 1) .

The term----- ----- refers to the general concept of a small block of data

134. Packet

135. Frame

136. Data

137. None of the given

Packet is a generic term that refers to small block of data. Packet have different format. Each hardware uses different packet format.

Question No: 5 (M - 1) .

----- scheme, which is designed to help detect transmissions errors, send one extra bit of information with each character

138. Parity

139. Checksums

140. CRC

141. None of given

A parity scheme, which is designed to help detect transmission errors, sends

one extra bit of information with each character. Although it allows receiver to determine if a single bit has been changed parity can not detect transmission errors that change an even number of bits.

Question No: 6 (M - 1) .

Local Talk is a LAN technology that employs -----

142. Bus topology

143. Ring topology

144. Star topology

145. None of the given

LOCAL TALK:

Apple invented the LAN technology that uses bus topology. Its interface is included with all Macintosh computers.

Question No: 7 (M - 1) .

Most LANs that employ ring topology use an access mechanism known as-----

146. CSMA/CD

147. CSMA/CA

148. TOKEN PASSING

149. None of the given

Most LANs that employ ring technology use an access mechanism known as *token passing*. A token ring operates as a single, shared medium. When a computer wants to send data, it must wait until it obtains the token, when it is in control of the ring

Question No: 8 (M - 1) .

Ethernet uses a ----- bit static addressing scheme in which each device is assigned a unique address by the manufacturer.

150. 64

151. 48

152. 32

153. 8

[Ethernet uses a 48-bit static addressing scheme](#)

Question No: 9 (M - 1) .

Formally named _____ informally known as the thick wire Ethernet or Thick net.

154. 10 Base 2

[155. 10 Base 5](#)

156. 10 Base T

157. None of the given

Thick Ethernet, officially known as 10 Base 5, is the oldest form of Ethernet. It was originally developed in the late 1970's by Digital Equipment Corporation, IBM, and [Xerox](#), and became an international standard (IEEE 802.3) in 1983.

Question No: 10 (M - 1) .

Formally named _____ informally known as the twisted pair Ethernet or TP Ethernet.

- 158. 10 Base 2
- 159. 10 Base 5
- 160. 10 Base T**
- 161. None of the given

[twisted pair ethernet](#)

[Formally called 10 Base -T](#)

- 162. [Also twisted pair Ethernet or simply TP Ethernet](#)

Question No: 11 (M - 1) .

The maximum size of an Ethernet segment is_____

- 163. 250 meters
- 164. [500 meters](#)**
- 165. 700 meters
- 166. None of the given

One Ethernet cable is sometimes called a segment. This segment is limited to 500 meters in length. The minimum separation between connections is 3 meters. (P# 27)

Question No: 12 (M - 1) .

A Bridge can _____

167. Filter a frame

168. Forward a frame

169. Extend a LAN

170. Do all the above

A bridge is a hardware device also used to connect two LAN segments to extend a LAN. It listens to all traffic and recognizes frame format. It also forwards only correct complete frames and discards the collided and error frames.

The most important task a bridge performs is frame filtering.

Question No: 13 (M - 1) .

_____ computes shortest paths in a graph by using weights on edges as a measure of distance.

- 171. Greedy algorithm
- 172. Distance vector algorithm

173. Dijkstra's algorithm

- 174. Non of the given

Dijkstras algorithm computes shortest paths in a graph by using weights on edg as a measure of distance. A path with the fewest number of edges may not be the path with least weight. (P# 173 e-books)

Question No: 14 (M - 1) .

_____is used for compressed audio and video where the data rate depends on the level of compression that can be achieved.

- 175. Constant Bit Rate (CBR) service

176. Variable Bit Rate (VBR) service

- 177. Available Bit Rate (ABR) service
- 178. None of the given

VARIABLE BIT RATE (VBR):

It is used for compressed audio and video where the data rate depends on the level of compression that can be achieved.

Question No: 15 (M - 1) .

Basic LAN technologies such as Ethernet, Token Ring, and FDDI use a _____.

179. Connectionless service paradigm

- 180. Connection-oriented service paradigm
- 181. Both Connectionless and Connection-oriented service paradigm
- 182. None of the given

Basic LAN technologies such **as Ethernet, Token Ring, and FDDI use a connectionless service paradigm.** Although a computer needs to wait for access to a shared medium before sending a packet

Question No: 16 (M - 1) .

The product of delay and throughput measures the _____ of data that can be present on the network.

183. Area

184. Volume

185. Length

186. None of the given

DELAY THROUPTUT PRODUCT:

It is computed as delay time multiplied by effective throughput. It measures

amount of data that can be present in the network (P# 80)

Question No: 17 (M - 1) .

A network with throughput T and delay D has a total of _____ bit in transit at any time.

187. T / D

188. $T \times D$

189. $T + D$

190. None of the given

A network with throughput T and delay D has a total of $T \times D$ bits in transit at

any time. (e-books)

Question No: 18 (M - 1) .

One repeater _____, two repeaters _____ the maximum cable length limitation.

191. doubles, triple

192. square roots, cube roots

193. and, triple

194. doubles, cancel

One repeater doubles, two repeaters triple the maximum cable length limitation. (P# 49)

Question No: 19 (M - 1) .

A network uses a ----- if all computers attach to a central point

195. Star Topology

196. Ring Topology

- 197. Bus Topology
- 198. None of the given

STAR TOPOLOGY:

In this topology, all computers are attached to a central point, which is sometimes called the "Hub" (P# 25)

Question No: 20 (M - 1) .

Computers attached to an ethernet use ----- in which a computer waits for the other to be idle before transmitting a frame.

199. CSMA/CD

- 200. CSMA/CA
- 201. TOKEN PASSING
- 202. None of the given

All computers attached to the Ethernet use CSMA/CD to co-ordinate their activities. A computer wishing to transmit checks for electrical activity on the cable, informally called a *carrier*. If there is no carrier, the computer can transmit. If a carrier is present, the computer waits for the sender to finish before proceeding.

_____ has a jitter zero

- None of the given
- Virtual Private Network
- Isochronous Network
- Asynchronous Network

Unlike Frame Relay and ATM, SMDS (Switched multi-megabit Data service) offers_____.

Connectionless service paradigm

Connection oriented service paradigm

Both Connectionless and Connection-oriented service paradigm

None of the given

ATM assigns each VC a _____ identifier that is divided two parts to produce a hierarchy.

21-bit

22-bit

23-bit

24-bit

The VC is identified by a 24-bit value formed from the VPI or Virtual Path Indicator (8-bit),

Most WAN systems include a mechanism that can be used to eliminate the common case of duplication routing is called_____

Hierarchical address

[Default route](#) Shortest path

None of the given

The next hop to which a packet is sent depends only on

[Packet's destination](#)

Packet's original source

Path the packet has taken

Non of the given

An interface for twisted pair Ethernet must have a _____ connector, and must generate signals according to the_____ specification.

[RJ-45, 10 Base T](#)

RJ-45, 10 Base 5

BNC, 10 Base 2

BNC, 10 Base T

(zh,vuzs,nov10)

When an application_____ data, it makes a copy of the data available to all

other computers on the network.

Broadcasting

Multicasting

Unicasting

None of the given

When an application broadcasts data, it makes a copy of the data available to all other computers on the network. Broadcast has many uses. For example, suppose a computer needs to find one of the printers on the network

A _____ provide a mechanism that a customer can use to set a physical address.

Static addressing scheme

Configurable addressing scheme

Dynamic addressing scheme

None of the given

One of the advantages of a configurable addressing scheme becomes apparent when network interface hardware fails and must be replaced: unlike hardware that uses a static assignment, a configurable interface can be replaced

-

FDDI can transmits data at a rate of _____

100 million bits per second

203. million bits per second

204. million bits per second

None of the given

FDDI: Fiber distributed data interconnect (FDDI) is another ring technology. Its most important features are:

It uses fiber optics between stations and transmits data at 100Mbps. It uses pair of fibers to form two concentric rings.

Computers attached to an ether use ----- in which a computer waits for the ether to be idle before transmitting a frame.

CSMA/CD

CSMA/CA

TOKEN PASSING

None of the given

_____ has advantages arisen from the size and ease of computation.

CRC

Parity

Checksums

None of given

The chief advantages arise from the size and ease of computation. Most networks that employ a checksum technique use a 76-bit or J2-bit checksum, and compute a single checksum for an entire packet.

The term _____ is used to denote the definition of a packet used with a specific type of network.

Packet

Frame

Data

None of the given

To help distinguish between the general idea of packet transmission and the specific definition of packets for a given hardware technology, we use the term frame to denote the definition of a packet used with a specific type of network

_____ has no way to determine the cause of the problem.

Ping

Trace route

ICMP

Non of the given

Ping has no way to determine the cause of the problem. Another reason why ping may fail to generate a response is less subtle: some companies configure their site to reject ping packets.

_____ Program sends a message to a remote computer and reports whether the computer responds.

Ping

Trace route

ICMP

Non of the given

In _____, network occupies the smaller area like a room a floor or a building.

LAN

WAN

MAN

None of the given

A typical port on an ATM switch operates at _____ or higher.

a. OC-2 speed (155Mbps)

b. OC-3 speed (100Mbps)

205. **OC-3 speed (155Mbps)**

a. OC-3 speed (155Gbps)

A typical port on an ATM switch operates at OC-3 speed (155 Mbps) or higher, with the largest and most expensive switches handling much higher capacities

The product of delay and throughput measures the _____ of data that can be present on the network.

- 206. Area
- 207. Volume
- 208. Length
- 209. None of the given

To summarize: The product of delay and throughput measures the volume of data that can be present on the network. A network with throughput T and delay D can have a total of $T \times D$ bits in transit at any time

_____ is used for compressed audio and video where the data rate depends on the level of compression that can be achieved.

Constant Bit Rate (CBR) service

Variable Bit Rate (VBR) service

Available Bit Rate (ABR) service

None of the given

VARIABLE BIT RATE (VBR):

It is used for compressed audio and video where the data rate depends on the level of compression that can be achieved.

Which of the following is a connecting device?

Bridge

Repeater

Hub

All the given

A bridges function in the _____ layers(s).

Physical (MAC)

Data link

Network

Physical (MAC) and Data link ►

IEEE LLC/SNAP header is -----, which is used to specify the type of data.

8 octets

210. bytes

211. bits

None of the given

X Packet Type (PT), the five values of which were listed previously, in Table 7-2. The IEEE 802. IX Packet Body Length (PBL) follows the Packet Type. Because

the LLC/SNAP header is eight octets long, and the IEEE 802

The third field of the header consists of ----- bit Ethernet frame type.

212.

32

16

An -----method, the network hardware designers specify how type information is included in the frame and the value use to identify various frame types.

Explicit frame type

Ideal frame type

Implicit frame type

None of the given

EXPLICIT FRAME TYPE:

In this type the identifying value is included with frame describes types of included data.

Local Talk is a LAN technology that employs _____

Star topology

Bus topology

Ring topology

None of the given

Almost altogether network technologies use different wiring schemes. e.g., local talk uses hubs

(physical star) to simulate a bus topology.

The Fast Ethernet hardware operates at a rate of _____

10 Mbps

100 Mbps

1000 Mbps

None of the given

The Ethernet speed was originally 3Mbps, and the current standard is 10Mbps
the fast Ethernet operates at 100Mbps. There are also gigabits Ethernet available now

_____scheme, which is designed to help detect transmissions errors, send

one extra bit of information with each character

Parity

Checksums

CRC

None of given www.vuzs.net

Computer networks are often called ----- because they use packet technology.

Ethernet

Switch networks

Packet networks

None of the given

Computer networks are often called packet networks or packet switching networks because they use packet technology

-

_____Program sends a message to a remote computer and reports whether the computer responds.

Ping

Trace route

ICMP

Non of the given

The ping program sends a message to a remote computer and reports whether the computer responds; the traceroute program identifies intermediate computers along a path to a remote destination. Ping and traceroute software is included in

In _____ network occupies larger areas like cities & countries.

LAN

WAN

MAN

None of the given

In -----, network occupies the smaller area like a room a floor or a building.

LAN

WAN

MAN

None of the given

In *Point-to-Point* topology there are two topologies.

Tree and Ring

Star and Ring

Star and Tree

None of the given

POINT-TO-POINT:

In Point-to-Point topology there are two topologies.

213. STAR topology

214. TREE topology

215. If a computer on the network shares resources for others to use, it is called ____

a. Server

b. Client

c. Mainframe

Answer : a

216. Terminators are used in _____ topology.

217. Bus

218. Star

Answer :

a

219. In _____ topology, if a computer's network cable is broken, whole network goes down.

a. Bus

b. Star

Answer : a

220. For large networks, _____ topology is used.

221. Bus

222. Star

223. Ring

Answer :

b

5. ISO stands for

a. International Standard Organization

b. International Student Organization

c. Integrated Services Organization

Answer : a

224. ISO OSI model is used in

a. Stand alone PC

b. Network environment

Answer : b

225. Network cable lies on _____ layer

a. Application

b. Network

c. Physical

Answer : c

226. _____ layer decides which physical pathway the data should take.

a. Application

b. Network

c. Physical

Answer : c

227. ISDN is an example of _____ network

a. Circuit switched

b. Packet switched

Answer : a

10. X.25 is an example of _____ network

- a. Circuit switched
- b. Packet switched

Answer : b

228. _____ allows LAN users to share computer programs and data.

- a. Communication server
- b. Print server
- c. File server

Answer : c

229. Print server uses _____ which is a buffer that holds data before it is send to the printer.

- a. Queue
- b. Spool
- c. Node

Answer : b

230. A standalone program that has been modified to work on a LAN by including concurrency controls such as file and record locking is an example of_____

- a. LAN intrinsic software
- b. LAN aware software
- c. Groupware

- d. LAN ignorant software

Answer : a

231. The _____ portion of LAN management software restricts access, records user activities and audit data etc.

- a. Configuration management
- b. Security management

- c. Performance management

Answer : b

232. What is the max cable length of STP?

- a. 100 ft
- b. 200 ft
- c. 100 m
- d. 200 m

Answer : d

233. What is the max data capacity of STP? a. 10 mbps

- b. 100 mbps c.
- 1000 mbps d.
- 10000 mbps

Answer : b

234. Which connector STP uses?

- 235. BNC
- 236. RJ-11
- 237. RJ-45

238. RJ-

69 **Answer**

: c

239. What is the central device in star topology? a. STP server

- b. Hub/switch
- c. PDC

d. Router

Answer : b

240. What is max data capacity for optical fiber cable? a. 10 mbps

b. 100 mbps c.
1000 mbps d.
10000 mbps

Answer : c

241. Which of the following architecture uses CSMA/CD access method? a. ARCnet

b. Ethernet

Answer : b

Question # 1 of CBR stands for.....

Constant Byte Rate

Constant Bit Rate

Connection break Recovery

None of the above

CONSTANT BIT RATE (CBR):

It is used for audio and video, since these have predefined maximum data rates.

Question # 2 of

The routing table in interior and exterior switches has which one of the following?

It should have next hop for each possible destination

The next hop in table must be on shortest path to destination

Both (a) and (b)

None of the above

Question # 3 of 10

Which one of the following is a method for computing routing table information?

Manual entry

Static routing

Boot time routing

None of the above

How to compute routing table information:

Static routing - at boot time

Dynamic routing - allow automatic updates by a program

Question # 4 of 10

Which one of the following is the responsibility of Dijkstra' algorithm?

To compute the shortest path between two nodes
To extract next hop information from path information
To insert next hop information into routing table
All of the above

While computing shortest path, first we assume graph representation of network

at each node then we use Dijkstra's algorithm to compute shortest path from each node to every other node. Then extract next hop information from resulting path information and insert next hop information into routing tables.

Question # 5 of 10

Routing table is used to keep the information about.....

Destination

Medium of transmission

Routers used in network

All of the above

The routing table contains list of destination networks and next hop for each destination.

Question # 6 of 10 (Start time: 04:48:43 PM) Total M - 1

Frame relay is.....

Connection oriented service

Connectionless service

It is typically ranges from 5Mbps to 1000Mbps.

None of the above

It is used for Telco service for delivering blocks of data. It is connection based service and must contract with Telco for circuit between two endpoints. It is typically 56kbps or 1.5Mbps and can run to 100Mbps.

Question # 7 of 10 (Start time: 04:49:43 PM) Total M - 1

Which of the following is not a guided medium?

Twisted-pair cable

Fiber-optic cable

Atmosphere

Coaxial cable

Question # 8 of 10

Which of the following statement is wrong regarding PVC?

PVC last as long as the customer pay the periodic fee for its use.

Forwarding tables are automatically restored after power of equipment failure.

Forwarding table entries for PVC's are dynamically configured.

All of the above are wrong

ATM can provide customers with virtual circuits that look like traditional leased

digital circuits. Such permanent virtual circuits (PVC) last as long as the customer pay the periodic fee for its use. The forwarding tables are automatically restored after power of equipment failure. The forwarding table entries for such permanent VC's are statically configured, the terms used by Telco's for this is provisioning

Question # 9 of 10

Which of the following statement is true regarding ATM?

Select correct option:

ATM is a single technology that is designed to meet the goals of both LANs and WANs.

ATM uses the concept of connection-oriented networking.

Both (a) and (b)

None of the above

ATM is a single technology that is designed

to meet the goals of both LANs and WANs.

ATM uses the concept of connection-oriented networking

Question # 10 of 10

Dijkstra's algorithm is used to compute the.....

shortest path between two nodes

number of nodes used in the network

distance between different nodes

none of the above

While computing shortest path, first we assume graph representation of network

at each node then we use Dijkstra's algorithm to compute shortest path from each node to every other node. Then extract next hop information from resulting path information and insert next hop information into routing tables.

Question 1

NAT is not useful at a residence with Cable Modem or DSL connectivity.

True

False

NAT is useful at a residence with Cable Modem or DSL connectivity

Question 2

_____ includes a 32-bits address mask with each address, which allows the address to be classful, classless, or subnetted.

RIP

OSPF

BGP

None of the given

OSPF includes a 32-bit address mask with each address, which allows the address to be classful, classless, or subnetted.

Question 3

NAPT stands for _____

Network address and protocol translation

Network address and port translation

Network address and packet translation

None of the given

But Network Address and Port Translation (NAPT) (which is another modified form of NAT)

Question 4

Which of the following are interior routing protocols?

RIP

OSPF

BGP

RIP and OSPF

The Routing Information Protocol (RIP) and Open Shortest Path First (OSPF) are the two most well-known interior routing protocols used in the TCP/IP suite

Question 5

NAT stands for _____

Network address translation

Network address transmission

Network address test

None of the given

Question 6

In IP routing, forwarding refers to _____ transfer.

Packet

Datagram

Message

None of the given

FORWARDING refers to datagram transfer.

Question 7

Basic NAT does not work well for communication initiated from the Internet.

True

False

Basic NAT does not work well for communication initiated from the Internet

Question 8

To save traffic, an EGP does not summarize routing information from the autonomous system before passing it to another autonomous system

True

False

OSPF has designed as an Interior Gateway Protocol used to pass routing information among routers within an autonomous system.

Each Autonomous System chooses a routing protocol to exchange routing information which is summarized before being passed to another group.

Question 9

OSPF is based on-----

Distance vector routing

Link state routing

Path vector routing

Distance vector and link state routing

LINK-STATE ALGORITHM:

OSPF uses link-state routing.

Question 10

The Border Gateway Protocol (BGP) uses_____ for all communication

UDP

TCP

UDP and TCP

None of the given

"It uses reliable transport (TCP).

BGP uses the TCP/IP protocol on port 179 to establish connections

Question 11

_____ identifies the application program that sent the data.

Destination

Source port

Logical address

None of the given (not sure)

(Reference not available)

Question 12

_____ can be used to propagate information about remote networks

Dynamic routing

Static routing

Address resolution

None of the given

Dynamic routing can be used to propagate information about remote networks.

Question 13

Which of the following protocol uses UDP for all message transmission?

OSPF

BGP

RIP

BGP AND RIP

RIP uses UDP for all message transmissions

Question 14

OSPF stands for _____

Open Source Path First Protocol

Open shortest path first

Open Secure Path First Protocol

open shortest path first

(OSPF) known as the Open Shortest Path First Protocol).

Question 15

Typical internet routing uses a combination of two metrics_____.

Mouthing metrics

Administrative cost and routing metrics

Hop cost

Administrative cost and hop count

Typical internet routing uses a combination of two metrics Administrative cost and hop count

Question 16

_____ is a current version of BGP.

2

3

4

None of these

The current version is four (BGP-4).

Question 17

OSPF doesn't allow a router to introduce routes learned from another means.

True

False

SPF allows a router to introduce routes learned from another means (e.g., from BGP).

Question 18

Preliminary version of IP was called _____.

Select correct option:

IP - New Generation (IPng)

IP - Next Generation (IPng)

IP - Net Generation (IPng)

None of the given

A preliminary version of IP was called IP- Next Generation (IPng).

Question 19

Source is responsible for fragmentation.

Select correct option:

IPV4

IPV6

IPv6 source (not intermediate routers) is responsible for fragmentation

Question 20

IPv6 address consists of _____

Select correct option:

242. Bits

243. Bits

244. Bits

none of the given

IPv6 addresses are 128 bits.

Question

Header contains all information needed to deliver datagram to the destination computer.

But which one of the following is not included:

Destination address

Source address

Rectifier

Other delivery information

Header contains all information needed to deliver datagram to the destination computer

Destination address

Source address

Identifier

Other delivery information

_____ field is used to identify a specific path through the network

Flow label

Traffic class

Both a and b

None of the given

Flow label is partitioned into a TRAFFIC CLASS field and a separate FLOW LABEL field used to identify a specific path thorough the network.

The Source can configure outgoing datagram's to avoid _____

Segmentation

Defragmentation

Fragmentation

Source can configure outgoing datagrams to avoid fragmentation

Question

Every hardware technology specification includes the definition of the maximum size of the frame data area, which is called the _____ Transmission Unit.

Least

Maximum

Fragment

Frame

Every hardware technology specification includes the definition of the maximum size of the frame data area, which is called the Maximum Transmission Unit (MTU).

Question

Although the ARP message format is sufficiently general to allow arbitrary protocol and hardware addresses. ARP is almost always used to bind a 32-bit IP address to a ____ Ethernet address.

16-bit

48-bit

64-bit

128-bit

ARP is almost always used to bind a 32-bit IP address to a 48-bit Ethernet address

Question

Fragmentation when using ICMP for path MTU should be avoided.

True

False

Question

End to End delivery Service of IP datagram is _____

Connection oriented

Connectionless

both a and b

none of the given

End-to-end delivery service is connection less.

Question

HEADER LEN field gives size of extension header.

False

True

Question

_____ Serve Same purpose in Internet as frames on LAN

Packets

Virtual Packets

Virtual Frames

Both b and c

VIRTUAL PACKETS:

These packets serve same purpose in Internet as frames on LAN.

Question

Header contains all information needed to deliver datagram to the destination computer. But which one of the following is not included

Destination address

Source address

Rectifier

Other delivery information

Header contains all information needed to deliver datagram to the destination computer

245. *Destination address*

246. • *Source address*

247. • *Identifier*

248. • *Other delivery information*

Question

_____ encapsulates IP datagram as data area in hardware frame.

Select correct option:

Network Interface Layer

Data link Layer

Network Layer

None of the given

Network interface layer encapsulates IP datagram as data area in hardware frame.

Question

_____ field is used to identify a specific path through the network

Select correct option:

Flow label

Traffic class

Both a and b

None of the given

Flow label is partitioned into a TRAFFIC CLASS field and a separate FLOW LABEL field used to identify a specific path thorough the network.

Question

IPV6 address with _____ leading zeros is interpreted to hold an IPV4 address.

Select correct option:

96

100

120

none of the given

IPv6 address with 96 leading zeros is interpreted to hold an IPv4 address.

Question

Although the ARP message format is sufficiently general to allow arbitrary protocol and hardware addresses. ARP is almost always used to bind a 32-bit IP address to a ____ Ethernet address.

48bit

Other options are missing

ARP is almost always used to bind a 32-bit IP address to a 48-bit Ethernet address

Question

For _____, information about forwarding is stored in a routing table, which is initialized at system initialization and must be updated as network topology changes.

Select correct option:

Efficiency

Security

Accuracy

Anomalies

For efficiency, information about forwarding is stored in a routing table, which is initialized at system initialization and must be updated as network topology changes

Question

_____ is a technique used to Limit datagram size to small MTU of any network

Select correct option:

Segmentation

Fragmentation

Encapsulation

none of the given

FRAGMENTATION:

One technique is to limit datagram size to smallest MTU of any network

Question

Class A mask is 255.0.0.0 which is used for _____

Select correct option:

Unicasting

Multicasting

Subnetting

All of the given

class A mask is 255.0.0.0 which is used for subnetting.

Question

When one computer sends an ARP message to another the message travels inside the hardware frame. Technically, placing a message inside a frame for transport is not called encapsulation. Select correct option:

True

False

*Technically, placing a message inside a frame for transport is called encapsulation***Question**
Which one of these is not a main feature of connectionless service:

It includes extension of LAN abstraction.

It has universal addressing and the data is delivered in packets frames), each with a header.

It combines collection of physical networks into a single virtual network.

It has universal addressing and the data is delivered in packets frames), without a header.

The main features of connectionless service are

It includes extension of LAN abstraction. It has universal addressing and the data is delivered in packets (frames), each with a header. It combines collection of physical networks into a single virtual network

Question

Which protocol is used to test different tools

ICMP

IGMP

TCP/IP

none of the given

ICMP can also be used to test different tools.

Question

_____ message is sent in response to incoming datagrams with problems.

Select correct option:

TCP/IP

IGMP

ICMP

none of the given

ICMP message is sent in response to incoming datagrams with problems. ICMP message is not sent for ICMP message.

Question

Which one is NOT the function of ping program

Traceability

Reach ability

Both a and b

None of the given

Ping program tests reachability

Question

IPV6 address with _____ leading zeros is interpreted to hold an IPV4 address.

Select correct option:

96

100

120

none of the given

IPv6 address with 96 leading zeros is interpreted to hold an IPv4 address

Question

A datagram cannot be larger than _____ of a network over which it is sent.

Select correct option:

MTU

Size

IP header

None of the given

Datagram encapsulated in a hardware frame must be smaller than the MTU for that hardware

Question

The Source can configure outgoing datagram's to avoid _____

Segmentation

Defragmentation

Fragmentation

None of the given

Fragmentation should be avoided. Source can configure outgoing datagrams to avoid fragmentation

Question

Every hardware technology specification includes the definition of the maximum size of the frame data area, which is called the _____ Transmission Unit. Select correct option:

Least

Maximum

Fragment

Frame

Every hardware technology specification includes the definition of the maximum size of the frame data area, which is called the Maximum Transmission Unit (MTU).

Question

_____ Protocol provides error reporting mechanism.

Select correct option:

IGMP

SNMP

ICMP

none of the given

Internet Control Message Protocol (ICMP) provides error reporting mechanism.

Question

_____ Source is responsible for fragmentation.

Select correct option:

IPV4

IPV6

IPv6 source (not intermediate routers) is responsible for fragmentation

MTU Stands for _____

Minimum transmission unit

Maximum transmission unit

Multicast transmission unit

None of the given

MTU (Maximum Transmission Unit).

Question No: 1 (Marks: 1) - Please choose one

In Point-to-Point topology there are two topologies.

Star and Tree

Tree and Ring

Star and Ring

Star and Tree

None of the given

Question No: 2 (Marks: 1) - Please choose one

----- Program sends a message to a remote computer and reports whether the computer responds.

Ping

Traceroute

ICMP

Non of the given

Question No: 3 (Marks: 1) - Please choose one

----- has no way to determine the cause of the problem.

ICMP

Non of the given

Ping

Trace route

Question No: 4 (Marks: 1) - Please choose one

The term----- refers to the general concept of a small block of data

Packet

Frame

Data

None of the given

Question No: 5 (Marks: 1) - Please choose one

----- scheme, which is designed to help detect transmissions errors, send one extra

bit of information with each character

Parity

Checksums

CRC

None of given

Question No: 6 (Marks: 1) - Please choose one

Local Talk is a LAN technology that employs -----

Bus topology

Ring topology

None of the given

Star topology

Question No: 7 (Marks: 1) - Please choose one

Most LANs that employ ring topology use an access mechanism known as-----

TOKEN PASSING

CSMA/CA

TOKEN PASSING

None of the given

Question No: 8 (Marks: 1) - Please choose one

Ethernet uses a ----- bit static addressing scheme in which each device is assigned a unique address by the manufacturer.

64

48

32

8

Question No: 9 (Marks: 1) - Please choose one

Formally named _____ informally known as the thick wire Ethernet or Thick net.

[b]10 Base T (not sure)[/b]

10 Base 2

10 Base 5

10 Base T

None of the given

Question No: 10 (Marks: 1) - Please choose one

Formally named _____ informally known as the twisted pair Ethernet or TP

Ethernet.

10 Base 2

10 Base 2

10 Base 5

10 Base T

None of the given

Question No: 11 (Marks: 1) - Please choose one

The maximum size of an Ethernet segment is _____

249. meters

250. meters

251. meters

252. meters

None of the given

Question No: 12 (Marks: 1) - Please choose one
A Bridge can _____

Do all the above

Filter a frame

Forward a frame

Extend a LAN

Do all the above

Question No: 13 (Marks: 1) - Please choose one

_____ computes shortest paths in a graph by using weights on edges as a measure of distance.

Dijkstra's algorithm

Greedy algorithm

Distance vector algorithm

Dijkstra's algorithm

Non of the given (not sure)

Question No: 14 (Marks: 1) - Please choose one

_____ is used for compressed audio and video where the data rate depends on the level of compression that can be achieved.

Constant Bit Rate (CBR) service

Variable Bit Rate (VBR) service

Available Bit Rate (ABR) service

None of the given

Question No: 15 (Marks: 1) - Please choose one

Basic LAN technologies such as Ethernet, Token Ring, and FDDI use a ____.

Connection-oriented service paradigm (not sure)

Connectionless service paradigm

Connection-oriented service paradigm

Both Connectionless and Connection-oriented service paradigm

None of the given

Question No: 1 (Marks: 1) - Please choose one

No error detection scheme is perfect because transmission errors can affect the additional
information as well as the data

True

False

Question No: 2 (Marks: 1) - Please choose one

----- Program sends a message to a remote computer and reports whether the computer
responds.

Ping

Ping

Traceroute

ICMP

Non of the given

Question No: 3 (Marks: 1) - Please choose one

----- was especially concerned about the lack of high powered computers.

ARPA

IEEE

EIA

Non of the given

Question No: 4 (Marks: 1) - Please choose one

The term ----- is used to denote the definition of a packet used with a specific type of network.

Packet

Frame

Data

None of the given

Question No: 5 (Marks: 1) - Please choose one

Computer networks are often called ----- because they use packet technology.

Ethernet

Switch networks

Packet networks



None of the given

Question No: 6 (Marks: 1) - Please choose one

----- have advantages arisen from the size and ease of computation.

CRC

Parity

Checksums

None of given

Question No: 7 (Marks: 1) - Please choose one

Most LANs that employ ring topology use an access mechanism known as-----

CSMA/CD

CSMA/CA

TOKEN PASSING

None of the given

Question No: 8 (Marks: 1) - Please choose one

IEEE LLC/SNAP header is -----, which is used to specify the type of data.

8 octets

8 bytes

8 bits

None of the given

Question No: 9 (Marks: 1) - Please choose one

Formally named _____ informally known as the twisted pair Ethernet or TP

Ethernet.

10 Base 2

10 Base 5

10 Base T

None of the given

Question No: 10 (Marks: 1) - Please choose one

An interface for twisted pair Ethernet must have an _____ connector , and must generate signals according to the _____ specification.

RJ-45, 10 Base T

RJ-45, 10 Base 5

BNC, 10 Base 2

BNC, 10 Base T

Question No: 11 (Marks: 1) - Please choose one

A bridges function in the _____ layers(s).

Physical (MAC)

Data link

Network

Physical (MAC) and Data link

Question No: 12 (Marks: 1) - Please choose one

A Bridge can _____

Filter a frame

Forward a frame

Extend a LAN

Do all the above

Question No: 13 (Marks: 1) - Please choose one

A Bridge forwards or filters a frame by comparing the information in its address table to the frame's _____

Layer 2 source address

Source node's physical address

Layer 2 destination address

Layer 3 destination address

[b]Question No: 14 (Marks: 1) - Please choose one

_____ computes shortest paths in a graph by using weights on edges as a measure of distance. [/b]

Greedy algorithm

Distance vector algorithm

Dijkstra's algorithm

Non of the given

Question No: 15 (Marks: 1) - Please choose one

_____ is used for audio and video, since these have predefined maximum data

rates

Constant Bit Rate (CBR) service

Variable Bit Rate (VBR) service

Available Bit Rate (ABR) service

None of the given

Question No: 10 (Marks: 1) - Please choose one

Formally named _____ informally known as the twisted pair Ethernet or TP Ethernet.

253. 10 Base 2

254. 10 Base 5

255. **10 Base T**

256. None of the given

Question No: 14 (Marks: 1) - Please choose one

_____ is used for compressed audio and video where the data rate depends on the level of compression that can be achieved.

257. Constant Bit Rate (CBR) service

258. **Variable Bit Rate (VBR) service**

259. Available Bit Rate (ABR) service

260. None of the given

Question No: 3 (Marks: 1) - Please choose one

ATM assigns each VC a _____ identifier that is divided two parts to produce a hierarchy.

21-bit ►

22-bit ►

23-bit ►

24-bit ►

Question No: 6 (Marks: 1) - Please choose one

An interface for twisted pair Ethernet must have an _____ connector , and must generate signals according to the _____ specification.

RJ-45, 10 Base T ►

RJ-45, 10 Base 5 ►

BNC, 10 Base 2 ►

BNC, 10 Base T ►

Question No: 8 (Marks: 1) - Please choose one

A ----- provide a mechanism that a customer can use to set a physical address.

Static addressing scheme ►

Configurable addressing scheme ►

Dynamic addressing scheme ►

None of the given ►

Question No: 15 (Marks: 1) - Please choose one

In -----, network occupies the smaller area like a room a floor or a building

- 261. LAN
- 262. WAN
- 263. MAN
- 264. None of the given

α. Which network topology uses a Hub?

- β. **Star**
- γ. Bus
- δ. Mesh
- ε. Ring

Which of the following is not a network adapter?

- 265. 3 com LP III
- 266. 3 com Etherlink III ISA
- 267. 3 com Etherlink II Adapter
- 268. **3 com 3C508 ISA 16 BIT**

Question No: 15 (Marks: 1

_____ is used for audio a
rates

- 269. Constant Bit Rate (CBR
- 270. Variable Bit Rate (VBR)
- 271. Available Bit Rate (ABR
- 272. None of the given

- 273. Twisted Pair
- 274. not given

- 275. Radio Frequency
- 276. Fiber

Q1= Network administrator wants to build an update routing which algorithm he use?

Q2=100mbps Ethernet NIC is connected to control Hub which supports 10mbps. What will be Data Rate in this scenario?

Q3=NIC 100mbps Ethernet NIC and other speed 10mbps. Are connected with each other What will be NIC Data Rate in this scenario?

Q4=two far away site offices want to connect which is better LAN and WAN?

and 2 I don't remember sorry

paper by Jawad

Question No: 10 (Marks: 1) - Please choose one

Formally named _____ informally known as the twisted pair Ethernet or TP Ethernet.

277. 10 Base 2

278. 10 Base 5

279. 10 Base T

280. None of the given

Question No: 14 (Marks: 1) - Please choose one

_____ is used for compressed audio and video where the data rate depends on the level of compression that can be achieved.

281. Constant Bit Rate (CBR) service

282. Variable Bit Rate (VBR) service

283. Available Bit Rate (ABR) service

284. None of the given

Question No: 3 (Marks: 1) - Please choose one

Which network topology uses a Hub?

285. **Star**

286. Bus

287. Mesh

288. Ring

Which of the following is not a network adapter?

- 289. 3 com LP III
- 290. 3 com Etherlink III ISA
- 291. 3 com Etherlink II Adapter
- 292. **3 com 3C508 ISA 16 BIT**

Question No: 1 (Marks: 1) - Please choose one

A typical port on an ATM switch operates at _____ or higher.

- 293. OC-2 speed (155Mbps)
- 294. OC-3 speed (100Mbps)
- 295. **OC-3 speed (155Mbps)**

296. OC-3 speed (155Gbps)

Question No: 2 (Marks: 1) - Please choose one

The product of delay and throughput measures the _____ of data that can be present on the network.

297. Area

298. Volume

299. Length

300. None of the given

Question No: 3 (Marks: 1) - Please choose one

_____ is used for compressed audio and video where the data rate depends on the level of compression that can be achieved.

Constant Bit Rate (CBR) service ►

Variable Bit Rate (VBR) service ►

Available Bit Rate (ABR) service ►

None of the given ►

Question No: 4 (Marks: 1) - Please choose one

Which of the following is a connecting device?

Bridge ►

Repeater ►

Hub ►

All the given ►

Question No: 5 (Marks: 1) - Please choose one

A bridges function in the _____ layers(s).

Physical (MAC) ►

Data link ►

Network ►

Physical (MAC) and Data link ►

Question No: 6 (Marks: 1) - Please choose one

IEEE LLC/SNAP header is -----, which is used to specify the type of data.

8 octets ►

8 bytes ►

8 bits ►

None of the given ►

Question No: 7 (Marks: 1) - Please choose one

The third field of the header consists of ----- bit Ethernet frame type.

48 ►

32 ►

16 ►

8 ►

Question No: 8 (Marks: 1) - Please choose one

An -----method, the network hardware designers specify how type information is included in the frame and the value use to identify various frame types.

Explicit frame type ►

Ideal frame type ►

Implicit frame type ►

None of the given ►

Question No: 9 (Marks: 1) - Please choose one

Local Talk is a LAN technology that employs -----

Star topology ►

Bus topology ►

Ring topology ►

None of the given ►

Question No: 10 (Marks: 1) - Please choose one

The Fast Ethernet hardware operates at a rate of -----

301. Mbps ►

302.Mbps ►

1000 Mbps ►

None of the given ►

Question No: 11 (Marks: 1) - Please choose one

----- scheme, which is designed to help detect transmissions errors, send one extra bit of information with each character

► **Parity**

Checksums ►

CRC ►

None of given ►

Question No: 12 (Marks: 1) - Please choose one

Computer networks are often called ----- because they use packet technology.

Ethernet ►

Switch networks ►

► **Packet networks**

None of the given ►

Question No: 13 (Marks: 1) - Please choose one

----- Program sends a message to a remote computer and reports whether the computer responds.

► **Ping**

Ping

Traceroute ►

ICMP ►

Non of the given ►

Question No: 14 (Marks: 1) - Please choose one

In -----, network occupies larger areas like cities & countries.

LAN ►

► **WAN**

MAN ►

None of the given ►

Question No: 15 (Marks: 1) - Please choose one

In -----, network occupies the smaller area like a room a floor or a building

► LAN

WAN ►

MAN ►

None of the given ►

Question No: 16 (Marks: 1) - Please choose one

In *Point-to-Point* topology there are two topologies.

Tree and Ring ►

Star and Ring ►

► **Star and Tree**

None of the given ►

A popular LAN Technology that uses a shared Bus Topology is called.

WAN CSMA **Ethernet** Multicast

An electronic device that connects LAN segments

Ethernet 10Baset **Bridge** 10Base5

A form of addressing; in which a set of computers is assigned one address.

Broadcast Multicast Packet Switch CRC Bandwidth

A network technology that uses Collision Avoidance

Bridge Broadcast WAN **Wireless LAN**

Formal name for Thick Ethernet..

TDM Bandwidth **Network Analyzer** Broadcast

_____ has a jitter zero

▶ None of the given ▶ **Virtual Private Network** ▶ Isochronous Network ▶ Asynchronous Network

Unlike Frame Relay and ATM, SMDS (Switched multi-megabit Data service) offers_____.

▶ Connectionless service paradigm

▶ Connection oriented service paradigm

▶ Both Connectionless and Connection-oriented service paradigm ▶ None of the given

ATM assigns each VC a _____ identifier that is divided two parts to produce a hierarchy.

► 21-bit ► 22-bit ► 23-bit ► **24-bit**

Most WAN systems include a mechanism that can be used to eliminate the common case of duplication routing is called _____

► Hierarchal address ► **Default route** ► Shortest path ► None of the given

The next hop to which a packet is sent depends only on

► **Packet's destination** ► Packet's original source ► Path the packet has taken ►
None of the given

An interface for twisted pair Ethernet must have a _____ connector, and must generate signals according to the _____ specification.

► RJ -45, 10 Base T ► RJ-45, 10 Base 5 ► BNC, 10 Base 2 ► BNC, 10 Base T

When an application _____ data, it makes a copy of the data available to all other computers on the network.

► **Broadcasting** ► Multicasting ► Unicasting ► None of the

given

A _____ provide a mechanism that a customer can use to set a physical address .

► Static addressing scheme ► Configurable addressing scheme

- ▶ Dynamic addressingscheme
- ▶ None of the given

FDDI can transmits data at a rate of _____



- ▶ 100 million bits per second
- ▶ 100million bits per second
- ▶ 100million bits per second
- ▶ None of the given

Computers attached to an ether use ----- in which a computer waits for the ether to be idle before transmitting a frame.

- ▶ CSMA/CD
- ▶ CSMA/CA
- ▶ TOKEN PASSING
- ▶ None of the given

_____ has advantages arisen from the size and ease of computation.

▶ **CRC**

▶ Parity

▶ Checksums

▶ None of given

The term _____ is used to denote the definition of a packet used with a specific type of network.

▶ Packet

▶ **Fram**

e▶ Data

▶ None of the given

_____ has no way to determine the cause of the problem.

▶ **Ping**▶ Trace route▶ ICMP

▶ Non of the given

_____ Program sends a message to a remote computer and reports whether the computer responds.

▶ **Pin g**

▶ Trace route

▶ ICMP

▶ Non of the given

In _____, network occupies the smaller area like a room a floor or a building.

▶ LAN

▶ WAN

▶ MAN

▶ None of the given

No error detection scheme is perfect because transmission errors can affect the additional information as well as the data.

▶ False

▶ True

A typical port on an ATM switch operates at _____ or higher.

▶ OC-2 speed (155Mbps) ▶ OC-3 speed (100Mbps) ▶ OC-3 speed (155Mbps)

▶ OC-3 speed (155Gbps)

The product of delay and throughput measures the _____ of data that can be present on the network.

- ▶ Area
- ▶ Volume
- ▶ Length
- ▶ **None of the given**

_____ is used for compressed audio and video where the data rate depends on the level of compression that can be achieved.

- ▶ Constant Bit Rate (CBR) service
- ▶ **Variable Bit Rate (VBR) service**
- ▶ Available Bit Rate (ABR) service
- ▶ None of the given

Which of the following is a connecting device?

- ▶ Bridge
- ▶ Repeater
- ▶ Hub
- ▶ **All the given**

A bridges function in the _____ layers(s).

- ▶ Physical (MAC)
- ▶ Data link
- ▶ Network
- ▶ Physical (MAC) and Data link

IEEE LLC/SNAP header is -----, which is used to specify the type of data.

- ▶ 8octets
- ▶ 8bytes
- ▶ 8 bits
- ▶ None of the given

The third field of the header consists of ----- bit Ethernet frame type.

► 48

► **32**

► 16

► 8

An -----method, the network hardware designers specify how type information is included in the frame and the value use to identify various frame types.

► **Explicit frame type** ► Ideal frame type ► Implicit frame type ► None of the given

Local Talk is a LAN technology that employs _____

► Startopology ► **Bus topology** ► Ring topology ► None of the given

The Fast Ethernet hardware operates at a rate of _____

- ▶ 10Mbps ▶ **100 Mbps** ▶ 1000 Mbps ▶ None of the given

_____scheme, which is designed to help detect transmissions errors, send one extra bit of

information with each character

- ▶ **Parity** ▶ Checksums ▶ CRC ▶ None of given

Computer networks are often called ----- because they use packet technology.

- ▶ Ethernet ▶ **Switch networks** ▶ Packet networks ▶ None of the given

_____Program sends a message to a remote computer and reports whether the computer responds.

► **Ping** ► Traceroute ► ICMP ► Non of the given

In _____ network occupies larger areas like cities & countries.

► LAN ► **WAN** ► MAN ► None of the given

In -----, network occupies the smaller area like a room a floor or a building.

► **LAN** ► WAN ► MAN ► None of the given

In *Point-to-Point* topology there are two topologies.

Tree and Ring ► Star and Ring ► Star and Tree ► None of the given

Computer networks are often called ----- because they use packet technology.

- ▶ Ethernet
- ▶ **Switch networks**
- ▶ Packet networks
- ▶ None of the given

A network uses a -----arranges for computers to be connected in a closed loop.

- ▶ Star Topology
- ▶ **Ring Topology**
- ▶ Bus Topology
- ▶ None of the given

An -----method, the network hardware designers specify how type information is included in the frame and the value use to identify various frame types.

- ▶ **Explicit frame type**
- ▶ Ideal frame type
- ▶ Implicit frame type
- ▶ None of the given

An interface for thin Ethernet must have a _____ connector, and must generate signals according to the _____ specification.

- ▶ **RJ-45, 10 Base T** ▶ RJ-45, 10 Base 5 ▶ BNC, 10 Base 2 ▶ BNC, 10 Base T

A Bridge forwards or filters a frame by comparing the information in its address table to the frame's _____

- ▶ Layer 2 source address ▶ **Source node's physical address**
- ▶ Layer 2 destination address ▶ Layer 3 destination address

Most WAN systems include a mechanism that can be used to eliminate the common case of duplication routing is called _____

- ▶ Hierarchal address ▶ **Default route** ▶ Shortest path ▶ None of the given

_____ of TCP/IP layering model corresponds to basic network hardware.

► Physical Layer ► **Network Interface Layer** ► Internet Layer ► Transport Layer

_____ Protocols of TCP/IP layering model specify how to ensure reliable transfer.



► Physical Layer ► Network Interface Layer ► Internet Layer ► **Transport Layer**

_____ is called an end-to-end protocol because it provides a connection directly from an application on one computer to an application on a remote computer.

- ▶ IP
- ▶ **UDP**
- ▶ TCP
- ▶ None of the given

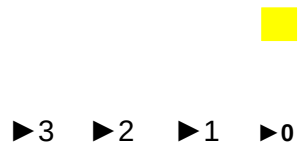
_____ uses distance vector approach to define routing

- ▶ BGP
- ▶ OSPF
- ▶ **RIP**
- ▶ None of the given

_____ is ideal in a situation where the group is small and all members are attached to contiguous Local Area Networks.

- ▶ **Flood-and -Prune**
- ▶ Configuration-and -Tunneling
- ▶ Core-Based Discovery
- ▶ None of the given

Router that decrements TTL to __ sends ICMP time exceeded message, with router's address as source address



Protocol addresses are abstractions provided by _____.



Although message exchange can be used to bind addresses, sending a request for each binding is hopelessly inefficient.

 **► True** ► False

ARP is almost always used to bind a ____-bit IP address to a ____-bit Ethernet address.



► 32, 48 ► 24, 32 ► 32, 64 ► 32, 128

In the 1970s large organizations began to acquire multiple networks. Each network in the organization formed island. Employees needed to choose a computer appropriate for each task. So they needed multiple screens, keyboards and computers.

► False ► **True**

In which method of Address Resolution Protocol the protocol address is determined by hardware address? Were "T" stands for Table lookup, "C" for Closed-form Computation and "D" for Data Exchange?

► T ► D ► **C** ► T, C

The general form of an IP datagram is with a header followed by data. The header contains information that controls where and how the datagram is to be sent.

► **True** ► False

To save traffic, an EGP does not summarize routing information from the autonomous system before passing it to another autonomous system.

► True ► **False**

Which of the following is a correct representation of the IPv6?

- ▶ 105.220.136.100.255.255.255.255.0.0.18.128.140.10.255.255
- ▶ 105.220.136.100.255.255.255.256.0.0.18.128.140.10.255.255
- ▶ 105.220.136.100.255.255.255.255.0.0.18.128.140.10.255.255.256
- ▶ 105.220.136.100.255.255.255.255.0.0.18.128.140.10.255

A _____ relies on the hardware manufacturer to assign a unique physical address to each network interface is called.

- ▶ Static addressing scheme
- ▶ Configurable addressing scheme
- ▶ Dynamic addressing scheme
- ▶ None of the given

A system with redundant bridges might have a problem with _____ in the system.

- ▶ Loop
- ▶ Filters
- ▶ Spanning Trees
- ▶ All given choices

A Bridge can _____

- ▶ Filter a frame
- ▶ Forward a frame
- ▶ Extend a LAN
- ▶ **Do all the above**

_____ is used for typical data applications (where the data rate may be unknown and bursty) and allows use of whatever bandwidth is available at a given time.

- ▶ Constant Bit Rate (CBR) service
- ▶ Variable Bit Rate (VBR) service

- ▶ **Available Bit Rate (ABR) service**
- ▶ None of the given

ATM assigns each VC a _____ identifier that is divided into two parts to produce a hierarchy.

- ▶ 21-bit
- ▶ 22-bit
- ▶ 23-bit
- ▶ **24-bit**

_____ of TCP/IP layering model corresponds to basic network hardware.

► **Physical Layer**

► Network Interface Layer

► Internet Layer

► Transport Layer

_____ places the boundary between the second and third octets



► **Class A**

► Class B

► Class C

► Class D

UDP and TCP are both _____ layer protocols



► Physical

► Data link

► Network

► **Transport**

Connection-oriented service, Point-to-point, Complete reliability, Full-duplex communication, Stream interface, Reliable connection startup and Graceful connection shutdown are the services provided by_____



- ▶ IP ▶ None of the given ▶ **TCP** ▶ UDP

_____protocols of TCP/IP layering model specify how to ensure reliable transfer.



- ▶ Physical Layer ▶ Network Interface Layer ▶ Internet Layer ▶ **Transport Layer**

_____identifies which application program on receiving computer should receive the data

Destination port

Source port

Logical address

None of the given

_____ identifies the application program that sent the data.

- ▶ Destination Port ▶ **Source port** ▶ Logical address ▶ None of the given

Which of the following are interior routing protocols?

- ▶ RIP ▶ OSPF ▶ BGP ▶ **RIP and OSPF**

The Border Gateway Protocol (BGP) uses_____ for all communication

- ▶ UDP ▶ **TCP** ▶ Both UDP and TCP ▶ None of the given

_____ measures distance in network hops, where each network between the source and

destination counts as single hop.



- ▶ BGP
- ▶ OSPF
- ▶ **RIP**
- ▶ None of the given

OSPF is based on-----

- ▶ Distance vector routing
- ▶ **Link state routing**
- ▶ Path vector routing
- ▶ Distance vector routing and Link state routing

_____ performs local multicast and uses IP-in-IP encapsulation to send multicast datagram's from one site on the Internet to another.

)

► Core Based Trees (CBT)

► Protocol Independent Multicast_ *Sparse Mode (PIM-SM)*

► Protocol Independent Multicast _ *Dense Mode (PIM-DM)*

The length of time required to send a variable length packet is variable and does not require a complicated interrupt scheme to detect completion of transmission.



► True ► False

NEXT HEADER field in the base header defines type of header and it appears at end of fixed-size base header.



► True ► False

Although message exchange can be used to bind addresses, sending a request for each binding is hopelessly inefficient.



► True ► False

Address mask defines how many bits of address are in suffix?



► True ► False

A computer attached to a given network can only communicate with other computers attached to the same network. Is this a problem with multiple networks?



► True ► False

The term self-identifying is used for Classful IP addresses because the class of the address can be computed from the address_____.



► **itsel** ► prefix ► suffix ► mask

In which method of Address Resolution Protocol the protocol address independent of hardware address?

Were "T" stands for Table lookup, "C" for Closed-form Computation and "D" for Data Exchange?



► T, C ► D ► **C** ► T, D

Reconstruction of original datagram is called reassembly.



► **True** ► False

A computer needs a complete stack of protocols to run either a client or a server.



▶ **True** ▶ False

TCP uses _____ mechanism to control the flow of data.



▶ Door ▶ **Window** ▶ Acknowledgment ▶ Retransmission

In CRC there is no error if the remainder at the receiver is

Equal to the Remainder of the sender **Zero** Non Zero The quotient at the sender

Router is _____ Device.

Layer one Intelligent Unintelligent **None of above**

The repeater works at _____ Layer of OSI model

Data Link layer **Physical layer** Transport Layer Network Layer

Bridges are _____ Devices

Networking Internetworking All of above None of above

The address field of a frame in HDLC protocol contains the address of the _____ station.

Primary Secondary Tertiary **Primary and secondary**

WDM stands for

Wave Division Multiplexing Wrong Division Multiplexing

Well Division Multiplexing

West Division Multiplexing

A traditional telephone line has a bandwidth of _____.

2000 Hz

4000 Hz

2000 MHz

4000 MHz

Time slots are dynamically allotted in _____ TDM technique.

FDM Asynchronous WDM **Synchronous**

A portion of the path that carries TX b/w a given pair of devices is known as _____.

Node Bridge **Channel** Access Point

T Lines are _____ lines designed for digital data

Analog **Digital** Composite Telephone

Multiplexing has long been used as an essential tool in _____.

Electronic industry **Telephone industry** Space science VLAN

Switched analog service is leased line.

True **False**

If a T-1 carries 8000 frames, then data rate will be _____

2.544 Mbps **1.544 Mbps** 1.544 kbps 1.544 Gbps

Given a 10-bit sequence 1010011110 and a divisor of 1011, CRC will be

001 000 100 **101**

Asynchronous TDM is efficient only when the size of the time slot is kept relatively _____

Large Small **Medium** Zero

FTTC can be implemented in _____.

Telephone system Cable System **Telephone and Cable System** Non of the above

Europeans use a version of T-lines called _____

F-lines **E-line** D-lines S-lines

A prism can deflect the light depending upon the angle of _____ and the frequency.

Deviation Incident Refraction **Reflection**

Which error detection method can detect a single-bit error?

Simple parity check Two-dimensional parity check CRC All of the given

Which of the following techniques takes data from one high speed line and breaks it into portions?

Inverse subtraction Inverse addition **multiplexing** inverse multiplexing

Telephone network is an example of a _____ network

Packet-switched

Circuit-switched

Message-switched

Tele-switched

FTTC stands for _____

fiber to the center

fiber to the cable

fiber to the curb

flexible to the curb

Cable that connects the subscriber telephone to the nearest end office in a local loop is called

Twisted-pair cable

Coaxial cable

Fiber-optic cable

Subscriber cable

In Synchronous TDM, Time slots are not same in number as input devices

True False

In Synchronous TDM, Time slots are not same in number as input devices.

True False

At the CRC checker, _____ means that the data unit is damaged

A string of 0s -- A string of 1s --A string of alternating 1s and 0s -**A nonzero-remainder**

Which of the following best describes a single-bit error?

A single bit is inverted **A single bit is inverted per data unit**

A single bit is inverted per transmission none of the given

Multiplexing is the set of techniques that allows simultaneous TX of multiple signals across a

_____ data link

Single

Double

Multi

None of the given

If the ASCII character G is sent and the character D is received, what type of error is this?

Single-bit

Multiple-bit

Burst

Recoverable

Which error detection method involves polynomials?

Simple parity check

Two-dimensional parity check

CRC

Checksum

For point-to-point configuration, addressing is needed

True **False**

Data link control is composed of _____ important functions.

2 **3** 4 5

The checksum of 0000 and 0000 is _____

1111 **0000** 1110 0111

_____ is used to attaché two autonomous system

BGP IGP **EGP** NONE OF

The border Gateway Protocol (BGP) uses_____ for all communication

UDP **TCP** UDP AND TCP Non of the given

Routing insert or changes value in _____

MAC address **routing table** both (a) and (b) non of the given

_____ does not depend on any particular unicast routing protocol

PIM-DM **PIM-SM** PIM-AM NON OF THE GIVEN

NAT software does not allow a PC to connect with the internet and act as a NAT device at the same time
.

True **false**

Each autonomous system used to communicate among autonomous systems by chooses an IGP.

TRUE. FALSE

Which of the following protocols provide the information at the autonomous system level.?

BGP OSPF **RIP** OSPF and RIP

Which of the following protocols allow the sender and receiver to enforce policies.

RIP SOPF **BGP** BGP and RIP

Interior Gateway Protocols (IGPs) and Exterior Gateway Protocols (EGPs) two broad classes of Internet Routing Protocol.

TRUE FALSE .

The computer uses_____ to inform local router about the last application when it leaves.

ICMP **IGMP** SNMP None of the given

We use the term _____ to refer to a measure of the path that routing software uses when choosing a route.

Routing path routing metric **routing** switching

_____ is a current version of BGP.

2 3 4 None of the given

OSPF is based on _____

Distance vector routing **link state routing** path vector routing Distance vector routing and link state routing

In _____ routing, the table is initialized when system boots and there are no further changes

Dynamic **static** both a and b none of the given

doesn't allow a router to introduce routes learned from another means

true **fales**

Twice NAT fails if an application uses the IP addresses instead of Domain Name.

TRUE FALSE

A one-to-many communication between a source and a specific group of hosts is classified as a _____

communication.

Unicast multicast **broadcast** unicast & multicast

_____ identifies the application program that sent the data.

Destination port source port logical port none of the given

Twice NAT is another variant of NAT. it is used with site that runs server. In this process NAT box is connected to Domain Name.

True false

OSPF includes _____ address mask with each address.

30bit **32bit** 34bit none of the given

Twice NAT allows a site to run servers

True false

OSPF has designed as an Exterior Gateway Protocol used to pass routing information among routers within an autonomous system.

True **false**

In IP routing, forwarding refers to _____ transfer

Packet datagram message none of the given

It allows _____ computer(s) to share _____ IP address(s).

Multiple , multiple **multiple , single** single , single none of the given

Network Address Translation (NAT) requires device to perform packet translation

True fales

_____ encapsulates IP datagram as data area in hardware frame.

Network interface layer datalink layer network layer none of the given

_____ field is used to identify a specific path through the network

Flow lable traffic class both a and b none of the given

IPV6 address with _____ leading zeros is interpreted to hold an IPV4 address.

96 100 120 none of the given

Although the ARP message format is sufficiently general to allow arbitrary protocol and hardware addresses. ARP is almost always used to bind a 32-bit IP address to a _____ Ethernet address.

48bit

For _____, information about forwarding is stored in a routing table, which is initialized at system initialization and must be updated as network topology changes.

Efficiency Security Accuracy Anomalies

_____ is a technique used to Limit datagram size to small MTU of any network

Segmentation **Fragmantation** Encapsulation none of the given

Class A mask is 255.0.0.0 which is used for _____

Unicast multicast **subnetting** all of the given

When one computer sends an ARP message to another the message travels inside the hardware frame.

Technically, placing a message inside a frame for transport is not called encapsulation.

True **false**

Which one of these is not a main feature of connectionless service:

It includes extension of LAN abstraction.

It has universal addressing and the data is delivered in packets frames), each with a header.

It combines collection of physical networks into a single virtual network.

It has universal addressing and the data is delivered in packets frames), without a header.

Which protocol is used to test different tools.

ICMP IGMP TCP/IP None of the given

_____ message is sent in response to incoming datagrams with problems.

TCP/IP IGMP **ICMP** none of the given

Which one is NOT the function of ping program

TRACABILITY REACH ABLITY BOTH a and b none of the given

IPV6 address with _____ leading zeros is interpreted to hold an IPV4 address

96 100 120 none of the

A datagram cannot be larger than _____ of a network over which it is sent.

MTU size IP header none of the

The Source can configure outgoing datagram's to avoid _____

Segmentation defragmentation **Fragmentation** none of the

Every hardware technology specification includes the definition of the maximum size of the frame data area, which is called the _____ Transmission Unit

Least **maximum** fragment frame

_____ Protocol provides error reporting mechanism

IGMP SNMP **ICMP** NONE OF THE

_____ Source is responsible for fragmentation.

IPV4 **IPV6**

MTU Stands for _____

Minimum transmission unit **Maximum transmission unit** Multi transmission unit none of the

TTL stands for _____

Time to Learn Time to Leave Time to Live **none of the given**

H.LEN shows the header length in units of ____ Bits

34 **32** 30 None of the given

There are _____ possibilities to detect the destination using Trace-route

1 **2** 3 None of the given

_____ Source is responsible for fragmentation.

IPV4 **IPV6**

_____ encapsulates IP datagram as data area in hardware frame.

Network Interface Layer Data link Layer Network Layer none of the given

IPV6 address consists of _____

32 Bits 64 Bits **128 Bits** none of the given

_____ shows senders preference for low latency, high Reliability.

:

TYPE **SERVICE TYPE** SERVICE PRIORITY None of the given

Class A mask is 255.0.0.0 which is used for _____

Unicasting Multicasting **Subnetting** All of the given

_____ Protocol provides error reporting mechanism.

IGMP SNMP **ICMP** none of the given

ICMP message transport is acted upon by getting ICMP _____ in IP.

De-encapsulated **Encapsulated** Segmented none of the given

Which protocol is used to test different tools.

ICMP IGMP TCP/IP none of the given

When one computer sends an ARP message to another the message travels inside the hardware frame. Technically, placing a message inside a frame for transport is not called encapsulation. True **False**

Preliminary version of IP was called _____.

IP - New Generation (IPng) **IP - Next Generation (IPng)** IP - Net Generation (IPng) None of the

given

Header contains all information needed to deliver datagram to the destination computer. But which one of the following is not included:

Destination address Source address **Rectifier** Other delivery information

_____ Source is responsible for fragmentation.

IPV4 **IPV6**

_____ layer Provides reliable delivery of datagram.

Network **Transport** Data link none of the given

Which one of these is not a main feature of connectionless service:

It includes extension of LAN abstraction.

It has universal addressing and the data is delivered in packets frames), each with a header.

It combines collection of physical networks into a single virtual network.

It has universal addressing and the data is delivered in packets frames), without a header.

The process of learning the path MTU is known as path MTU discovery.

True False

IPv6 _____ is responsible for fragmentation. Routers simply drop datagram's larger than
network

Destination Intermediate routers **Source** Medium

IP datagram can contains _____ octets.

0-65,535 **1-65,535** 1-65,536 none of the given

_____ shows senders preference for low latency, high Reliability.

TYPE **SERVICE TYPE** SERVICE PRIORITY None of the given

Computer networks are often called ----- because they use packet technology.

► Ethernet ► **Switch networks** ► Packet networks ► None of the given

A network uses a -----arranges for computers to be connected in a closed loop.

Ring Topology

► Star Topology ► **Ring Topology** ► Bus Topology ► None of the given

An -----method, the network hardware designers specify how type information is included in the frame and the value use to identify various frame types.

- **Explicit frame type** ► Ideal frame type ► Implicit frame type ► None of the given

Most WAN systems include a mechanism that can be used to eliminate the common case of duplication routing is called _____

- Hierarchal address ► **Default route** ► Shortest path ► None of the given

_____ of TCP/IP layering model corresponds to basic network hardware.

► Physical Layer ► **Network Interface Layer** ► Internet Layer ► Transport Layer

_____ Protocols of TCP/IP layering model specify how to ensure reliable transfer.



► Physical Layer ► Network Interface Layer ► Internet Layer ► **Transport Layer**

_____ is called an end-to-end protocol because it provides a connection directly from an

application on one computer to an application on a remote computer.

► IP  ► TCP ► None of the given

_____ uses distance vector approach to define routing



► BGP ► OSPF ► **RIP** ► None of the given

_____ is ideal in a situation where the group is small and all members are attached to contiguous

Local Area Networks.



► **Flood-and -Prune** ► Configuration-and -Tunneling ► Core-Based Discovery ► None of the given

Router that decrements TTL to __ sends ICMP time exceeded message, with router's address as source address



▶3 ▶2 ▶1 ▶0

Protocol addresses are abstractions provided by _____.



▶ Hardware ▶ **Software** ▶ Operating system ▶ Internet

Although message exchange can be used to bind addresses, sending a request for each binding is hopelessly inefficient.



▶ True ▶ False

ARP is almost always used to bind a ____-bit IP address to a ____-bit Ethernet address.



- ▶ **32, 48**
- ▶ 24, 32
- ▶ 32, 64
- ▶ 32, 128

To save traffic, an EGP does not summarize routing information from the autonomous system before passing it to another autonomous system.



- ▶ True
- ▶ **False**

When an application_____data, it makes a copy of the data available to all other computers on the network.

- ▶ **Broadcasting**
- ▶ Multicasting
- ▶ Unicasting
- ▶ None of the given

Ethernet uses a _____ bit static addressing scheme in which each device is assigned a unique address by the manufacturer.



▶ 64 ▶ 48 ▶ 32 ▶ 8

The product of delay and throughput measures the _____ of data that can be present on the network.



▶ Area ▶ Volume ▶ Length ▶ None of the give

Connectionless service, Message-Oriented protocol, best effort delivery service, arbitrary interaction and operating system independent are the characteristics of _____



▶ TCP ▶ UDP ▶ IP ▶ None of the given

Connection-oriented service, Point-to-point, Complete reliability, Full-duplex communication, Stream interface, Reliable connection startup and Graceful connection shutdown are the services provided by_____

- 
- ▶ None of the given ▶ **TCP** ▶ UDP ▶ IP

The process of using a routing table to select a next hop for a given datagram is called_____

- 
- ▶ Encapsulation ▶ Reassembling ▶ **Routing or forwarding** ▶ None of the given

_____uses distance vector approach to define routing

- 
- ▶ BGP ▶ OSPF ▶ **RIP** ▶ None of the given

A multicast routing scheme in which the protocol software builds a delivery tree from a central point is called_____

► **Distance Vector Multicast Routing Protocol (DVMRP)**

► Core Based Trees (CBT)

► Protocol Independent Multicast_ *Sparse Mode (PIM-SM)*

► Protocol Independent Multicast_ *Dense Mode (PIM-DM)*

One repeater _____ two repeaters _____ the maximum cable length limitation.



► doubles, cancel ► **doubles, triple** ► square roots, cube roots ► and, triple

Whenever it handles a packet, IP software needs to separate the destination address into a

_____ and _____.

► postfix, infix ► non of these ► Infix, prefix ☒ x

End-to-end delivery service is connection oriented.

☐
► True ► False

A single networking technology is best for all needs.

☐
► True ► False

We use the term _____ to refer to a measure of the path that routing software use when choosing a route.

☐

► routing path ► routing metric ► routing ► switching

No error detection scheme is perfect because transmission errors can affect the additional information as well as the data.

303. ☒ True ► ☐ False

----- Program sends a message to a remote computer and reports whether the

computer responds.

- ▶ **Ping**
- ▶ Traceroute
- ▶ ICMP
- ▶ Non of the given

----- was especially concerned about the lack of high powered computers.

- ▶ **ARPA**
- ▶ IEEE
- ▶ EIA
- ▶ Non of the given

The term ____ is used to denote the definition of a packet used with a specific type of network.

- ▶ Packet
- ▶ **Frame**
- ▶ Data
- ▶ None of the given

Computer networks are often called ----- because they use packet technology.

- ▶ Ethernet
- ▶ Switch networks
- ▶ Packet networks
- ▶ **None of the given**

----- have advantages arisen from the size and ease of computation.

- ▶ CRC
- ▶ Parity
- ▶ Checksums
- ▶ None of given

Most LANs that employ ring topology use an access mechanism known as-----

- ▶ CSMA/CD
- ▶ CSMA/CA
- ▶ **TOKEN PASSING**
- ▶ None of the given

IEEE LLC/SNAP header is -----, which is used to specify the type of data.

- ▶ **8 octets**
- ▶ 8 bytes
- ▶ 8 bits
- ▶ None of the given

Formally named _____ informally known as the twisted pair Ethernet or TP Ethernet.

- ▶ 10 Base 2
- ▶ 10 Base 5
- ▶ **10 Base T**
- ▶ None of the given

twisted pair ethernet

304. Formally called 10 Base -T

305. Also twisted pair Ethernet or simply TP Ethernet

An interface for twisted pair Ethernet must have an _____ connector , and must generate signals according to the _____ specification.

- ▶ **RJ-45, 10 Base T**
- ▶ RJ-45, 10 Base 5
- ▶ BNC, 10 Base 2
- ▶ BNC, 10 Base T
- ▶ A

bridges function in the _____ layers(s).

- ▶ Physical (MAC)
- ▶ **Data link**
- ▶ Network
- ▶ Physical (MAC) and Data link

A Bridge can _____

- ▶ Filter a frame
- ▶ Forward a frame
- ▶ Extend a LAN
- ▶ **Do all the above**

A Bridge forwards or filters a frame by comparing the information in its address table to

the frame's _____

- ▶ Layer 2 source address
- ▶ Source node's physical address

► **Layer 2 destination address** ► Layer 3 destination address

_____ computes shortest paths in a graph by using weights on edges as a measure of distance.

306. Greedy algorithm ► Distance vector algorithm ► **Dijkstra's algorithm**

a. Non of the given

_____ is used for audio and video, since these have predefined maximum data rates

► **Constant Bit Rate (CBR) service** ► Variable Bit Rate (VBR) service

► Available Bit Rate (ABR) service ► None of the given

Unlike Frame Relay and ATM, SMDS (Switched multi-megabit Data service) offers_____ .

► **Connectionless service paradigm** ► Connection oriented service paradigm

307. Both Connectionless and Connection-oriented service paradigm ► None of the given

A network with throughput T and delay D has a total of _____ bit in transit at any time.

► T / D ► **$T \times D$** ► $T + D$ ► None of the given

ATM is designed to work on_____.

► Twisted Pair ► Coaxial ► Radio Frequency ► **Fiber**

ATM is designed to work on fiber (but can be used with twisted pair). (P# 72)

Computers attached to an ethernet use ----- in which a computer waits for the ether to be idle before transmitting a frame.

308. CSMA/CD ► **CSMA/CA** ► **TOKEN PASSING** ► **None of the**
given **FDDI** can transmits data at a rate of -----

309. 100 million bits per second
► 10 million bits per second ► 1000 million bits per second ► None of the given

FDDI: Fiber distributed data interconnect (FDDI) is another ring technology. Its most important features are: It uses fiber optics between stations and transmits data at 100Mbps.

A typical port on an ATM switch operates at _____ or higher.

► OC-2 speed (155Mbps) ► **OC-3 speed (100Mbps)**
► **OC-3 speed (155Mbps)** ► OC-3 speed (155Gbps)

ATM DETAILS:

310. It transmits data at over 100Mbps.
a. It uses fiber optics to connect computer to switch.
b. Each connection includes two fibers.
c. It is also shown in figure.

The product of delay and throughput measures the _____ of data that can be present on the network.

► Area

► Volume

► Length

► None
of the
given

Ans is amount
of data

DELAY THROUPTUT PRODUCT:

It is computed as delay time multiplied by effective throughput. It measures amount of data that can be present in the network.

_____ is used for compressed audio and video where the data rate depends on the level of compression that can be achieved.

Constant Bit Rate (CBR) service ► **Variable Bit Rate (VBR) service** ►

Available Bit Rate (ABR) service ► None of the given ►

VARIABLE BIT RATE (VBR):

It is used for compressed audio and video where the data rate depends on the level of compression that can be achieved.

Which of the following is a connecting device?

Bridge ► Repeater ► Hub ► **All the given ►**

A bridges function in the _____ layers(s).

Physical (MAC) ► **Data link ►** Network ► Physical (MAC) and Data link

Switches and bridges use MAC addressing to make networking decisions and therefore these types of equipment function on the data link layer.

Question No: 6 (Marks: 1) - Please choose one

IEEE LLC/SNAP header is -----, which is used to specify the type of data.

8 octets ► 8 bytes ► **8 bits ►** None of the given

Question No: 7 (Marks: 1) - Please choose one

The third field of the header consists of ----- bit Ethernet frame type.

48 ► 32 ► 16 ► **8 ►**

Question No: 8 (Marks: 1) - Please choose one

An ----- method, the network hardware designers specify how type information is included in the frame and the value use to identify various frame types.

Explicit frame type ► Ideal frame type ► Implicit frame type ► None of the given ►

EXPLICIT FRAME TYPE:

In this type the identifying value is included with frame describes types of included data.

Local Talk is a LAN technology that employs -----

Star topology ► **Bus topology** ► Ring topology ► None of the given ►

Almost altogether network technologies use different wiring schemes. e.g., local talk uses hubs (physical star) to simulate a bus topology.

The Fast Ethernet hardware operates at a rate of -----

311. Mbps ► **100 Mbps** ► 1000 Mbps ► None of the given ►

The Ethernet speed was originally 3Mbps, and the current standard is 10Mbps
the fast Ethernet operates at 100Mbps. There are also gigabits Ethernet available now

----- scheme, which is designed to help detect transmissions errors, send one extra bit of
information with each character

► Parity Checksums ► CRC ► None of given

Computer networks are often called ----- because they use packet technology.

Ethernet **Switch networks** Packet networks None of the given

-----Program sends a message to a remote computer and reports whether the computer responds.

Ping Traceroute ICMP Non of the given

Ping program tests reachability. It sends datagram from B to A, that echoes back to B. it uses ICMP echorequest and echo reply messages.

In -----, network occupies larger areas like cities & countries.

► LAN ► **WAN** ► MAN ► None of the given

WIDE AREA NETWORK (WAN):

It is used for a country level networking and even for continents.

In -----, network occupies the smaller area like a room a floor or a building

► **LAN** ► WAN ► MAN ► None of the given

LOCAL AREA NETWORK (LAN): It is used for a single building.

In *Point-to-Point* topology there are two topologies.

► Tree and Ring

► Star and Ring

► **Star and Tree**

► None of the given

_____ has a jitter zero

► None of the given ► **Virtual Private Network** ► Isochronous Network ►

Asynchronous Network

Unlike Frame Relay and ATM, SMDS (Switched multi-megabit Data service) offers_____ .

Connectionless service paradigm ► **Connection oriented service paradigm**

Both Connectionless and Connection-oriented service paradigm ► None of the given ►

ATM assigns each VC a _____ identifier that is divided two parts to produce a hierarchy.

21-bit ► 22-bit ► 23-bit ► **24-bit** ►

Most WAN systems include a mechanism that can be used to eliminate the common case of duplication routing is called_____

Hierarchical address ► **Default route** ► Shortest path ► None of the given ►

The next hop to which a packet is sent depends only on

► **Packet's destination** ► Packet's original source ► Path the packet has taken ► Non of the given ►

An interface for twisted pair Ethernet must have an _____ connector , and must generate signals according to the_____ specification.

RJ-45, 10 Base T ► RJ-45, 10 Base 5 ► BNC, 10 Base 2 ► BNC, 10 Base T ►

When an application----- data, it makes a copy of the data available to all other computers on the network.

Broadcasting ► Multicasting ► Unicasting ► None of the given ►

A ----- provide a mechanism that a customer can use to set a physical address.

Static addressing scheme ► **Configurable addressing scheme** ► Dynamic addressing scheme ► None of the given ►

FDDI can transmits data at a rate of ----- **100 million bits per second** -----

100 million bits per second ► 100 million bits per second ► 100 million bits per second ► None of the given ►

Computers attached to an ether use ----- in which a computer waits for the ether to be idle before transmitting a frame.

► **CSMA/CD** CSMA/CA ► TOKEN PASSING ► None of the given ►

----- have advantages arisen from the size and ease of computation.

► **CRC** Parity ► Checksums ► None of given ►

The term ----- is used to denote the definition of a packet used with a specific type of network.

► Packet ► **Frame** ► Data ► None of the given

----- has no way to determine the cause of the problem.

► **Ping** ► Trace route ► **ICMP** ► Non of the given

----- Program sends a message to a remote computer and reports whether the computer responds.

Ping ► Traceroute ► ICMP ► Non of the given

In -----, network occupies the smaller area like a room a floor or a building

► **LAN** ► WAN ► MAN ► None of the given

No error detection scheme is perfect because transmission errors can affect the additional information as well as the data.

True **False**

312. **In Point-to-Point topology there are two topologies.**

► Tree and Ring ► Star and Ring ► **Star and Tree** ► None of the given

In *Point-to-Point* topology there are two topologies.

313. STAR topology

314. TREE topology

----- **Program sends a message to a remote computer and reports whether the computer responds.**

► **Ping** (Not Sure) ► Traceroute ► ICMP ► Non of the given

Ping program tests reach ability. It sends datagram from B to A, that echoes back to B. it uses ICMP echo request and echo reply messages.

----- has no way to determine the cause of the problem.

► ICMP ► **Ping** ► Trace route ► Non of the given

Ping has no way to determine the cause of the problem. Tracing A Route Network administrators use another tool, trace route.

The term----- ----- refers to the general concept of a small block of data

► **Packet** ► Frame ► Data ► None of the given

Packet is a generic term that refers to small block of data. Packet have different format.

Each hardware uses different packet format.

----- scheme, which is designed to help detect transmissions errors, send

one extra bit of information with each character

► **Parity** ► Checksums ► CRC ► None of given

A parity scheme, which is designed to help detect transmission errors, sends one extra bit of information with each character. Although it allows receiver to determine if a single bit has been changed parity can not detect transmission errors that change an even number of bits.

Local Talk is a LAN technology that employs -----

► **Bus topology** ► Ring topology ► Star topology ► None of the given

LOCAL TALK:

Apple invented the LAN technology that uses bus topology. Its interface is included with all Macintosh computers.

Most LANs that employ ring topology use an access mechanism known as-----

-

315. CSMA/CD ► CSMA/CA ► **TOKEN PASSING** ► None of the given

Most LANs that employ ring technology use an access mechanism known as *token passing*. A token ring operates as a single, shared medium. When a computer wants to send data, it must wait until it obtains the token, when it is in control of the ring

Ethernet uses a ----- bit static addressing scheme in which each device is assigned a unique address by the manufacturer.

► 64 ► **48** ► 32 ► 8

Ethernet uses a 48-bit static addressing scheme

Formally named _____ informally known as the thick wire Ethernet or Thick net.

► 10 Base 2 ► 10 Base 5 ► 10 Base T ► **None of the given** Thick

Ethernet Wiring informally called thick wire Ethernet or Thick net

Formally named _____ informally known as the twisted pair Ethernet or TP Ethernet.

► 10 Base 2 ► 10 Base 5 ► **10 Base T** ► None of the given

twisted pair ethernet

316. Formally called 10 Base –T

317. Also twisted pair Ethernet or simply TP Ethernet

The maximum size of an Ethernet segment is_____

- 250 meters ► **500 meters** ► 700 meters ► None of the given

One Ethernet cable is sometimes called a segment. This segment is limited to 500 meters in length. The minimum separation between connections is 3 meters. (P# 27)

A Bridge can _____

- Filter a frame ► Forward a frame ► Extend a LAN ► **Do all the above**

A bridge is a hardware device also used to connect two LAN segments to extend a LAN. It listens to all traffic and recognizes frame format. It also forwards only correct complete frames and discards the collided and error frames.

The most important task a bridge performs is frame filtering.

_____ computes shortest paths in a graph by using weights on edges as a measure of distance.

318. Greedy algorithm ► Distance vector algorithm ► **Dijkstra's algorithm** ► Non of the given

Dijkstras algorithm computes shortest paths in a graph by using weights on edge as a measure of distance. A path with the fewest number of edges may not be the path with least weight. (P# 173 e-books)

_____is used for compressed audio and video where the data rate depends on the level of compression that can be achieved.

320. Available Bit Rate (ABR) service

319. Constant Bit Rate (CBR) service

321. Variable Bit Rate (VBR) service

322. None of the given

VARIABLE BIT RATE (VBR):

It is used for compressed audio and video where the data rate depends on the level of compression that can be achieved.

Basic LAN technologies such as Ethernet, Token Ring, and FDDI use a _____.

323. Connectionless service paradigm ► Connection-oriented service paradigm

324. Both Connectionless and Connection-oriented service paradigm

325. None of the given

The product of delay and throughput measures the _____ of data that can be present on the network.

► Area ► **Volume** ► Length ► None of the given

DELAY THROUPTUT PRODUCT:

It is computed as delay time multiplied by effective throughput. It measures amount of data that can be present in the network (P# 80)

A network with throughput T and delay D has a total of _____ bit in transit at any time.

► T / D ► **T x D** ► T + D ► None of the given

A network with throughput T and delay D has a total of T x D bits in transit at any time. (e-books)

One repeater _____, two repeaters _____ the maximum cable length limitation.

► **doubles, triple** ► square roots, cude roots ► and, triple ► doubles, cancel

One repeater doubles, two repeaters triple the maximum cable length limitation. (P# 49)

A network uses a ----- if all computers attach to a central point

► **Star Topology** ► Ring Topology ► Bus Topology ► None of the given

STAR TOPOLOGY:

In this topology, all computers are attached to a central point, which is sometimes called the "Hub" (P# 25)

Computers attached to an ethernet use ----- in which a computer waits for the other to be idle before transmitting a frame.

► **CSMA/CD** ► CSMA/CA ► TOKEN PASSING ► None of the given

LAN traffic congestion generally caused by

- A. Low bandwidth, Too many hosts in a broadcast domain, Multicasting, Broadcast Storms
- B. High Bandwidth
- C. None of above

—

2.

Which among following is not a router function

- A. Packet Switching
- B. Packet Filtering
- C. Internetwork communication
- D. Forwarding broadcasts by default

—

Explanation:

It is advantage of router that :

They do not forward broadcasts by default

They can filter the network based on layer 3 information

3.

OSI reference model has how many layers

-
-
-
-

7
4
3
5

-Answer

4.

Framing is done on which layer

- A. Datalink Layer
- B. Physical Layer
- C. Transport Layer
- D. Application Layer

—

5.

Segments are made on which layer

-
-
-



- A. Session Layer
- B. Transport Layer
- C. Application Layer
- D. Network Layer

6.

Which layer deals with TCP and UDP

- A. Transport Layer
- B. Physical Layer
- C. Network Layer
- D. None of above

—

7.

Which layer deals with Flow control

- A. Session Layer
- B. Network Layer
- C. Transport Layer
- D. Application Layer

—

Explanation:

Data integrity is ensured at the Transport layer by maintaining [flow control](#) and by allowing users to request reliable data transport between systems.

8.

Windows are used to

- A. control the amount of unacknowledged data segments
- B. creating new frames
- C. creating segments
- D. none of above

Explanation:

Windows are used to control the the amount of outstanding, unacknowledged data segments. This process is also known as windowing

9.

Which layer deals with Packets

- A. Physical Layer
- B. Network Layer
- C. Transport Layer
- D. Application Layer

10.

Which layer deals with Network topology

-
- ●
-

- A. Physical Layer
- B. Datalink Layer
- C. Network Layer
- D. Session Layer

—

Explanation:

Datalink layer deals with network topology and ensures the data to be delivered on the intended device on the LAN

11.

Binary value of 128 is

- 10000000
- 10100000
- 11000000
- 10100100
-

—

12.

Which layer sends and receives bits

- A. Session Layer
- B. Physical Layer
- C. Datalink Layer
- D. None of above

13.

Which mode has no collisions

- A. Half Duplex
- B. Full Duplex
- C. All of above
- D. None of above

14.

A dedicated switch port is required for each

- A. Full duplex node
- B. Half duplex node
- C. Both of above
- D. None of above

15.

MAC address is of how many bits

- A. 24 bit
- B. 32 bit
- C. 48 bit
- D. 128 bit

Explanation:

MAC address means Media Access Control Address, which is 48 bit address in hexadecimal format burned on NETWORK INTERFACE CARD(NIC).

16.

MAC address is written in which format

- A. Binary format
- B. Simple numeric format
- C. Hexa decimal format
- D. none of above

17.

Which of following is not an ethernet cable type

-
- 326. straight-through cable
- 327. crossover cable
- 328. rolled cable
- 329. array cable

18.

Straight cable is used to connect

- A. Host to switch or hub
- B. Host to Router
- C. Both of above
- D. None of above

19.

Crossover cable is used to connect

- A. Switch to router
- B. Host to host
- C. Hub to router
- D. none of above

20.

Rolled cable is used to

- A. Connect switch to switch
- B. Connect Hub to Hub
- C. Connect Router to Router
- D. Connect Host to Router console serial communication port

21.

Which network will not be susceptible to EMI

-
- A. Thicknet coax
- B. Thinnnet coax
- C. **Fiber optic cable**
- D. none of above

22.

Which type of RJ45 UTP calbe is used between switches

- straight-through cable
- **Crossover cable**
- Both can be used
- None can be used
-

TCP/IP suite was created by

- IEEE
- Department of defence
- Open source
- None of above
-

330. TCP/IP reference model contains how many layers

- 3
- 4
- 6
- 7
-

Explanation:

TCP/IP reference model has four layer

Application Layer

Host to Host

Internet Layer

Network Access

331. Telnet

- A. allows user to connect client machine
- B. transferring files
- C. sharing files
- D. none of above

Which protocol deals with emails

- FTP
- MTP
- LPD
- X window
-

Which protocol deals with resolving domain names

- X-Window
- SMTP
- DNS
- FTP
-

Which protocol ensures reliable delivery

- TCP
- UDP
- Both of above
- None of above
-

Which protocol uses window flow system

- UDP
- TCP
- FTP
- None of above
-

ICMP works on which layer

- A. Physical Layer
- B. Datalink Layer
- C. Network layer
- D. Transport Layer

Which protocol finds the MAC address from IP address

- SMTP
- ICMP
- ARP
- RARP
-

Which IP is reserved for loopback testing

- 192.168.0.0
- 10.255.255.255
- 255.0.0.0
- None of above
-

Explanation:

127.0.0.1 is reserved for loopback testing

332. Which protocol uses both TCP and UDP

-
-
-
-

SMTP
Telnet
FTP
DNS

Which layer of TCP/IP stack is equivalent to transport layer of OSI

- A. Application layer
- B. Host to Host layer
- Internet
- 333. Network access

—

Which layer will be used while transmitting data using FTP or Telnet

- Presentation
- Application
- Session
- Transport
-

Which company developed TCP/IP protocol for networking

- IBM
- DEC
- DARPA
- Xerox
-

What is disadvantage of NAT

- A. Conserves the legally registered addresses
- B. Loss of end to end IP traceability
- C. Eliminates address renumbering as network changes
- D. None of above

Which term is not related to NAT

- A. inside local
- B. outside local
- C. inside global
- D. external global

—

Meaning of Outside local

- A. Name of inside source address before translation
- B. Name of destination host before translation
- C. Name of inside host after translation
- D. Name of outside destination host after translation

Which command will allow you to see real time translation on your router

- A. show ip nat translations
- B. show ip nat [statistics](#)
- C. debug ip nat
- D. clear ip nat translations

-

Which command will show you all the translations active on your router

- A. show ip nat translations
- B. show ip nat statistics
- C. debug ip nat
- D. clear ip nat translations

Which command will show you the summary of NAT configuration

●

-
- A. show ip nat translations
- B. show ip nat statistics
- C. debug ip nat
- D. clear ip nat translations

Which command would you place on the interface on a private network

- A. ip nat inside
- B. ip nat outside
- C. ip outside global
- D. ip inside local

Pat address translation is also termed as

- A. NAT Fast
- B. NAT Static
- C. Overloading Static

D. NAT Overload

-

. IPV6 is of how many bits

- A. 32 bits
- B. 64 bits
- C. 128 bits
- D. 256 bits

334. IPV6 is in which format

- A. Hexadecimal format
- B. Octal format
- C. Alphanumeric format
- D. Numeric format

—

EIGRPv6 use which multicast address?

-
- FF02::A
- FF02::5
- FF02::6
- FF02::9

What multicast address does RIPng use

- FF02::A
- FF02::8
- FF02::9
- FF02::5
-

When IPV6 launched

- A. June 2, 2012
- B. June 4, 2012
- C. June 5, 2012
- D. June 6, 2012

Which of the following is not a feature of Anycast addressing in IPv6?

- A. Same IPv6 address assigned to multiple nodes
- B. [Routing protocol](#) makes the "nearest" determination
- C. Supports many future potential applications
- D. Provides an any-to-many communication model

Which is not a valid type of routing

- A. Static routing
- B. [Default](#) routing
- C. Dynamic routing
- D. Custom routing

—

335. Where we should use default routing

- A. on stub networks- which have only one exit path out of the network
- B. which have more than one exit path out of the network
- C. minimum five exit paths out of the network
- D. None of above

Default administrative distance of EIGRP

- 0
- 1
- 110
- 90
-

Default administrative distance of Static Route

- 0
- 90
- 100
- 1
-

Default administrative distance of RIP

- 0
- 90
- 120
- 130
-

Which among following have administrative distance 0

- Unknown
- RIP
- Static route
- Connected interface
-

Which protocol uses an autonomous system number for activation

- IGRP
- RIP
- Both of above
- None of above
-

Which protocol gives a full route table update every 30 seconds

- IEGRP
- RIP
- both of above
- none of above
-

336. Which statement is true regarding classless routing protocol

- A. The use of discontinuous networks is not allowed
- B. Use of variable length subnet masks is premitted
- C. RIPv1 is a classless routing protocol
- D. IGRP supports classes routing within the same autonomous system

Which command displays RIP routing updates

- A. show ip route
- B. debug ip rip
- C. show protocols
- D. debug ip route

337. Which is true regarding RIPv2

- A. It has lower administrative distance than RIPv1
- B. It converges faster than RIPv1
- C. It is harder to configure than RIPv1
- D. It has the same timers as RIPv1

Meaning of IOS is

- A. Integrated operating system
- B. Intergrated open system
- C. Internetworking operating system
- D. None of above

First IOS was written in

- 1984
- 1985
- 1986
- 1987
-

Which is not a function of IOS

- A. Carrying network protocols and functions
- B. Connecting high speed traffic between devices
- C. Providing scalability for ease of network growth and redundancy
- D. Creating session between hosts in LAN

What happens first after starting a router

-
-
-
-

memory

- A. Load IOS from flash
B. Power on self test of router
C. Searching by IOS of a valid configuration
D. None of above

—

Explanation:

When you bring up a router, it firstly run a Power on self test(POST), If it passes then it will load IOS from flash memory then after that IOS looks for a valid configuration.

CLI in IOS terminology is short form of

- A. Caller line interface
- B. Common line interface
- C. Command link interface
- D. None of above

338. Subinterfaces allows to you to

- A. Create logical interfaces within the router
- B. Get into previleged mode
- C. Get into user execution mode
- D. None of above

339. Which mode provides interactive configuration dialog

- A. User EXEC mode
- B. Privileged mode
- C. Global configuration mode
- D. Setup mode

340. Which user mode has limited monitoring commands in IOS

-
-

●

● A. User EXEC mode

●

B. Privileged EXEC mode

C. Specific configuration mode

D. Setup mode

341. Which mode provide access to all router commands

● A. User EXEC mode

● B. Privileged mode

●

C. Global configuration mode

●

D. Setup mode

342. Which mode commands affects interfaces/processes only

α. A. User EXEC mode

- B. Privileged mode
- C. Specific configuration mode
- D. Setup mode

Which command will show all commands available from that particular mode

- config
- enable
- interface
- ?
-

Explanation: ? is called help command. It shows all commands available from mode in which you are [login](#)

343. Which command will take cursor to beginning of line

-
-
-
-

344. **Ctrl+E**
C
trl Ctrl+B
+
A Ctrl+F

345. Which command will take cursor to the end of line

- Ctrl+A
- Ctrl+E
- Ctrl+B
- Ctrl+F
-

Which command will take cursor to one word back

- Ctrl+A
- **Ctrl+B**
- Ctrl+F
- Ctrl+D
-

Which command will end configuration mode and returns to EXEC

- Ctrl+A
- Ctrl+B
- Ctrl+F
- Ctrl+Z
-

Command to find the broadcast address from router

- show running-config
- show startup-config
- show interfaces
- show protocols
-

—

Command to show if DTE or DCE cable is plugged into serial 0

-
- ●
-

- B. sh int serial 0
- A.
sh
int
s0
- C. show serial 0 controllers
- D. show controllers s 0

How many simultaneous sessions a Cisco router support by default

- 3
- 4
- 5
- 6
-

346. Command to delete NVRAM on a router

- A. delete NVRAM
- delete running-config
- erase NVRAM
- erase start

347. Which layer of OSI model will have problem if you get message "Serial1 is down, line protocol is down" after typing "show interface serial 1"

- A. Physical Layer
- B. Datalink Layer
- C. Network layer
- D. It is router problem

If you delete contents of NVRAM and reboot router, in which mode you will enter

- A. Privileged mode
- B. Global mode
- C. Setup mode
- D. There will be error

348. How to display configurable parameters and statistics of all interfaces on router

- show running-config
- show startup-config
- show versions
- **show interfaces**
-

. Routing is done on which layer

- A. Datalink Layer
- B. **Network Layer**

- Session
- Presentation

Which performs modulation and demodulation

- Modem
- Fiber optic
- satellite
- coaxial cable
-

Routing is done on which layer

- A. Datalink Layer
- B. Session Layer
- C. Network layer
- D. Transport Layer

Dialogue control and token management are responsibilities of

- A. Network layer
- B. Session Layer
- C. Transport Layer
- D. None of above

349. X.25 standard specifies

- technique for start-stop data
- technique for dial access
- DTE/DCE interface
- data bit rate
-

350. First network which was made available

- A. IBMm Token Ring 1985
- B. Novell Netware
- C. IBM PC Network 1984
- D. DECNet 1980

351. Most commonly used transmission speeds in BPS is

- 2400
- 9600
- 1200
- 4800

352. A central computer surrounded by one or more computers is called

- A. ring network
- B. bus network
- C. star network
- D. mesh network

D. all of above

—

353. Which among following is correct

- A. Protocol converters are the same as multiplexers
- B. Protocol converters are the same as TDMs
- C. Protocol converters are usually not operated in pairs
- D. None of above

-Answer

Answer: Option C

354. A network which need human intervention for routing signal is

- A. Bus network
- B. ring network
- C. star network
- D. T switched network

-Answer

Answer: Option **D**

355. Videotax is combination of

- communication
- television
- computer technology
- all of above

356. Name of the topology in which there are bidirectional links between each possible nodes

- Ring
- Mesh
- Star
- Tree

-Answer

Answer: Option **B**

357. Communication mode which supports data in both directions at the same time is called simplex

- Full duplex

Half duplex

multiplex

—

358. Frequency range 300 kHz to 3 MHz is used for

- A. AM radio transmission
- B. FM radio transmission
- C. TV radio transmission
- D. None of above

359. **Sink means**

A. sending unit

B. receiving unit

c. broadcasting

d. none of above

360. At data link layer error detection is achieved by

A. Cyclic Redundancy Code

B. Hamming codes

C. Bit stuffing

D. Equalization

361. Loss in signal power as light travels down the fiber is called

- Interruption
- Propagation
- Scattering
- Attenuation

362. During networking, the process of processor asking each host that if it want to send message is called

- Querying
- sharing
- communicating
- polling

363. In data transmission the bit coding scheme used to represent a byte is typically

- EBCDIC
- SNA
- ASCII
- Hexadecimal

364. A communication network which is used over regional, national or global area is

LAN

WAN

MAN

VAN

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Question No: 1 (Marks: 1)

No error detection scheme is perfect because transmission errors can affect the additional information as well as the data.

► True (Computer Networks and Internets, page82)

365. False

Question No: 2 (Marks: 1)

----- Program sends a message to a remote computer and reports whether the computer responds.

► Ping (Computer Networks and Internets, page9)

1 Traceroute

- 2 ICMP
- 3 Non of the given

Question No: 3 (M a r k s: 1)

----- was especially concerned about the lack of high powered computers.

► [ARPA](#) (Computer Networks and Internets, page5)

- 2. IEEE
- 3. EIA
- 4. Non of the given

Question No: 4 (M a r k s: 1)

The term ----- is used to denote the definition of a packet used with a specific type of network.

- 1 Packet

► [Frame](#) (Computer Networks and Internets, page 73)

- 3. Data
- 4. None of the given

Question No: 5 (M a r k s: 1)

Computer networks are often called ----- because they use packet technology.

- I. Ethernet
- J. Switch networks

► [Packet networks](#) (Computer Networks and Internets, page 73)

- II. None of the given

Question No: 6 (M a r k s: 1)

----- have advantages arisen from the size and ease of computation.

- a. CRC

► [Parity](#) (Page 19)

- 1 Checksums
- 2 None of given

Question No: 7 (M a r k s: 1)

Most LANs that employ ring topology use an access mechanism known as-----

- 5. CSMA/CD
- 6. CSMA/CA

► [TOKEN PASSING](#) (Computer Networks and Internets, page95)

- 6. None of the given

Question No: 8 (M a r k s: 1)

IEEE LLC/SNAP header is -----, which is used to specify the type of data.

- 1 [8 octets](#) (Computer Networks and Internets, page112)
- 2 8 bytes

- 3 8 bits
- 4 None of the given

Question No: 9 (M a r k s: 1)

Formally named _____ informally known as the twisted pair Ethernet or TP Ethernet.

- 7. 10 Base 2
- 8. 10 Base 5

► [10 Base T](#) (Page 43)

- 1 None of the given

Question No: 10 (M a r k s: 1)

An interface for twisted pair Ethernet must have an _____ connector , and must generate signals according to the _____ specification.

8. RJ-45, 10 Base T(Computer Networks and Internets, page 126)

- 9. RJ-45, 10 Base 5
- 10. BNC, 10 Base 2
- 11. BNC, 10 Base T

Question No: 11 (M a r k s: 1)

A bridges function in the _____ layers(s).

► Physical (MAC) [Click here for Detail](#)

- 1 Data link
- 2 Network
- 3 Physical (MAC) and Data link

Question No: 12 (M a r k s: 1)

A Bridge can _____

- 9. Filter a frame
- 10. Forward a frame
- 11. Extend a LAN

► Do all the above [Click here for Detail](#)

Question No: 13 (M a r k s: 1)

A Bridge forwards or filters a frame by comparing the information in its address table to the frame's _____

- b. Layer 2 source address
- c. Source node's physical address

► [Layer 2 destination address](#) [Click here for Detail](#)

- c. Layer 3 destination address

Question No: 14 (Marks: 1)

_____ computes shortest paths in a graph by using weights on edges as a measure of distance.

- c. Greedy algorithm
- d. Distance vector algorithm

► [Dijkstra's algorithm](#) (Computer Networks and Internets, page112)

- b. Non of the given

Question No: 15 (M a r k s: 1)

_____ is used for audio and video, since these have predefined maximum data rates

► [Constant Bit Rate \(CBR\) service](#) (Page 71)

- a. Variable Bit Rate (VBR) service
- b. Available Bit Rate (ABR) service
- c. None of the given

Question No: 16 (M a r k s: 1)

Unlike Frame Relay and ATM, SMDS (Switched multi-megabit Data service) offers_____ .

► [Connectionless service paradigm](#) (Computer Networks and Internets, page112)

- c. Connection oriented service paradigm
- d. Both Connectionless and Connection-oriented service paradigm
- e. None of the given

Question No: 17 (M a r k s: 1)

A network with throughput T and delay D has a total of _____ bit in transit at any time.

- a. T / D

b. $T \times D$ (Computer Networks and Internets, page203)

- c. $T + D$
- d. None of the given

Question No: 18 (M a r k s: 1)

ATM is designed to work on_____.

- a. Twisted Pair
- b. Coaxial
- c. Radio Frequency

► [Fiber](#) (Computer Networks and Internets, page 72)

Question No: 19 (Marks: 1)

Computers attached to an Ethernet use ----- in which a computer waits for the ether to be idle before transmitting a frame.

► [CSMA/CD](#) (Computer Networks and Internets, page 93)

- c. CSMA/CA
- d. TOKEN PASSING
- e. None of the given

Question No: 20 (M a r k s: 1)

FDDI can transmits data at a rate of -----

► [100 million bits per second](#) (Page 314)

- 20. 10 million bits per second
- 21. 1000 million bits per second
- 22. None of the given

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Question No: 1 (M a r k s: 1)

In Point-to-Point topology there are two topologies.

- c. Star and Tree
- d. Tree and Ring
- e. Star and Ring

► [Star and Tree](#) (Page 5)

- c. None of the given

Question No: 2 (M a r k s: 1)

----- Program sends a message to a remote computer and reports whether the computer responds.

- c. [Ping\(Computer Networks and Internets, page9\)](#)
- d. Traceroute

- e. ICMP
- f. Non of the given

Question No: 3 (M a r k s: 1)

----- has no way to determine the cause of the problem.

- c. ICMP

d. Ping(Computer Networks and Internets, page 11)

- e. Trace route
- f. Non of the given

Question No: 4 (M a r k s: 1)

The term----- refers to the general concept of a small block of data

c. Packet (Page 16)

- d. Frame
- e. Data
- f. None of the given

Question No: 5 (M a r k s: 1)

----- scheme, which is designed to help detect transmissions errors, send one extra bit of information with each character

a. Parity(Computer Networks and Internets, page 77)

- b. Checksums
- c. CRC
- d. None of given

Question No: 6 (M a r k s: 1)

Local Talk is a LAN technology that employs -----

► Bus topology (Page 30)

- b. Ring topology
- c. Star topology
- d. None of the given

Question No: 7 (M a r k s: 1)

Most LANs that employ ring topology use an access mechanism known as-----

- a. CSMA/CD
- b. CSMA/CA

► [TOKEN PASSING \(Computer Networks and Internets, page95\)](#) rep ►

None of the given

Question No: 8 (M a r k s: 1)

Ethernet uses a ----- bit static addressing scheme in which each device is assigned a unique address by the manufacturer.

a. 64

b. 48 (Computer Networks and Internets, page 109)

c. 32

d. 8

Question No: 9 (M a r k s: 1)

Formally named _____ informally known as the thick wire Ethernet or Thick net.

c. 10 Base 2

d. 10 Base 5 Computer Networks and Internets, page 120)

e. 10 Base T

f. None of the given

Question No: 10 (M a r k s: 1)

Formally named _____ informally known as the twisted pair Ethernet or TP Ethernet.

► 10 Base 2

► 10 Base 5

► 10 Base T (Page 43) rep

► None of the given

Question No: 11 (M a r k s: 1)

The maximum size of an Ethernet segment is _____

a. 250 meters

b. 500 meters (Page 27)

c. 700 meters

d. None of the given

Question No: 12 (M a r k s: 1)

A Bridge can _____

a. Filter a frame

b. Forward a frame

c. Extend a LAN

► [Do all the above](#) rep [Click here For Detail](#)

Question No: 13 (M a r k s: 1)

_____ computes shortest paths in a graph by using weights on edges as a measure of distance.

c. Greedy algorithm

d. Distance vector algorithm

► [Dijkstra's algorithm](#) (Computer Networks and Internets, page112) rep

► Non of the given

Question No: 14 (M a r k s: 1)

_____ is used for compressed audio and video where the data rate depends on the level of compression that can be achieved.

► Constant Bit Rate (CBR) service

► [Variable Bit Rate \(VBR\) service](#) (Page 71)

- c. Available Bit Rate (ABR) service
- d. None of the given

Question No: 15 (M a r k s: 1)

Basic LAN technologies such as Ethernet, Token Ring, and FDDI use a _____.

- a. [Connectionless service paradigm](#) (Computer Networks and Internets, page112)
- b. Connection-oriented service paradigm
- c. Both Connectionless and Connection-oriented service paradigm
- d. None of the given

Question No: 16 (M a r k s: 1)

The product of delay and throughput measures the _____ of data that can be present on the network.

- b. Area

► [Volume](#) (Page 80)

- a. Length
- b. None of the given

Question No: 17 (M a r k s: 1)

A network with throughput T and delay D has a total of _____ bit in transit at any time.

- a. T / D

b. [T x D](#)(Computer Networks and Internets, page203) rep

- c. $T + D$

d. None of the given

Question No: 18 (M a r k s: 1)

One repeater _____, two repeaters _____ the maximum cable length limitation.

► doubles, triple (Page 49)

- a. square roots, cude roots
- b. and, triple
- c. doubles, cancel

Question No: 19 (Marks: 1)

A network uses a ----- if all computers attach to a central point

► Star Topology (Page 25)

- a. Ring Topology
- b. Bus Topology
- c. None of the given

Question No: 20 (Marks: 1)

Computers attached to an Ethernet use ----- in which a computer waits for the other to be idle before transmitting a frame.

► CSMA/CD (Computer Networks and Internets, page 93) rep

- b. CSMA/CA
- c. TOKEN PASSING
- d. None of the given

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Question No: 1 (Marks: 1) - Please choose one

A typical port on an ATM switch operates at _____ or higher.

- c. OC-2 speed (155Mbps)
- d. OC-3 speed (100Mbps)

► OC-3 speed (155Mbps) (Page 72)

- a. OC-3 speed (155Gbps)

Question No: 2 (Marks: 1) - Please choose one

The product of delay and throughput measures the _____ of data that can be present on the network.

c. Area

d. Volume (Page 80) rep

e. Length

f. None of the given

Question No: 3 (Marks: 1) - Please choose one

_____is used for compressed audio and video where the data rate depends on the level of compression that can be achieved.

▶ Constant Bit Rate (CBR) service

▶ Variable Bit Rate (VBR) service(Page 71) rep

▶ Available Bit Rate (ABR) service

▶ Nonegivenofthe

Question No: 4 (Marks: 1) - Please choose one

Which of the following is a connecting device?

- ▶ Bridge
- ▶ Repeater
- ▶ Hub

▶ All the given [Click here for Detail](#)

Question No: 5 (Marks: 1) - Please choose one

A bridges function in the _____ layers(s).

▶ Physical (MAC) rep [Click here for Detail](#)

- ▶ Data link
- ▶ Network
- ▶ Physical (MAC) and Data link

Question No: 6 (Marks: 1) - Please choose one

IEEE LLC/SNAP header is -----, which is used to specify the type of data.

▶ 8 (Computer networks and Internets, page112) rep

- ▶ 8 bytes
- ▶ 8 bits
- ▶ None of the given

Question No: 7 (Marks: 1) - Please choose one

The third field of the header consists of ----- bit Ethernet frame type.

- ▶ 48
- ▶ 32

▶ 16 (Computer Networks and Internets, page110)

- ▶ 8

Question No: 8 (Marks: 1) - Please choose one

An -----method, the network hardware designers specify how type information is included in the frame and the value use to identify various frame types.

► Explicit frame(Pagtype 35)

- Ideal frame type
- Implicit frame type
- None of the given

Question No: 9 (Marks: 1) - Please choose one Local Talk is a LAN technology that employs -----

- Star topology

► Bus topology(Page 30) rep

- Ring topology
- None of the given

Question No: 10 (Marks: 1) - Please choose one

The Fast Ethernet hardware operates at a rate of -----

- ▶ 10 Mbps
- ▶ **100 Mbps(Page 27)**

- ▶ 1000 Mbps
- ▶ None of the given

Question No: 11 (Marks: 1) - Please choose one

----- scheme, which is designed to help detect transmissions errors, send one extra bit of information with each character

- ▶ **Parity(Computer Networks and Internets, page 77) rep**

- ▶ Checksums
- ▶ CRC
- ▶ None of given

Question No: 12 (Marks: 1) - Please choose one

Computer networks are often called ----- because they use packet technology.

- ▶ Ethernet
- ▶ Switch networks
- ▶ **Packet networks(Computer Networks and Internets, page 73) rep**

- ▶ None of the given

Question No: 13 (Marks: 1) - Please choose one

----- Program sends a message to a remote computer and reports whether the computer responds.

- ▶ **Ping (Computer Networks and Internets, page 9) rep**

- ▶ Traceroute

▶ ICMP

▶ Non of the given

Question No: 14 (Marks: 1) - Please choose one

In -----, network occupies larger areas like cities & countries.

▶ LAN

▶ WAN (Page 4)

▶

▶ MAN

▶ None of the given

Question No: 15 (Marks: 1) - Please choose one

In -----, network occupies the smaller area like a room a floor or a building

▶ LAN(Page 4)

▶ WAN

▶ MAN

▶ None of the given

Question No: 16 (Marks: 1) - Please choose one

In Point-to-Point topology there are two topologies.

- ▶ Tree and Ring
- ▶ Star and Ring

▶ Star and Tree(Page 5)

- ▶ None of the given

CS610- Computer Network MIDTERM EXAMINATION

Fall 2008

Question No: 1 (Marks: 1) - Please choose one

_____ has a jitter zero

- a. None of the given
- b. Virtual Private Network
- c. Isochronous Network

d. Asynchronous Network (Page 66)

Question No: 2 (Marks: 1) - Please choose one

Unlike Frame Relay and ATM, SMDS Switched(multi-megabit Data service) offers_____ .

► **Connectionless service paradigm (Computer Networks and Internets, page112)**

► Connection oriented service paradigm

► Both Connectionless and Connection-oriented service paradigm
► None of theiveng

Question No: 3 (Marks: 1) - Please choose one

ATM assigns each VC a _____ identifier that is divided two parts to produce ahierarchy.

► 21-bit

► 22-bit

► 23-bit

► **24-bit (Page 67)**

Question No: 4 (Marks: 1) - Please choose one

Most WAN systems include a mechanism that can be used to eliminate the common case of duplication routing is called_____

▶ Hierarchal address

▶ **Default route** (Computer Networks and Internets, page172)

▶ Shortest path

▶ None of the given

Question No: 5 (Marks: 1) - Please choose one

The next hop to which a packet is sent depends only on

▶ **Packet's destination** (Computer Networks and Internets, page169)

▶ Packet's original source

▶ Path the packet has taken

▶ Non of the given

Question No: 6 (Marks: 1) - Please choose one

An interface for twisted pair Ethernet must have an _____ connector , and must generate signals according to the_____ specification.

▶ **RJ-45, 10 Base T** (Computer Networks and Internets, page 126) rep

▶ ~~RJ-45, 10 Base 5~~

▶ BNC, 10 Base 2

▶ BNC, 10 Base T

Question No: 7 (Marks: 1) - Please choose one

When an application----- data, it makes a copy of the data available to all other computers on the network.

► [Broadcasting](#) [Click here for Detail](#)

- Multicasting
- Unicasting
- None of the given

Question No: 8 (Marks: 1) - Please choose one

A----- provide a mechanism that a customer can use to set a physical address.

- Static addressing scheme

► [Configurable addressing scheme \(Page 34\)](#)

- Dynamic addressing scheme
- None of the given

Question No: 9 (Marks: 1) - Please choose one

FDDI can transmits data at a rate of -----

▶ **100 million bits per second (Page 314) rep**

- ▶ 100 million bits per second
- ▶ 100 million bits per second
- ▶ None of the given

Question No: 10 (Marks: 1) - Please choose one

Computers attached to an ether use----- in which a computer waits for the ether to be idle before transmitting a frame.

▶ **CSMA/CD (Computer Networks and Internets, page 93) rep**

- ▶ CSMA/CA
- ▶ TOKEN PASSING
- ▶ None of the given

Question No: 11 (Marks: 1) - Please choose one

----- have advantages arisen from the size and ease of computation.

- ▶ CRC
- ▶ **Parity (Page 19) rep**

▶ **Checksums**

- ▶ None of given

Question No: 12 (Marks: 1) - Please choose one

The term----- is used to denote the definition of a packet used with a specific type of network.

▶ Packet

▶ [Frame \(Computer Networks and Internets, page 73\)](#)

▶ Data

▶ None of the given

Question No: 13 (Marks: 1) - Please choose one

----- has no way to determine the cause of the problem.

a. [Ping\(Computer Networks and Internets, page 11\)](#) rep

▶ Trace route ▶
ICMP ▶ Non of the
given

Question No: 14 (Marks: 1) - Please choose one

----- Program sends a message to a remote computer and reports whether the computer responds.

► [Ping \(Computer Networks and Internets, page9\) rep](#) ► Traceroute

- ICMP
- Non of the given

Question No: 15 (Marks: 1) - Please choose one

In-----, network occupies the smaller area like a room a floor or a building

► [LAN \(Page 4\) rep](#)

- WAN
- MAN
- None of the given

Question No: 16 (Marks: 1) - Please choose one

No error detection scheme is perfect because transmission errors can affect the additional information as well as the data.

► [True \(Computer Networks and Internets, page82\)](#)

- a. False

_____ is used for compressed audio and video where the data rate depends on the level of compression that can be achieved.

Constant Bit Rate (CBR) service

Variable Bit Rate (VBR) service

Available Bit Rate (ABR) service

None of the given

L_19, p71

VARIABLE BIT RATE (VBR)

It is used for compressed audio and video where the data rate depends on the level of compression that can be achieved.

A Bridge forwards or filters a frame by comparing the information in its address table to the frame's_____

Layer 2 source address

Source node's physical address

Layer 2 destination address

Layer 3 destination address

L_13, p50

A bridge keeps a list for each segment that consist of physical addresses of the computer attached to that segment. In this way a bridge knows on which segment a destination computer is attached.

Most bridges are self learning bridges. As soon as a frame arrives to a bridge, it extracts a source address from its header and automatically adds it in the list for that segment. In this way a bridge builds up address lists.

_____uses distance vector approach to define routing

BGP

OSPF

RIP

None of the given

L_17, p64

COMPERASION

DISTANCE-VECTOR ROUTING

It is very simple to implement

Packet switch updates its own routing table first.

It is used in RIP.

Computers attached to an ethernet use _____ in which a computer waits for the other to be idle before transmitting a frame.

CSMA/CD

CSMA/CA

TOKEN PASSING

none of the given

L_8, p28

After collision is detected computers stop transmitting. So Ethernet uses CSMA/CD to coordinate transmission. RECOVERY FROM COLLISION

Computers that detects a collision sends special signal to force all other interfaces to detect collision. A computer then waits for other to be idle before transmission.

_____ device is used for multicasting.

Hub

Switch

Router

None of the given

L_12, p43

HUBS:

They are used for extension of connection with multiplexing concept.

A bridge uses _____ to determine which computer is connected to which segment, and uses the _____ to determine whether to forward a copy of frame.

Source address, destination address.

Destination address, destination address.

Destination address, source address.

source address, source address.

L_13, p50

FRAME FILTERING:

If both the source and destination are on the same segment, it does not forward the frame to the other segment. A frame is forwarded to the other segment, if it is destined to that segment.

A bridge keeps a list for each segment that consists of physical addresses of the computer attached to that segment. In this way a bridge knows on which segment a destination computer is attached.

A _____ provide a mechanism that a customer can use to set a physical address.

Static addressing scheme

Configurable addressing scheme

Dynamic addressing scheme

None of the given

L_9, p34

CONFIGURABLE

In this category, the address can be set by the end user either manually

In Direct point to point communication adding the Nth computer requires_____ new connections.

N^2

N-1

$N^2 - N)/2$

None of the given

L_6, p23

Another disadvantage is that adding a new computer to the network requires N-1 new connections

A form of addressing; in which a set of computers is assigned one address

Broadcast

Multicast

Packet Switch

CRC

Bandwidth

L_9, p34

CONFIGURABLE

A network uses a _____arranges for computers to be connected in a closed loop.

Star Topology

Ring Topology

Bus Topology

None of the given

L_7, p25

RING TOPOLOGY

In this topology of network the computers are connected to each other in closed loop.

The term _____ is used to denote the definition of a packet used with a specific type of network.

L_4, p16

FRAME:

A frame or hardware frame denotes a packet of a specific format on a specific hardware technology.

A network uses a _____ if all computers attach to a central point

Star Topology

Ring Topology

Bus Topology

None of the given

l_7, p25

Star Topology

In this topology, all computers are attached to a central point, which is sometimes called the Hub.

A_____ provides mechanism that automatically assigns a physical address to a station when the station first boots.

L_9, p34

DYNAMIC:

In this category the interface automatically assigns physical address each time it is powered up.

In *Point-to-Point* topology there are two topologies.

Tree and Ring

Star and Ring

Star and Tree

None of the given

L_1, p5

In *Point-to-Point* topology there are two topologies.

- 366. Star topology
- 367. Tree topology

Local Talk is a LAN technology that employs _____

Star topology

Bus topology

Ring topology

None of the given

L_8, p30

Apple invented the LAN technology that uses bus topology.

In _____ network occupies larger areas like cities & countries.

LAN

WAN

MAN

None of the given

L_1, p4

In WAN, network occupies larger areas like cities & countries.

_____ Program sends a message to a remote computer and reports whether the computer responds.

Ping

Traceroute

ICMP

Non of the given

L_2, p11

Most NICs contain _____ circuitry that allows the NIC to operate independent of the CPU.

L_11, p40

NIC AND CPU PRCESSING:

NIC contains sufficient hardware to process data independent of system CPU. In which some NICs contain separate microprocessor. In addition to this it also include analog circuitry interface to system bus, buffering and processing.

Missing eot indicates sending computer crashed in frame format.

True

False

L_4, p16

FRAMING IN PRACTICE

_____ scheme, which is designed to help detect transmissions errors, send one extra bit of information with each character Parity

Checksums

CRC

none of given

L_5, p18

Parity checking

To detect the error there are different schemes in which parity checking is also common used. In parity checking, parity refers to the number of bits set to 1 in the data item. A parity bit is an extra bit transmitted with data item chose to give the resulting bit even or odd parity.

CRC can detect more errors than a simple shecksum.

True

False

L_5, P20

To enable a network system to detect more error without increasing the amount of information in each packet another most successful approach is mad which is called CRC.

The checksum of 0000 and 0000 is _____

1111

0000

1110

0111

L 5, p 19

In this procedure data is treated as sequence integers and their arithmetic sum is computed and the carry bits are added to the final sum.

```
0000
+0000
-----
0000
```

A Bridge can_____

Filter a frame

Forward a frame

Extend a LAN

Do all the above

L_13, p50

BRIDGES

A bridge is a hardware device also used to connect two LAN segments to extend a LAN.

FRAME FILTERING

The most important task a bridge performs is frame filtering. If both the source and destination are on the same segment, it does not forward the frame to the other segment. A frame is forwarded to the other segment, if it is destined to that segment.

Formally named _____ informally known as the twisted pair Ethernet or TP Ethernet.

10 Base 2

10 Base 5

10 Base T

None of the given

L_12, p46

10Base-T version of twisted pair Ethernet operated at 10Mbps. 10Base-T twisted pair Ethernet operates at 100Mbps.

The Gigabit Ethernet hardware operates at a rate of _____

4 Mbps

5. Mbps
1000 Mbps None
of the given

L_7, p27

ETHERNET SPEEDS:

The Ethernet speed was originally 3Mbps, and the current standard is 10Mbps the fast Ethernet operates at 100Mbps. There are also gigabits Ethernet available now.

The original Ethernet hardware operated at a rate of _____

2 mbps

5. mbps

1000 mbps

None

L_7, p27

ETHERNET SPEEDS:

The Ethernet speed was originally 3Mbps, and the current standard is 10Mbps the fast Ethernet operates at 100Mbps. There are also gigabits Ethernet available now.

The term_____ refers to the general concept of a small block of data

Packet

Frame

Data

L_4, p16

Packet is a generic term that refers to small block of data.

_____was especially concerned about the lack of high powered computers.

ARPA

IEEE

EIA

None of the given

L_2, p9

The efforts of ARPA was to enable all its research groups have access to latest computers.

Which error detection method involves polynomials?

Simple parity check

Two-dimensional parity check

CRC

Checksum

L_5, p20

To understand the concept of CRC considers data in a message as co-efficient of a polynomial. Their co-efficient set is divided by a known polynomial.

In broadcast topology there are two categories_____.

L_1, p5

In broadcast topology there are further two categories

K.SATELLITE\ RADIO

L. RING TOPOLOGY

The term_____ refers to the general concept of a small block of data

L_4, p16

Packet is a generic term that refers to small block of data.

The original Ethernet hardware operated at a rate of _____

JJ. mbps

b. mbps

1000 mbps

None

L_7, p27

ETHERNET SPEEDS:

The Ethernet speed was originally 3Mbps, and the current standard is 10Mbps the fast Ethernet operates at 100Mbps. There are also gigabits Ethernet available now.

_____ computes shortest paths in a graph by using weights on edges as a measure of distance.

Greedy algorithm

Distance vector algorithm

Dijkstra's algorithm

None of the given

L_17, p62

Dijkstra's algorithm can accommodate weights on edges in graph. The shortest path is then the path with lowest total weight.

_____ sense a signal on one cable and then transmits an amplified copy on the other cable.

Repeater

Bridge

Hub

None of the given

L_13, p49

A repeater is used to increase the signal strength. It amplifies the weakening signal received from one segment and then retransmits onto another segment.

CRC can detect more errors than a simple checksum

True

False

L_5, p20

CYCLIC REDUNDANCY CHECK (CRC):

To enable a network system to detect more error without increasing the amount of information in each packet another most successful approach is made which is called CRC

A network technology that uses Collision Avoidance

Bridge

Broadcast

WAN

Wireless LAN

L_8, p29

CSMA/CA

Wireless uses collision avoidance rather than collision detection.

A network with throughput T and delay D has a total of _____ bit in transit at any time.

T / D

$T \times D$

$T + D$

None of the given

L_22, p80

It is computed as delay time multiplied by effective throughput.

A network with throughput T and delay D has a total of $T \times D$ bits in transit at any time. (e-books)

_____ is used for audio and video, since these have predefined

maximum data rates

Constant Bit Rate (CBR) service

Variable Bit Rate (VBR) service

Available Bit Rate (ABR) service

None of the given

L_19, p71

CONSTANT BIT RATE (CBR):

It is used for audio and video, since these have predefined maximum data rates.

Most NICs contain _____ circuitry that allows the NIC to operate independent of the CPU.

L_11, p40

NIC AND CPU PROCESSING:

NIC contains sufficient hardware to process data independent of system CPU. In which some NICs contain separate microprocessor. In addition to this it also include analog circuitry interface to system bus, buffering and processing.

The next hop to which a packet is sent depends only on_____

L_15, p57

NEXT HOP FORWARDING:

- 3 If the destination is local computer, packet switch delivers computer port.
- 4 If the destination is attached another switch, this packet switch forwards to next hop through connection to another switch.
- 5 The choice of another switch is based on destination address in packet.

Most LANs that employ ring topology use an access mechanism known as _____

L_8, p30

Many LAN technologies that are ring topology use token passing for synchronized access to the ring

If the ASCII character G is sent and the character D is received, what type of error is this?

Single-bit

Multiple-bit

Burst

Recoverable

L 6, p22

When a small set of bits changes near a specific location due to lighting or electric motor starting nearby etc. then these types of errors are called burst errors.

A_____ relies on the hardware manufacturer to assign a unique physical

address to each network interface.

Static addressing scheme

Dynamic addressing scheme

Configurable addressing scheme

None of the given

L_9, p34

STATIC:

In this category the hardware manufacturer assigns permanent physical address to each network interface and manufacturer must ensure that every interface has a unique address.

The length of time required to send a variable length packet is variable and does not require a complicated interrupt scheme to detect completion of transmission.

True

False

L_20, p72

CELLS VS PACKETS:

The length of time required to send a variable length packet is variable and requires complicated interrupt scheme to detect completion of transmission.

LAN that use ATM technology have a -----

Star topology

Bus topology

Ring topology

None of the given

L_8, p30

Many LAN technologies that are ring topology use token passing for synchronized access to the ring.

Formally named _____ informally known as the twisted pair Ethernet or

TP Ethernet.

10 Base 2

10 Base 5

10 Base T

none of the given

L_12, p43

10 Base-T

This is another standard of wiring scheme. It is commonly called 10 Base-T, twisted pair or TP Ethernet.

ATM was designed for _____ and use a_____

L_17, p65

ATM (ASYNCHRONOUS TRANSFER MODE):

It was designed as a single technology for voice, video and data and has low jitter (variance in delivery time) and high capacity. It uses fixed size, small cells, 48 octet's data and 5 octets header. It can also connect multiple ATM switches into a network.

In which type of topology computers are connected to a central hub?

Tree

Mesh

Bus

Star

L_1, p5

In *star* topology each computer is connected to a central hub.

Computers attached to an ether use ----- in which a computer waits for the ether to be idle before transmitting a frame.

CSMA/CD

CSMA/CA

TOKEN PASSING

None of the given

L_8, p28

After collision is detected computers stop transmitting. So Ethernet uses CSMA/CD to coordinate transmission. **RECOVERY FROM COLLISION:**

Computer that detects a collision sends special signal to force all other interfaces to detect collision. Computer then waits for other to be idle before transmission.

OSPF is based on_____

Distance vector routing

Link state routing

Path vector routing

Distance vector routing and Link state routing

L_17, p64

LINK STATE ALGORITHM

It is used in OSPF

The network that uses a _____, usually consist of a single long cable to which computer attach.

Star topology

Bus topology

Ring topology

None of the given

L_7, p26

BUS TOPOLOGY:

In a bus topology all computers are attached to a single long cable and any computer can send data to any other computer.

The maximum size of an Ethernet segment is_____

7. meters

7. meters

5 meters

None of
the given

L_7, p27

EXAMPLE BUS NETWORK: ETHERNET

One Ethernet cable is sometimes called a segment. This segment is limited to 500 meters in length. The minimum separation between connections is 3 meters.

A network uses a _____ if all computers attach to a central point

Star Topology

Ring Topology

Bus Topology

None of the given

l_7, p25

Star Topology

In this topology, all computers are attached to a central point, which is sometimes called the Hub.

A network uses a _____arranges for computers to be connected in a closed loop.

Star Topology

Ring Topology

Bus Topology

None of the given

L_7, p25

RING TOPOLOGY

In this topology of network the computers are connected to each other in closed loop.

The Fast Ethernet hardware operates at a rate of _____

9. Mbps

2 Mbps

1000 Mbps

None of
the given

L_12, p46

10Base-T version of twisted pair Ethernet operated at 10Mbps. 10Base-T twisted pair Ethernet operates at 100Mbps.

A popular LAN Technology that uses a shared Bus Topology is called.

WAN

CSMA

Ethernet

Multicast

L_7, p27

EXAMPLE BUS NETWORK; ETHERNET:

Ethernet is a widely used LAN technology. -----

-----The Ethernet uses bus topology.

L_8, p30

LOCAL TALK

Apple invented the LAN technology that uses bus topology.

Local Talk is a LAN technology that employs_____

Bus topology

Ring topology

Star topology

None of the given

L_8, p30

Apple invented the LAN technology that uses bus topology.

In _____ network occupies larger areas like cities & countries.

LAN

WAN

MAN

None of the given

L_1, p4

In WAN, network occupies larger areas like cities & countries.

Most LANs that employ ring topology use an access mechanism known as_____

CSMA/CD

CSMA/CA

TOKEN PASSING

None of the given

L_8, p30

Many LAN technologies that are ring topology use token passing for synchronized access to the ring.

Ethernet uses a ____ bit static addressing scheme in which each device is assigned a unique address by the manufacturer.

64

48

32

8

12. Ethernet uses a 48-bit static addressing scheme

Computers attached to an ethernet use _____ in which a computer waits for the other to be idle before transmitting a frame.

CSMA/CD

CSMA/CA

TOKEN PASSING

none of the given

L_8, p28

After collision is detected computers stop transmitting. So Ethernet uses CSMA/CD to coordinate transmission. RECOVERY FROM COLLISION

Computers that detects a collision sends special signal to force all other interfaces to detect collision. A computer then waits for other to be idle before transmission.

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CSMA

Ethernet

Multicast

L_7, p27

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Ethernet is a widely used LAN technology. -----

-----The Ethernet uses bus topology.

L_8, p30

LOCAL TALK

Apple invented the LAN technology that uses bus topology.

A network technology that uses Collision Avoidance

Bridge

Broadcast

WAN

Wireless LAN

L_8, p29

CSMA/CA

Wireless uses collision avoidance rather than collision detection.

FDDI can transmit data at a rate of _____

100 million bits per second

100 million bits per second

100 million bits per second

None of the given

L_8, p31

FDDI: Fiber distributed data interconnect (FDDI) is another ring technology. Its most important features are

It uses fiber optics between stations and transmits data at 100Mbps.

Most LANs that employ ring topology use an access mechanism known as _____

L_8, p30

Many LAN technologies that are ring topology use token passing for synchronized access to the ring

A network that permits many sources to take turns accessing a shared communication resource is providing a form of _____

Packet Division Multiplexing

Time Division Multiplexing

Frequency Division Multiplexing

Wave Division Multiplexing

L_4, p15

In TDM each packet leaves the source and is switched on the shared communication channel through a multiplexer.

Local Talk is a LAN technology that employs _____

Star topology

Bus topology

Ring topology

None of the given

L_8, p30

Apple invented the LAN technology that uses bus topology.

For point-to-point configuration, addressing is needed

True

False

L_8, p31

Computer gets point-to-point connections. Data from transmitters is routed directly through hub switches to destination.

The Gigabit Ethernet hardware operates at a rate of _____

4 Mbps
368. Mbps

1000 Mbps None
of the given

L_7, p27

ETHERNET SPEEDS:

The Ethernet speed was originally 3Mbps, and the current standard is 10Mbps the fast Ethernet operates at 100Mbps. There are also gigabits Ethernet available now.

When an application _____ data, it makes a copy of the data available

to all other computers on the network?

Broadcasting

Multicasting

Unicasting

None of the given

L_9, p34

**Some applications want to broadcast messages to all stations on the LAN.
For this purpose shared communication channel can make broadcast efficient
in such a way that message is delivered to all stations.**

**A special broadcast address is used to identify broadcast message, which
are captured by all stations**

A _____ provide a mechanism that a customer can use to set a physical address.

Static addressing scheme

Configurable addressing scheme

Dynamic addressing scheme

None of the given

L_9, p34

CONFIGURABLE

In Direct point to point communication adding the Nth computer requires _____ new connections.

N^2

N-1

$N^2 - N)/2$

None of the given

L_6, p23

Another disadvantage is that adding a new computer to the network requires N-1 new connections

Unlike Frame Relay and ATM, SMDS (Switched multi-megabit Data service) offers_____.

Connectionless service paradigm

Connection oriented service paradigm

Both Connectionless and Connection-oriented service paradigm

None of the given

L_17, p65

SMDS:

Switched Multi megabit Data Service (SMDS) is also a Telco service. It is a connection less service. Any SMDS station can send information to any station on the same SMDS cloud. It is typically ranges from 1.5Mbps to 1000Mbps.

_____ have advantages arisen from the size and ease of computation.

CRC

Parity

Checksums

None of given

L_5, P20

To enable a network system to detect more error without increasing the amount of information in each packet another most successful approach is mad which is called CRC.

The third field of the header consists of _____ bit Ethernet frame type.

48

32

16

8

L_10, p36

Figure 10.1 Ethernet frame format

Header's 3rd field is frame type 2byts, 16bits

The term_____ refers to the general concept of a small block of data

Packet

Frame

Data

None of the given

L4, p14

Network systems divide data in small blocks or junks called packets, which they send individually.

An electronic device that connects LAN segments

Ethernet

10Baset

Bridge

10Base5

L_13, p50

BRIDGES:

A bridge is a hardware device also used to connect two LAN segments to extend a LAN.

A Bridge can _____

Filter a frame

Forward a frame

Extend a LAN

Do all the above

L_13, p50

A bridge is a hardware device also used to connect two LAN segments to extend a LAN.

It listens to all traffic and recognizes frame format. It also forwards only correct complete frames and discards the collided and error frames. **FRAME FILTERING**

The most important task a bridge performs is frame filtering.

In _____, network occupies the smaller area like a room a floor or a building.

LAN

WAN

MAN

None of the given

L_1, p4

In LAN network occupies the smaller area like a room a floor or a building.

FDDI can transmits data at a rate of _____

100 million bits per second

100 million bits per second

100 million bits per second

None of the given

L_8, p31

**FDDI: Fiber distributed data interconnect (FDDI) is another ring technology.
It most important features are**

It uses fiber optics between stations and transmits data at 100Mbps.

_____ has no way to determine the cause of the problem.

ICMP

Ping

Trace route

None of the given

L_2, p11

If ping does not review any responses from host computer it can't tell the reason of problem.

The number of connections needed for N computer in direct point to point communication is equal to:

$(N^2 - N) / 2$

$$\frac{N^2 - N}{2}$$

None of the given

L_6, p23

For 'n' computers connections = $(n^2 - n) / 2$.

Hardware that calculates a CRC uses two simple components.

AND unit and XOR unit

Shift register and XOR unit

Shift register and AND unit

None of the given

L_5, p20

HARDWARE COMPONENTS USED IN CRC:

CRC uses just two hardware components:

370. Shift register

371. Exclusive OR (XOR unit)

The term _____ is used to denote the definition of a packet used with a specific type of network.

Packet

Frame

Data

None of the given

L_4, p16

A frame or hardware frame denotes a packet of a specific format on a specific hardware technology.

The Gigabit Ethernet hardware operates at a rate of_____

372. Mbps

373. Mbps

1000 Mbps None

of the given

L_7, p27

ETHERNET SPEEDS:

The Ethernet speed was originally 3Mbps, and the current standard is 10Mbps the fast Ethernet operates at 100Mbps. There are also gigabits Ethernet available now.

_____ computes shortest paths in a graph by using weights on edges as a measure of distance.

Greedy algorithm

Distance vector algorithm

Dijkstra's algorithm

None of the given

L_17, p62

Dijkstra's algorithm can accommodate weights on edges in graph. The shortest path is then the path with lowest total weight.

Most LANs that employ ring topology use an access mechanism known as_____

CSMA/CD

CSMA/CA

TOKEN PASSING

None of the given

L_8, p30

Many LAN technologies that are ring topology use token passing for synchronized access to the ring.

An _____ method, the network hardware designers specify how type information is included in the frame and the value use to identify various frame types.

Explicit frame type

Ideal frame type

Implicit frame type

None of the given

L_10, p35

In this type the identifying value is included with frame describes types of included data.

_____ has advantages arisen from the size and ease of computation.

CRC

Parity

Checksums

None of given

L_5, P20

To enable a network system to detect more error without increasing the amount of information in each packet another most successful approach is mad which is called CRC.

ATM is designed to work on_____.

Twisted Pair

Coaxial

Radio Frequency

Fiber

L_20, p72

ATM is designed to work on fiber (but can be used with twisted pair). A typical port on an ATM switch operates at OC-3 speed (155Mbps) or higher.

The term_____ refers to the general concept of a small block of data

Packet

Frame

Data

L_4, p16

Packet is a generic term that refers to small block of data.

IEEE LLC/SNAP header is _____, which is used to specify the type of data.

374. octets

375. bytes

376. bits

None of the given

L_10, p37

Figure 10.3

The checksum of 0000 and 0000 is _____

1111

0000

1110

0111

L 5, p 19

In this procedure data is treated as sequence integers and their arithmetic sum is computed and the carry bits are added to the final sum.

0000

+0000

0000

The third field of the header consists of _____ bit Ethernet frame type.

48

32

16

8

L_10, p36

Figure 10.1 Ethernet frame format

Header's 3rd field is frame type 2bytes, 16bits

FDDI can transmits data at a rate of _____

100 million bits per second

100 million bits per second

100 million bits per second

None of the given

L_8, p31

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It uses fiber optics between stations and transmits data at 100Mbps.

The term_____ refers to the general concept of a small block of data

L_4, p16

Packet is a generic term that refers to small block of data.

A _____ provide a mechanism that a customer can use to set a physical address.

Static addressing scheme

Configurable addressing scheme

Dynamic addressing scheme

None of the given

L_9, p34

CONFIGURABLE

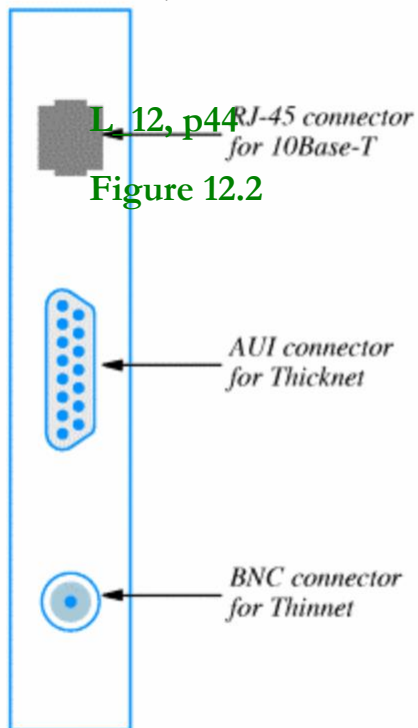
An interface for thin Ethernet must have a _____ connector, and must generate signals according to the _____ specification.

RJ-45, 10 Base T

RJ-45, 10 Base 5

BNC, 10 Base 2

BNC, 10 Base T



Which of the following is a connecting device?

Bridge

Repeater

Hub

All the given

L_13, p50

A bridge is a hardware device also used to connect two LAN segments to extend a LAN.

P49

Repeaters are used when copper wire communication is carries out.
Figure 13.2

One repeater doubles, two repeaters triple the maximum cable length limitation.

L_12, p42

They are used for extension of connection with multiplexing concept.

..... it can be connected into larger Ethernet.

A popular LAN Technology that uses a shared Bus Topology is called.

WAN

CSMA

Ethernet

Multicast

L_7, p27

EXAMPLE BUS NETWORK; ETHERNET:

Ethernet is a widely used LAN technology. -----

-----The Ethernet uses bus topology.

L_8, p30

LOCAL TALK

Apple invented the LAN technology that uses bus topology.

FDDI can transmits data at a rate of _____

100 million bits per second

100 million bits per second

100 million bits per second

None of the given

L_8, p31

FDDI: Fiber distributed data interconnect (FDDI) is another ring technology. It most important features are

It uses fiber optics between stations and transmits data at 100Mbps.

A_____ relies on the hardware manufacturer to assign a unique physical address to each network interface.

Static addressing scheme

Dynamic addressing scheme

Configurable addressing scheme

None of the given

L_9, p34

STATIC:

In this category the hardware manufacturer assigns permanent physical address to each network interface and manufacturer must ensure that every interface has a unique address.

_____was especially concerned about the lack of high powered computers.

ARPA

IEEE

EIA

None of the given

L_2, p9

The efforts of ARPA was to enable all its research groups have access to latest computers.

_____ scheme, which is designed to help detect transmissions errors, send one extra bit of information with each character

Parity

Checksums

CRC

none of given

L_5, p18

Parity checking

To detect the error there are different schemes in which parity checking is also common used. In parity checking, parity refers to the number of bits set to 1 in the data item. A parity bit is an extra bit transmitted with data item chose to give the resulting bit even or odd parity.

Formally named _____ informally known as the twisted pair Ethernet or TP Ethernet.

10 Base 2

10 Base 5

10 Base T

None of the given

L_12, p46

10Base-T version of twisted pair Ethernet operated at 10Mbps. 10Base-T twisted pair Ethernet operates at 100Mbps.

Missing eot indicates sending computer crashed in frame format.

True

False

L_4, p16

FRAMING IN PRACTICE

The Fast Ethernet hardware operates at a rate of _____

377. Mbps

378. **Mb**

ps 1000

Mbps

None of

the given

L_12, p46

10Base-T version of twisted pair Ethernet operated at 10Mbps. 10Base-T

twisted pair Ethernet operates at 100Mbps.

_____ was especially concerned about the lack of high powered computers.

ARPA

IEEE

EIA

None of the given

L_2, p9

The efforts of ARPA was to enable all its research groups have access to latest computers.

When an application_____ data, it makes a copy of the data available to all other computers on the network.

Broadcasting

Multicasting

Unicasting

None of the given

L_9, p34

Some applications want to broadcast messages to all stations on the LAN. For this purpose shared communication channel can make broadcast efficient in such a way that message is delivered to all stations.

A special broadcast address is used to identify broadcast message, which are captured by all stations

One repeater _____two repeaters _____ the maximum cable length limitation.

Doubles, cancel

Doubles, triple

Square roots, cube roots

And, triple

L_13, p49

REPEATERS:

One repeater doubles two repeaters triple the maximum cable length limitation.

An interface for thin Ethernet must have an _____ connector, and must generate signals according to the _____ specification.

RJ-45, 10 Base T

RJ-45, 10 Base 5

BNC, 10 Base 2

BNC, 10 Base T

L_12, p44

Figure 12.2

_____ device is used for multicasting.

Hub

Switch

Router

None of the given

L_12, p43

HUBS:

They are used for extension of connection with multiplexing concept.

In _____ network occupies larger areas like cities & countries.

LAN

WAN

MAN

None of the given

L_1, p4

In WAN, network occupies larger areas like cities & countries.

ATM is designed to work on_____.

Twisted Pair

Coaxial

Radio Frequency

Fiber

L_20, p72

ATM is designed to work on fiber (but can be used with twisted pair). A typical port on an ATM switch operates at OC-3 speed (155Mbps) or higher.

The Fast Ethernet hardware operates at a rate of _____

379. Mbps

380. Mb

ps 1000

Mbps

None of

the given

L_12, p46

10Base-T version of twisted pair Ethernet operated at 10Mbps. 10Base-T twisted pair Ethernet operates at 100Mbps.

A Bridge can _____

Filter a frame

Forward a frame

Extend a LAN

Do all the above

L_13, p50

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It listens to all traffic and recognizes frame format. It also forwards only correct complete frames and discards the collided and error frames. FRAME FILTERING

The most important task a bridge performs is frame filtering.

The checksum of 0000 and 0000 is _____

1111

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1110

0111

L 5, p 19

In this procedure data is treated as sequence integers and their arithmetic sum is computed and the carry bits are added to the final sum.

0000

+0000

0000

CRC can detect more errors than a simple shecksum.

True

False

L_5, P20

To enable a network system to detect more error without increasing the amount of information in each packet another most successful approach is mad which is called CRC.

Hardware that calculates a CRC uses two simple components.

AND unit and XOR unit

Shift register and XOR unit

Shift register and AND unit

None of the given

L_5, p20

HARDWARE COMPONENTS USED IN CRC:

CRC uses just two hardware components:

- 381. Shift register
- 382. Exclusive OR (XOR unit)

The network that uses a _____, usually consist of a single long cable to which computer attach.

Star topology

Bus topology

Ring topology

None of the given

L_7, p26

BUS TOPOLOGY:

In a bus topology all computers are attached to a single long cable and any computer can send data to any other computer.

A bridge uses _____ to determine which computer is connected to which segment, and uses the _____ to determine whether to forward a copy of frame.

Source address, destination address.

Destination adress, destination address.

Destination adress, source address.

source adress, source address.

L_13, p50

FRAME FILTERING:

If both the source and destination are on the same segment, it does not forward the frame to the other segment. A frame is forwarded to the other segment, if it is destined to that segment.

A bridge keeps a list for each segment that consists of physical addresses of the computer attached to that segment. In this way a bridge knows on which segment a destination computer is attached.

Local Talk is a LAN technology that employs _____

Star topology

Bus topology

Ring topology

None of the given

L_8, p30

Apple invented the LAN technology that uses bus topology.

Which of the following is a connecting device?

Bridge

Repeater

Hub

All the given

L_13, p50

A bridge is a hardware device also used to connect two LAN segments to extend a LAN.

P49

Repeaters are used when copper wire communication is carries out.
Figure 13.2

One repeater doubles, two repeaters triple the maximum cable length limitation.

L_12, p42

They are used for extension of connection with multiplexing concept.

..... it can be connected into larger Ethernet.

Which error detection method involves polynomials?

Simple parity check

Two-dimensional parity check

CRC

Checksum

L_5, p20

To understand the concept of CRC considers data in a message as co-efficient of a polynomial. Their co-efficient set is divided by a known polynomial.

The maximum size of an Ethernet segment is_____

383. meters

384. meters

385. mete

rs None of
the given

L_7, p27

EXAMPLE BUS NETWORK: ETHERNET

One Ethernet cable is sometimes called a segment. This segment is limited to 500 meters in length. The minimum separation between connections is 3 meters.

The Gigabit Ethernet hardware operates at a rate of _____

386. Mbps

387. Mbps

1000 Mbps None
of the given

L_7, p27

ETHERNET SPEEDS:

The Ethernet speed was originally 3Mbps, and the current standard is 10Mbps the fast Ethernet operates at 100Mbps. There are also gigabits Ethernet available now.

One repeater _____ two repeaters _____ the maximum cable length limitation.

Doubles, cancel

Doubles, triple

Square roots, cube roots

And, triple

L_13, p49

REPEATERS:

One repeater doubles two repeaters triple the maximum cable length limitation.

_____ have advantages arisen from the size and ease of computation.

CRC

Parity

Checksums

None of given

L_5, P20

To enable a network system to detect more error without increasing the amount of information in each packet another most successful approach is mad which is called CRC.

Unlike Frame Relay and ATM, SMDS (Switched multi-megabit Data service) offers_____.

Connectionless service paradigm

Connection oriented service paradigm

Both Connectionless and Connection-oriented service paradigm

None of the given

L_17, p65

SMDS:

Switched Multi megabit Data Service (SMDS) is also a Telco service. It is a connection less service. Any SMDS station can send information to any station on the same SMDS cloud. It is typically ranges from 1.5Mbps to 1000Mbps.

_____ Program sends a message to a remote computer and reports whether the computer responds.

Ping

Traceroute

ICMP

Non of the given

L_2, p11

_____ is used for audio and video, since these have predefined maximum data rates

Constant Bit Rate (CBR) service

Variable Bit Rate (VBR) service

Available Bit Rate (ABR) service

None of the given

L_19, p71

CONSTANT BIT RATE (CBR):

It is used for audio and video, since these have predefined maximum data rates.

The original Ethernet hardware operated at a rate of _____

388. mbps
389. mbps
1000 mbps
None

L_7, p27

ETHERNET SPEEDS:

The Ethernet speed was originally 3Mbps, and the current standard is 10Mbps the fast Ethernet operates at 100Mbps. There are also gigabits Ethernet available now.

Most NICs contain _____ circuitry that allows the NIC to operate independent of the CPU.

L_11, p40

NIC AND CPU PROCESSING:

NIC contains sufficient hardware to process data independent of system CPU. In which some NICs contain separate microprocessor. In addition to this it also include analog circuitry interface to system bus, buffering and processing.

The next hop to which a packet is sent depends only on _____

L_15, p57

NEXT HOP FORWARDING:

- 390. If the destination is local computer, packet switch delivers computer port.
- 391. If the destination is attached another switch, this packet switch forwards to next hop through connection to another switch.
- 392. The choice of another switch is based on destination address in packet.

An electronic device that connects LAN segments

Ethernet

10Baset

Bridge

10Base5

L_13, p50

BRIDGES:

A bridge is a hardware device also used to connect two LAN segments to extend a LAN.

An _____ method, the network hardware designers specify how type information is included in the frame and the value use to identify various frame types.

Explicit frame type

Ideal frame type

Implicit frame type

None of the given

L_10, p35

In this type the identifying value is included with frame describes types of included data.

A network uses a _____arranges for computers to be connected in a closed loop.

Star Topology

Ring Topology

Bus Topology

None of the given

L_7, p25

RING TOPOLOGY

In this topology of network the computers are connected to each other in closed loop.

A Bridge forwards or filters a frame by comparing the information in its address table to the frame's_____

Layer 2 source address

Source node's physical address

Layer 2 destination address

Layer 3 destination address

L_13, p50

A bridge keeps a list for each segment that consist of physical addresses of the computer attached to that segment. In this way a bridge knows on which segment a destination computer is attached.

Most bridges are self learning bridges. As soon as a frame arrives to a bridge, it extracts a source address from its header and automatically adds it in the list for that segment. In this way a bridge builds up address lists.

A Bridge can_____

Filter a frame

Forward a frame

Extend a LAN

Do all the above

L_13, p50

BRIDGES

A bridge is a hardware device also used to connect two LAN segments to extend a LAN.

FRAME FILTERING

The most important task a bridge performs is frame filtering. If both the source and destination are on the same segment, it does not forward the frame to the other segment. A frame is forwarded to the other segment, if it is destined to that segment.

_____ sense a signal on one cable and then transmits an amplified copy on the other cable.

Repeater

Bridge

Hub

None of the given

L_13, p49

A repeater is used to increase the signal strength. It amplifies the weakening signal received from one segment and then retransmits onto another segment.

IEEE LLC/SNAP header is _____, which is used to specify the type of data.

393. octets

394. bytes

8 bits

None of the given

L_10, p37

Figure 10.3

A Bridge forwards or filters a frame by comparing the information in its address table to the frame's_____

Layer 2 source address

Source node's physical address

Layer 2 destination address

Layer 3 destination address

L_13, p49

The most important task a bridge performs is frame filtering. A

frame is forwarded to the other segment, if it is destined to that segment.

Which of the following is a connecting device?

Bridge

Repeater

Hub

All the given

L_13, p50

A bridge is a hardware device also used to connect two LAN segments to extend a LAN.

P49

Repeaters are used when copper wire communication is carries out.
Figure 13.2

One repeater doubles, two repeaters triple the maximum cable length
limitation.

L_12, p42

They are used for extension of connection with multiplexing concept.

..... it can be connected into larger Ethernet.

The Gigabit Ethernet hardware operates at a rate of_____

395. Mbps

396. Mbps

1000 Mbps None
of the given

L_7, p27

ETHERNET SPEEDS:

The Ethernet speed was originally 3Mbps, and the current standard is 10Mbps the fast Ethernet operates at 100Mbps. There are also gigabits Ethernet available now.

Computers attached to an ethernet use _____in which a computer waits for the other to be idle before transmitting a frame.

CSMA/CD

CSMA/CA

TOKEN PASSING

none of the given

L_8, p28

After collision is detected computers stop transmitting. So Ethernet uses CSMA/CD to coordinate transmission. **RECOVERY FROM COLLISION**

Computers that detects a collision sends special signal to force all other interfaces to detect collision. A computer then waits for other to be idle before transmission.

CRC can detect more errors than a simple checksum

True

False

L_5, p20

CYCLIC REDUNDANCY CHECK (CRC):

To enable a network system to detect more error without increasing the amount of information in each packet another most successful approach is made which is called CRC

The next hop to which a packet is sent depends only on

Packet's destination

Packet's original source

Path the packet has taken

None of the given

L_15, p57

NEXT HOP FORWARDING

Packet switch must choose outgoing connection for forwarding the packet. There are two cases

If the destination is local computer, packet switch delivers computer port.
If the destination is attached another switch, this packet switch forwards to next hop through connection to another switch.

The choice of another switch is based on destination address in packet.

An interface for thin Ethernet must have an _____ connector, and must generate signals according to the _____ specification.

RJ-45, 10 Base T

RJ-45, 10 Base 5

BNC, 10 Base 2

BNC, 10 Base T

L_12, p44

Figure 12.2

_____was especially concerned about the lack of high powered computers.

ARPA

IEEE

EIA

None of the given

L_2, p9

The efforts of ARPA was to enable all its research groups have access to latest computers.

_____ computes shortest paths in a graph by using weights on edges as a measure of distance.

Greedy algorithm

Distance vector algorithm

Dijkstra's algorithm

Non of the given

L_17, p62

Dijkstra's algorithm can accommodate weights on edges in graph. The shortest path is then the path with lowest total weight.

In broadcast topology there are two categories_____.

L_1, p5

In broadcast topology there are further two categories

397. SATELLITE\ RADIO

398. RING TOPOLOGY

_____uses distance vector approach to define routing

BGP

OSPF

RIP

None of the given

L_17, p64

DISTANCE-VECTOR ROUTING

It is very simple to implement

Packet switch updates its own routing table first

It is used in RIP

L_41, p138

ROUTING INFORMATION PROTOCOL (RIP)

It uses distance vector algorithm.

OSPF is based on_____

Distance vector routing

Link state routing

Path vector routing

Distance vector routing and Link state routing

L_17, p64

LINK STATE ALGORITHM

It is used in OSPF

_____ was especially concerned about the lack of high powered computers.

ARPA

IEEE

EIA

None of the given

L_2, p9

The efforts of ARPA was to enable all its research groups have access to latest computers.

ATM assigns each VC a _____ identifier that is divided two parts to produce a hierarchy.

21-bit

22-bit

23-bit

24-bit

L_18, p67

The VC is identified by a 24 bit value formed from the VPI (8 bits) and VCI (16 bits)

_____ is used for typical data applications (where the data rate may be unknown and bursty) and allows use of whatever bandwidth is available at a given time.

Constant Bit Rate (CBR) service

Variable Bit Rate (VBR) service

Available Bit Rate (ABR) service

None of the given

L_19, p71

AVAILABLE BIT RATE (ABR)

It is used for typical data applications (where the data rate may be unknown and bursty) and allows use of whatever bandwidth is available at a given time.

_____ device is used for multicasting.

Hub

Switch

Router

None of the given

L_12, p43

HUBS:

They are used for extension of connection with multiplexing concept.

The term_____ refers to the general concept of a small block of data

L_4, p16

Packet is a generic term that refers to small block of data.

The term _____ is used to denote the definition of a packet used with a specific type of network.

Packet

Frame

Data

None of the given

L_4, p16

A frame or hardware frame denotes a packet of a specific format on a specific hardware technology.

A typical port on an ATM switch operates at _____ or higher.

OC-2 speed (155Mbps)

OC-3 speed (100Mbps)

OC-3 speed (155Mbps)

OC-3 speed (155Gbps)

L_20, p72

ATM SPEED

A typical port on an ATM switch operates at OC-3 speed (155Mbps) or higher

The term_____ refers to the general concept of a small block of data

Packet

Frame

Data

L_4, p16

Packet is a generic term that refers to small block of data.

The length of time required to send a variable length packet is variable and does not require a complicated interrupt scheme to detect completion of transmission.

True

False

L_20, p72

CELLS VS PACKETS

The length of time required to send a variable length packet is variable and requires complicated interrupt scheme to detect completion of transmission.

The product of delay and throughput measures the _____ of data that can be present on the network.

Area

Volume

Length

None of the given

L_22, p80

It measures amount of data that can be present in the network.

The checksum of 0000 and 0000 is _____

1111

0000

1110

0111

L 5, p 19

In this procedure data is treated as sequence integers and their arithmetic sum is computed and the carry bits are added to the final sum.

```
0000
+0000
-----
0000
```

_____ is used for compressed audio and video where the data rate depends on the level of compression that can be achieved.

Constant Bit Rate (CBR) service

Variable Bit Rate (VBR) service

Available Bit Rate (ABR) service

none of the given

L_19, p71

VARIABLE BIT RATE (VBR):

It is used for compressed audio and video where the data rate depends on the level of compression that can be achieved.

The product of delay and throughput measures the _____ of data that can be present on the network.

Area

Volume

Length

None of the given

1_22, P80

DELAY THROUPUT PRODUCT

It is computed as delay time multiplied by effective throughput. It measures amount of data that can be present in the network.

One repeater _____ two repeaters _____ the maximum cable length limitation.

Doubles, cancel

Doubles, triple

Square roots, cube roots

And, triple

L_13, p49

REPEATERS:

One repeater doubles two repeaters triple the maximum cable length limitation.

Which error detection method involves polynomials?

Simple parity check

Two-dimensional parity check

CRC

Checksum

L_5, p20

To understand the concept of CRC considers data in a message as co-efficient of a polynomial. Their co-efficient set is divided by a known polynomial.

A network with throughput T and delay D has a total of _____ bit in transit at any time.

T / D

$T \times D$

$T + D$

None of the given

L_22, p80

It is computed as delay time multiplied by effective throughput.

A network with throughput T and delay D has a total of $T \times D$ bits in transit at any time. (e-books)

The product of delay and throughput measures the _____ of data that can be present on the network.

L_22, p80

It measures amount of data that can be present in the network.

The maximum size of an Ethernet segment is_____

399. meters

400. meters

401. mete

rs None of

the given

L_7, p27

EXAMPLE BUS NETWORK: ETHERNET

One Ethernet cable is sometimes called a segment. This segment is limited to 500 meters in length. The minimum separation between connections is 3 meters.

Formally named _____ informally known as the twisted pair Ethernet or TP Ethernet.

10 Base 2

10 Base 5

10 Base T

None of the given

L_12, p46

10Base-T version of twisted pair Ethernet operated at 10Mbps. 10Base-T twisted pair Ethernet operates at 100Mbps.

Missing eot indicates sending computer crashed in frame format.

True

False

L_4, p16

FRAMING IN PRACTICE

A network technology that uses Collision Avoidance

Bridge

Broadcast

WAN

Wireless LAN

L_8, p29

CSMA/CA

Wireless uses collision avoidance rather than collision detection.

A typical port on an ATM switch operates at _____ or higher.

L_20, p72

A typical port on an ATM switch operates at OC-3 speed (155Mbps) or higher.

A _____ provides mechanism that automatically assigns a physical address to a station when the station first boots.

L_9, p34

DYNAMIC:

In this category the interface automatically assigns physical address each time it is powered up.

By size there are _____ classifications of networks.

L_1, p4

1) BY SIZE:

According to their size there are two classifications of networks.

- 402. Local Area Network. (LAN)
- 403. Wide Area Network (WAN)

In *Point-to-Point* topology there are two topologies.

Tree and Ring

Star and Ring

Star and Tree

None of the given

L_1, p5

In *Point-to-Point* topology there are two topologies.

404. Star topology

405. Tree topology

In _____, network occupies the smaller area like a room a floor or a building.

LAN

WAN

MAN

None of the given

L_1, p4

In LAN network occupies the smaller area like a room a floor or a building.

The product of delay and throughput measures the _____ of data that can be present on the network.

Area

Volume

Length

None of the give

L_22, p80

DELAY THROUGHPUT PRODUCT

It is computed as delay time multiplied by effective throughput. It measures amount of data that can be present in the network. In fast network with long delay times, sending computer can generate large amounts of data before destination receives first bit.

When an application _____ data, it makes a copy of the data available to all

other computers on the network.

Broadcasting

Multicasting

Unicasting

None of the given

L_9, p34

Some applications want to broadcast messages to all stations on the LAN. For this purpose shared communication channel can make broadcast efficient in such a way that message is delivered to all stations.

A special broadcast address is used to identify broadcast message, which are captured by all stations

CRC can detect more errors than a simple checksum

True

False

L_5, p20

CYCLIC REDUNDANCY CHECK (CRC):

To enable a network system to detect more error without increasing the amount of information in each packet another most successful approach is made which is called CRC

A _____ provide a mechanism that a customer can use to set a physical address.

Static addressing scheme

Configurable addressing scheme

Dynamic addressing scheme

None of the given

L_9, p34

CONFIGURABLE

_____ scheme, which is designed to help detect transmissions errors, send one extra bit of information with each character

Parity

Checksums

CRC

none of given

L_5, p18

Parity checking

To detect the error there are different schemes in which parity checking is also common used. In parity checking, parity refers to the number of bits set to 1 in the data item. A parity bit is an extra bit transmitted with data item chose to give the resulting bit even or odd parity.

_____ has no way to determine the cause of the problem.

ICMP

Ping

Trace route

None of the given

L_2, p11

If ping does not review any responses from host computer it can't tell the reason of problem.

A form of addressing; in which a set of computers is assigned one address.

Broadcast

Multicast

Packet Switch

CRC

Bandwidth

L_9, p34

CONFIGURABLE

A Bridge can _____

Filter a frame

Forward a frame

Extend a LAN

Do all the above

L_13, p50

A bridge is a hardware device also used to connect two LAN segments to extend a LAN.

It listens to all traffic and recognizes frame format. It also forwards only correct complete frames and discards the collided and error frames. **FRAME FILTERING**

The most important task a bridge performs is frame filtering.

Formally named _____ informally known as the thick wire Ethernet or Thick net.

10 Base 2

10 Base 5

10 Base T

None of the given

L_11, p41

THICK ETHERNET WIRING

It uses thick coax cable. AUI cable connects from NIC to transceiver.

L_12, p42

10 Base-T

It replaces AUI cable with twisted pair cable and thick coax with hub.

The number of connections needed for N computer in direct point to point communication is equal to:

$$\frac{(N^2 - N)}{2}$$

$$\frac{406 \cdot (N - 1)}{2}$$

None of the given

L_6, p23

For 'n' computers connections = $(n^2 - n)/2$.

A bridge uses _____ to determine which computer is connected to which segment, and uses the _____ to determine whether to forward a copy of frame.

Source address, destination address.

Destination address, destination address.

Destination address, source address.

source address, source address.

L_13, p50

FRAME FILTERING:

If both the source and destination are on the same segment, it does not forward the frame to the other segment. A frame is forwarded to the other segment, if it is destined to that segment.

A bridge keeps a list for each segment that consists of physical addresses of the computer attached to that segment. In this way a bridge knows on which segment a destination computer is attached.

_____ scheme, which is designed to help detect transmissions errors, send one extra bit of information with each character

Parity

Checksums

CRC

None of given

L_5, p18

PARITY CHECKING:

PARITY BIT:

A parity bit is an extra bit transmitted with data item chose to give the resulting bit even or odd parity.

Ethernet uses a ____ bit static addressing scheme in which each device is assigned a unique address by the manufacturer.

64

48

32

8

407. *Ethernet uses a 48-bit static addressing scheme* www.mutah.edu.jo/e-courses/ch9.ppt

In which type of topology computers are connected to a central hub?

Tree

Mesh

Bus

Star

L_1, p5

In *star* topology each computer is connected to a central hub.

A_____ relies on the hardware manufacturer to assign a unique physical address to each network interface.

Static addressing scheme

Dynamic addressing scheme

Configurable addressing scheme

None of the given

L_9, p34

STATIC:

In this category the hardware manufacturer assigns permanent physical address to each network interface and manufacturer must ensure that every interface has a unique address.

Basic LAN technologies such as Ethernet, Token Ring, and FDDI use a _____.

Connectionless service paradigm

Connection-oriented service paradigm

Both Connectionless and Connection-oriented service paradigm

None of the given

L_21, p77

CONNECTIONLESS:

An example of service paradigm is shown in the figure below.

Figure 21.1

Local Talk is a LAN technology that employs_____

Bus topology

Ring topology

Star topology

None of the given

L_8, p30

Apple invented the LAN technology that uses bus topology.

ATM is designed to work on_____.

Twisted Pair

Coaxial

Radio Frequency

Fiber

L_20, p72

ATM is designed to work on fiber (but can be used with twisted pair). A typical port on an ATM switch operates at OC-3 speed (155Mbps) or higher.

_____ Program sends a message to a remote computer and reports whether the computer responds.

Ping

Traceroute

ICMP

Non of the given

L_2, p11

A popular LAN Technology that uses a shared Bus Topology is called.

WAN

CSMA

Ethernet

Multicast

L_7, p27

EXAMPLE BUS NETWORK; ETHERNET:

Ethernet is a widely used LAN technology. -----

-----The Ethernet uses bus topology.

L_8, p30

LOCAL TALK

Apple invented the LAN technology that uses bus topology.

Formally named _____ informally known as the thick wire Ethernet or Thick net.

10 Base 2

10 Base T

None of the given

L_11, p41

THICK ETHERNET WIRING

It uses thick coax cable. AUI cable connects from NIC to transceiver.

L_12, p42

10 Base-T

It replaces AUI cable with twisted pair cable and thick coax with hub.

The original Ethernet hardware operated at a rate of _____

408. mbps

409. mbps

1000 mbps

None

L_7, p27

ETHERNET SPEEDS:

The Ethernet speed was originally 3Mbps, and the current standard is 10Mbps the fast Ethernet operates at 100Mbps. There are also gigabits Ethernet available now.

For point-to-point configuration, addressing is needed

True

False

L_8, p31

Computer gets point-to-point connections. Data from transmitters is routed directly through hub switches to destination.

_____was especially concerned about the lack of high powered computers.

ARPA

IEEE

EIA

None of the given

L_2, p9

The efforts of ARPA was to enable all its research groups have access to latest computers.

Most NICs contain _____ circuitry that allows the NIC to operate independent of the CPU.

DMA(Direct Memory Access)

Multiplexer

Transceiver

None of the given

L_11, p40

NIC contains sufficient hardware to process data independent of system CPU. In which some NICs contain separate microprocessor. In addition to this it also include analog circuitry interface to system bus, buffering and processing.

_____ have advantages arisen from the size and ease of computation.

CRC

Parity

Checksums

None of given

L_5, P20

To enable a network system to detect more error without increasing the amount of information in each packet another most successful approach is mad which is called CRC.

Unlike Frame Relay and ATM, SMDS (Switched multi-megabit Data service) offers_____.

Connectionless service paradigm

Connection oriented service paradigm

Both Connectionless and Connection-oriented service paradigm

None of the given

L_17, p65

SMDS:



Switched Multi megabit Data Service (SMDS) is also a Telco service. It is a connection less service. Any SMDS station can send information to any station on the same SMDS cloud. It typically ranges from 1.5Mbps to 1000Mbps.

A network uses a _____ if all computers attach to a central point

Star Topology

Ring Topology

Bus Topology

None of the given

L_7, p25

Star Topology

In this topology, all computers are attached to a central point, which is sometimes called the Hub.

OSPF is based on _____

Distance vector routing

Link state routing

Path vector routing

Distance vector routing and Link state routing

L_17, p64

LINK STATE ALGORITHM

It is used in OSPF

LAN that use ATM technology have a -----

Star topology

Bus topology

Ring topology

None of the given

L_8, p30

Many LAN technologies that are ring topology use token passing for synchronized access to the ring.

An interface for thin Ethernet must have a _____ connector, and must generate signals according to the _____ specification.

RJ-45, 10 Base T

RJ-45, 10 Base 5

BNC, 10 Base 2

BNC, 10 Base T

L_12, p44

Figure 12.2

The term_____ refers to the general concept of a small block of data

Packet

Frame

Data

None of the given

L4, p14

Network systems divide data in small blocks or junks called packets, which they send individually.

Missing eot indicates sending computer crashed in frame format.

True

False

L_4, p16

FRAMING IN PRACTICE

CRC can detect more errors than a simple checksum.

True

False

L_5, P20

To enable a network system to detect more error without increasing the amount of information in each packet another most successful approach is mad which is called CRC.

A Bridge forwards or filters a frame by comparing the information in its address table to the frame's_____

Layer 2 source address

Source node's physical address

Layer 2 destination address

Layer 3 destination address

L_13, p49

The most important task a bridge performs is frame filtering. A frame is forwarded to the other segment, if it is destined to that segment.

When an application_____ data, it makes a copy of the data available to all other computers on the network.

Broadcasting

Multicasting

Unicasting

None of the given

L_9, p34

Some applications want to broadcast messages to all stations on the LAN. For this purpose shared communication channel can make broadcast efficient in such a way that message is delivered to all stations.

A special broadcast address is used to identify broadcast message, which are captured by all stations

The product of delay and throughput measures the _____ of data that can be present on the network.

L_22, p80

It measures amount of data that can be present in the network.

By size there are _____ classifications of networks.

L_1, p4

1) BY SIZE:

According to their size there are two classifications of networks.

- 410. Local Area Network. (LAN)
- 411. Wide Area Network (WAN)

A _____ provide a mechanism that a customer can use to set a physical address.

Static addressing scheme

Configurable addressing scheme

Dynamic addressing scheme

None of the given

L_9, p34

CONFIGURABLE

In this category, the address can be set by the end user either manually

Computers attached to an ether use ----- in which a computer waits for the ether to be idle before transmitting a frame.

CSMA/CD

CSMA/CA

TOKEN PASSING

None of the given

L_8, p28

After collision is detected computers stop transmitting. So Ethernet uses CSMA/CD to coordinate transmission. **RECOVERY FROM COLLISION:**

Computer that detects a collision sends special signal to force all other interfaces to detect collision. Computer then waits for other to be idle before transmission.

The term _____ is used to denote the definition of a packet used with a specific type of network.

L_4, p16

FRAME:

A frame or hardware frame denotes a packet of a specific format on a specific hardware technology.

The term_____ refers to the general concept of a small block of data

Packet

Frame

Data

L_4, p16

Packet is a generic term that refers to small block of data.

_____ has advantages arisen from the size and ease of computation.

CRC

Parity

Checksums

None of given

L_5, P20

To enable a network system to detect more error without increasing the amount of information in each packet another most successful approach is mad which is called CRC.

A Bridge can_____

Filter a frame

Forward a frame

Extend a LAN

Do all the above

L_13, p50

BRIDGES

A bridge is a hardware device also used to connect two LAN segments to extend a LAN.

FRAME FILTERING

The most important task a bridge performs is frame filtering. If both the source and destination are on the same segment, it does not forward the frame to the other segment. A frame is forwarded to the other segment, if it is destined to that segment.

In _____ network occupies larger areas like cities & countries.

LAN

WAN

MAN

None of the given

L_1, p4

In WAN, network occupies larger areas like cities & countries.

The next hop to which a packet is sent depends only on_____

L_15, p57

NEXT HOP FORWARDING:

- 412. If the destination is local computer, packet switch delivers computer port.
- 413. If the destination is attached another switch, this packet switch forwards to next hop through connection to another switch.
- 414. The choice of another switch is based on destination address in packet.

The Gigabit Ethernet hardware operates at a rate of _____

415. Mbps

416. Mbps

1000 Mbps None
of the given

L_7, p27

ETHERNET SPEEDS:

The Ethernet speed was originally 3Mbps, and the current standard is 10Mbps the fast Ethernet operates at 100Mbps. There are also gigabits Ethernet available now.

_____ is used for audio and video, since these have predefined maximum data rates

Constant Bit Rate (CBR) service

Variable Bit Rate (VBR) service

Available Bit Rate (ABR) service

None of the given

L_19, p71

CONSTANT BIT RATE (CBR):

It is used for audio and video, since these have predefined maximum data rates.

_____ is used for typical data applications (where the data rate may be unknown and bursty) and allows use of whatever bandwidth is available at a given time.

Constant Bit Rate (CBR) service

Variable Bit Rate (VBR) service

Available Bit Rate (ABR) service

None of the given

L_19, p71

AVAILABLE BIT RATE (ABR)

It is used for typical data applications (where the data rate may be unknown and bursty) and allows use of whatever bandwidth is available at a given time.

_____ has no way to determine the cause of the problem.



ICMP

Ping

Trace route

None of the given

L_2, p11

If ping does not review any responses from host computer it can't tell the reason of problem.

A Bridge forwards or filters a frame by comparing the information in its address table to the frame's _____

Layer 2 source address

Source node's physical address

Layer 2 destination address

Layer 3 destination address

L_13, p50

A bridge keeps a list for each segment that consist of physical addresses of the computer attached to that segment. In this way a bridge knows on which segment a destination computer is attached.

Most bridges are self learning bridges. As soon as a frame arrives to a bridge, it extracts a source address from its header and automatically adds it in the list for that segment. In this way a bridge builds up address lists.

In broadcast topology there are two categories_____.

L_1, p5

In broadcast topology there are further two categories

417. SATELLITE\ RADIO

418. RING TOPOLOGY

When an application_____ data, it makes a copy of the data available to all other computers on the network?

Broadcasting

Multicasting

Unicasting

None of the given

L_9, p34

Some applications want to broadcast messages to all stations on the LAN. For this purpose shared communication channel can make broadcast efficient in such a way that message is delivered to all stations.

A special broadcast address is used to identify broadcast message, which are captured by all stations

A typical port on an ATM switch operates at _____ or higher.

L_20, p72

A typical port on an ATM switch operates at OC-3 speed (155Mbps) or higher.

A network that permits many sources to take turns accessing a shared communication resource is providing a form of _____

Packet Division Multiplexing

Time Division Multiplexing

Frequency Division Multiplexing

Wave Division Multiplexing

L_4, p15

In TDM each packet leaves the source and is switched on the shared communication channel through a multiplexer.

When an application_____data, it makes a copy of the data available to all other computers on the network.

Broadcasting

Multicasting

Unicasting

None of the given

L_9, p34

Some applications want to broadcast messages to all stations on the LAN. For this purpose shared communication channel can make broadcast efficient in such a way that message is delivered to all stations.

A special broadcast address is used to identify broadcast message, which are captured by all stations

ATM assigns each VC a _____ identifier that is divided two parts to produce a hierarchy.

21-bit

22-bit

23-bit

24-bit

L_18, p67

The VC is identified by a 24 bit value formed from the VPI (8 bits) and VCI (16 bits)

Basic LAN technologies such as Ethernet, Token Ring, and FDDI use a _____.

Connectionless service paradigm

Connection-oriented service paradigm

Both Connectionless and Connection-oriented service paradigm

None of the given

L_21, p77

CONNECTIONLESS:

An example of service paradigm is shown in the figure below.

Figure 21.1

☐ ICMP

☐ Non of the given

Question No: 3 (Marks: 1) - Please choose one

----- was especially concerned about the lack of high powered computers.

☐ [ARPA \(Search by Internet\)](#) [click here to see](#)

☐ IEEE

☐ EIA

☐ Non of the given

Question No: 4 (Marks: 1) - Please choose one

The term ----- is used to denote the definition of a packet used with a specific type of network.

☐ Packet

☐ [Frame \(p-16\)](#)

☐ Data

☐ None of the given

Question No: 5 (Marks: 1) - Please choose one

Computer networks are often called ----- because they use packet technology.

☐ Ethernet

☐ Switch networks

☐ **Packet networks** ([Search by Internet](#)) [Click Here to see](#)

☐ None of the given

Question No: 6 (Marks: 1) - Please choose one

----- have advantages arisen from the size and ease of computation.

☐ **CRC** (p-20)

☐ Parity

☐ Checksums

☐ None of given

Question No: 7 (Marks: 1) - Please choose one

Most LANs that employ ring topology use an access mechanism known as-----

☐ CSMA/CD

☐ CSMA/CA

☐ **TOKEN PASSING** (p-30)

☐ None of the given

Question No: 8 (Marks: 1) - Please choose one

IEEE LLC/SNAP header is -----, which is used to specify the type of data.

☒ 8 octets (p-37)

☐ 8 bytes

☐ 8 bits

☐ None of the given

Question No: 9 (Marks: 1) - Please choose one

Formally named _____ informally known as the twisted pair Ethernet or TP Ethernet.

☐ 10Base 2

☐ 10Base 5

☒ 10Base T (p-46)

☐ None of the given

Delivered By:

Question No: 10 (Marks: 1) - Please choose one

An interface for twisted pair Ethernet must have an _____ connector , and must generate signals according to the _____ specification.

☒ **RJ-45, 10 Base T (p-47)**

☐ RJ-45, 10 Base 5

☐ BNC, 10 Base 2

☐ BNC, 10 Base T

Question No: 11 (Marks: 1) - Please choose one

A bridges function in the _____ layers(s).

☐ Physical (MAC)

☐ Data link

☒ **Network (not sure)**

☐ Physical (MAC) and Data link

Question No: 12 (Marks: 1) - Please choose one

A Bridge can _____

☐ Filter a frame

☐ Forward a frame

☐ Extend a LAN

☒ **Do all the above (p-50)**

Question No: 13 (Marks: 1) - Please choose one

A Bridge forwards or filters a frame by comparing the information in its address table to the frame's _____

- ☐ Layer 2 source address
- ☐ Source node's physical address
- ☒ **Layer 2 destination address (not sure)**
- ☐ Layer 3 destination address

Question No: 14 (Marks: 1) - Please choose one

_____ computes shortest paths in a graph by using weights on edges as a measure of distance.

- ☐ Greedy algorithm
- ☐ Distance vector algorithm
- ☒ **Dijkstra's algorithm (p-62)**
- ☐ Non of the given

Question No: 15 (Marks: 1) - Please choose one

_____ is used for audio and video, since these have predefined maximum data rates

☒ **Constant Bit Rate (CBR) service (p-71)**

☐ Variable Bit Rate (VBR) service

☐ Available Bit Rate (ABR) service

☐ None of the given

Question No: 16 (Marks: 1) - Please choose one

Unlike Frame Relay and ATM, SMDS (Switched multi-megabit Data service) offers_____ .

☒ **Connectionless service paradigm (p-152)**

☐ Connection oriented service paradigm

☐ Both Connectionless and Connection-oriented service paradigm

☐ None of the given

Question No: 17 (Marks: 1) - Please choose one

A network with throughput T and delay D has a total of _____ bit in transit at any time.

☐ T/D

☒ **T x D (p-80)**

☐ T+D

☐ None of the given

Question No: 18 (Marks: 1) - Please choose one

ATM is designed to work on_____.

☐ Twisted Pair

☐ Coaxial

☐ Radio Frequency

☒ **Fiber (p-32)**

Question No: 19 (Marks: 1) - Please choose one

Computers attached to an Ethernet use ----- in which a computer waits for the ether to be idle before transmitting a frame.

☒ **CSMA/CD (p-43)**

☐ CSMA/CA

☐ TOKEN PASSING

☐ None of the given

Question No: 20 (Marks: 1) - Please choose one

FDDI can transmits data at a rate of -----

_ 100 million bits per second (p-31)

_ 10 million bits per second

_ 1000 million bits per second

_ None of the given

Question No: 1 (M a r k s: 1)

In Point-to-Point topology there are two topologies.

419. Tree and Ring

420. Star and Ring

421. Star and Tree (Reference is given below)

422. None of the given

In Point-to-Point topology there are two topologies.

1) STAR topology 2) TREE topology

Question No: 2 (M a r k s: 1)

----- Program sends a message to a remote computer and reports whether the computer responds.

423. Ping (Reference is given below)

- 424. Traceroute
- 425. ICMP
- 426. Non of the given

Ping program tests reach ability. It sends datagram from B to A, that echoes back to B. it uses ICMP echo request and echo reply messages.

Question No: 3 (M a r k s: 1)

----- has no way to determine the cause of the problem.

- 427. ICMP
- 428. Ping** (Reference is given below)
- 429. Trace route
- 430. Non of the given

Ping has no way to determine the cause of the problem. Tracing A Route Network administrators use another tool, trace route.

Question No: 4 (M a r k s: 1)

The term----- refers to the general concept of a small block of data

431. Packet (Reference is given below)

432. Frame

433. Data

434. None of the given

Packet is a generic term that refers to small block of data. Packet have different format.

Each hardware uses different packet format.

Question No: 5 (M a r k s: 1)

----- scheme, which is designed to help detect transmissions errors, send one

extra bit of information with each character

435. Parity (Reference is given below)

436. Checksums

437. CRC

438. None of given

A parity scheme, which is designed to help detect transmission errors, sends one extra bit of information with each character. Although it allows receiver to determine if a single bit has been changed parity can not detect transmission errors that change an even number of bits.

Question No: 6 (Marks: 1)

Local Talk is a LAN technology that employs -----

439. Bus topology (Reference is given below)

440. Ring topology

441. Star topology

442. None of the given

LOCAL TALK:

Apple invented the LAN technology that uses bus topology. Its interface is included with all Macintosh computers.

Question No: 7 (M a r k s: 1)

Most LANs that employ ring topology use an access mechanism known as-----

443. CSMA/CD

444. CSMA/CA

445. TOKEN PASSING (Reference is given below)

446. None of the given

Most LANs that employ ring technology use an access mechanism known as token passing. A token ring operates as a single, shared medium. When a computer wants to send data, it must wait until it obtains the token, when it is in control of the ring

Question No: 8 (M a r k s: 1)

Ethernet uses a ----- bit static addressing scheme in which each device is assigned a unique address by the manufacturer.

447. 64

448. 48 (Reference is given below)

449. 32

450. 8

Ethernet uses a 48-bit static addressing scheme

Question No: 9 (M a r k s: 1)

Formally named _____ informally known as the thick wire Ethernet or Thick net.

451. 10 Base 2

452. 10 Base 5 (Reference is given below)

453. 10 Base T

454. None of the given

Thick Ethernet, officially known as 10 Base 5, is the oldest form of Ethernet. It was originally developed in the late 1970's by Digital Equipment Corporation, IBM, and Xerox, and became an international standard (IEEE 802.3) in 1983. Taken from <http://www.maznets.com/tech/10base-5.htm>

Question No: 10 (M a r k s: 1)

Formally named _____ informally known as the twisted pair Ethernet or TP Ethernet.

- 455. 10 Base 2
- 456. 10 Base 5
- 457. 10 Base T (Reference is given below)**
- 458. None of the given

twisted pair ethernet

Formally called 10 Base –T

Also twisted pair Ethernet or simply TP Ethernet

Question No: 11 (M a r k s: 1)

The maximum size of an Ethernet segment is_____

- 459. 250 meters
- 460. 500 meters** (Reference is given below)
- 461. 700 meters
- 462. None of the given

One Ethernet cable is sometimes called a segment. This segment is limited to 500 meters in length. The minimum separation between connections is 3 meters. (P# 27)

Question No: 12 (M a r k s: 1)

A Bridge can _____

- 463. Filter a frame
- 464. Forward a frame
- 465. Extend a LAN
- 466. Do all the above** (Reference is given below)

Delivered

A bridge is a hardware device also used to connect two LAN segments to extend a LAN. It listens to all traffic and recognizes frame format. It also forwards only correct complete frames and discards the collided and error frames.

The most important task a bridge performs is frame filtering.

Question No: 13 (M a r k s: 1)

_____ computes shortest paths in a graph by using weights on edges as a measure of distance.

- 467. Greedy algorithm
- 468. Distance vector algorithm
- 469. Dijkstra's algorithm** (Reference is given below)
- 470. Non of the given

Dijkstras algorithm computes shortest paths in a graph by using weights on edge as a measure of distance. A path with the fewest number of edges may not be the path with least weight. (P# 173 e-books)

Question No: 14 (M a r k s: 1)

_____ is used for compressed audio and video where the data rate depends on the level of compression that can be achieved.

► Constant Bit Rate (CBR) service

► **Variable Bit Rate (VBR) service** (Reference is given below)

471. Available Bit Rate (ABR) service

472. None of the given

VARIABLE BIT RATE (VBR):

It is used for compressed audio and video where the data rate depends on the level of compression that can be achieved.

Question No: 15 (Marks: 1)

Basic LAN technologies such as Ethernet, Token Ring, and FDDI use a ____.

473. Connectionless service paradigm (Reference is given below)

474. Connection-oriented service paradigm

475. Both Connectionless and Connection-oriented service paradigm

476. None of the given

Basic LAN technologies such as Ethernet, Token Ring, and FDDI use a connectionless service paradigm. Although a computer needs to wait for access to a shared medium before sending a packet

(zh,vuzs,nov10)

Question No: 16 (Marks: 1)

The product of delay and throughput measures the ____ of data that can be present on the network.

477. Area

478. Volume

479. Length

480. None of the given (Reference is given below)

DELAY THROUPTUT PRODUCT:

It is computed as delay time multiplied by effective throughput. It measures amount of data that can be present in the network (P# 80)

Question No: 17 (M a r k s: 1)

A network with throughput T and delay D has a total of _____ bit in transit at any time.

481. T / D

482. T x D (Reference is given below)

483. T + D

484. None of the given

A network with throughput T and delay D has a total of $T \times D$ bits in transit at any time. (e-books)

Question No: 18 (M a r k s: 1)

One repeater _____, two repeaters _____ the maximum cable length limitation.

485. doubles, triple (Reference is given below)

486. square roots, cube roots

487. and, triple

488. doubles, cancel

One repeater doubles, two repeaters triple the maximum cable length limitation. (P# 49)

Question No: 19 (M a r k s: 1)

A network uses a ----- if all computers attach to a central point

489. Star Topology (Reference is given below)

490. Ring Topology

491. Bus Topology

492. None of the given

STAR TOPOLOGY:

In this topology, all computers are attached to a central point, which is sometimes called the “Hub” (P# 25)

Question No: 20 (M a r k s: 1)

Computers attached to an Ethernet use ----- in which a computer waits for the other to be idle before transmitting a frame.

493. CSMA/CD (Reference is given below)

494. CSMA/CA

495. TOKEN PASSING

496. None of the given

All computers attached to the Ethernet use CSMA/CD to co-ordinate their activities. A computer wishing to transmit checks for electrical activity on the cable, informally called a carrier. If there is no carrier, the computer can transmit. If a carrier is present, the computer waits for the sender to finish before proceeding.

MIDTERM EXAMINATION

Fall 2008

CS610- Computer Network (Session - 2)

Time: 60 min

Marks: 38

Question No: 1 (Marks: 1) - Please choose one

A typical port on an ATM switch operates at _____ or higher.

☐ OC-2 speed (155Mbps)

☐ OC-3 speed (100Mbps)

☒ **OC-3 speed (155Mbps) (p-72)**

☐ OC-3 speed (155Gbps)

Question No: 2 (Marks: 1) - Please choose one

The product of delay and throughput measures the _____ of data that can be present on the network.

☐ Area

☐ Volume

☐ Length

☒ **None of the given (p-80)**

Question No: 3 (Marks: 1) - Please choose one

_____ is used for compressed audio and video where the data rate depends on the level of compression that can be achieved.

☐ Constant Bit Rate (CBR) service

☐ Variable Bit Rate (VBR) service (p-71)

☐ Available Bit Rate (ABR) service

☐ None of the given

Question No: 4 (Marks: 1) - Please choose one
Which of the following is a connecting device?

☐ Bridge

☐ Repeater

☐ Hub

☐ All the given (p-54)

Question No: 5 (Marks: 1) - Please choose one

A bridges function in the _____ layers(s).

☐ Physical (MAC)

☐ Data link

___Network (not sure)

___Physical (MAC) and Data link

Question No: 6 (Marks: 1) - Please choose one

IEEE LLC/SNAP header is -----, which is used to specify the type of data.

___8 octets (p-37)

___8 bytes

___8 bits

___None of the given

Question No: 7 (Marks: 1) - Please choose one

The third field of the header consists of ----- bit Ethernet frame type.

___48

___32

___16 (p-36)

___8

Question No: 8 (Marks: 1) - Please choose one

An ----- method, the network hardware designers specify how type information is included in the frame and the value use to identify various frame types.

___Explicit frame type (not sure)

___Ideal frame type

___Implicit frame type

___None of the given

Question No: 9 (Marks: 1) - Please choose one

Local Talk is a LAN technology that employs -----

___Star topology

☒ **Bus topology (p-30)**

☐ Ring topology

☐ None of the given

Question No: 10 (Marks: 1) - Please choose one

The Fast Ethernet hardware operates at a rate of -----

☐ 10 Mbps

☒ **100 Mbps (p-27)**

☐ 1000 Mbps

☐ None of the given

Question No: 11 (Marks: 1) - Please choose one

----- scheme, which is designed to help detect transmissions errors, send one extra bit of information with each character

☐ Parity

☒ Checksums (p-19)

☐ CRC

☐ None of given

Question No: 12 (Marks: 1) - Please choose one

Computer networks are often called ----- because they use packet technology.

☐ Ethernet

☐ Switch networks

☒ Packet networks (repeated)

☐ None of the given

Question No: 13 (Marks: 1) - Please choose one

----- Program sends a message to a remote computer and reports whether the computer responds.

☒ Ping (p-11)

☐ Traceroute

☐ ICMP

☐ Non of the given

Question No: 14 (Marks: 1) - Please choose one

In -----, network occupies larger areas like cities & countries.

☐ LAN

☒ WAN (p-55)

☐ MAN

☐ None of the given

Question No: 15 (Marks: 1) - Please choose one

In -----, network occupies the smaller area like a room a floor or a building

☒ LAN (p-55)

☐ WAN

☐ MAN

☐ None of the given

Question No: 16 (Marks: 1) - Please choose one In Point-to-Point topology there are two topologies.

☐ Tree and Ring

☐ Star and Ring

☒ **Star and Tree (p-2)**

☐ None of the given

MIDTERM EXAMINATION

Fall 2008

CS610- Computer Network (Session - 2)

Ref No: 109816

Time: 60 min

Marks: 38

Question No: 1 (Marks: 1) - Please choose one

_____ has a jitter zero

☐ None of the given

☐ Virtual Private Network

☐ Isochronous Network

☒ **Asynchronous Network (p-65)**

Question No: 2 (Marks: 1) - Please choose one

_Unlike Frame Relay and ATM, SMDS (Switched multi-megabit Data service) offers_____ .

___Connectionless service paradigm (p-77)

___Connection oriented service paradigm

___Both Connectionless and Connection-oriented service paradigm

___None of the given

Question No: 3 (Marks: 1) - Please choose one

_ATM assigns each VC a _____ identifier that is divided two parts to produce a hierarchy.

___21-bit

___22-bit

___23-bit

___24-bit (p-67)

Question No: 4 (Marks: 1) - Please choose one

_Most WAN systems include a mechanism that can be used to eliminate the common case of duplication routing is called_____

___Hierarchal address

___**Default route (p-60)**

___Shortest path

___None of the given

Question No: 5 (Marks: 1) - Please choose one

_The next hop to which a packet is sent depends only on

___**Packet's destination (p-102)**

___Packet's original source

___Path the packet has taken

___Non of the given

Question No: 6 (Marks: 1) - Please choose one

_An interface for twisted pair Ethernet must have an _____ connector , and must generate signals according to the_____ specification.

___**RJ-45, 10 Base T (p-46)**

___RJ-45, 10 Base 5

___BNC, 10 Base 2

___BNC, 10 Base T

Question No: 7 (Marks: 1) - Please choose one

_When an application----- data, it makes a copy of the data available to all other computers on the network.

___Broadcasting (p-53)

___Multicasting

___Unicasting

___None of the given

Question No: 8 (Marks: 1) - Please choose one

_A ----- provide a mechanism that a customer can use to set a physical address.

___Static addressing scheme

___Configurable addressing scheme (p-34)

___Dynamic addressing scheme

___None of the given

Question No: 9 (Marks: 1) - Please choose one

_FDDI can transmits data at a rate of -----

☒ **100 million bits per second ----- (p-31)**

☐ 100 million bits per second

☐ 100 million bits per second

☐ 100 million bits per second

☐ None of the given

Question No: 10 (Marks: 1) - Please choose one

_Computers attached to an ether use ----- in which a computer waits for the ether to be idle before transmitting a frame.

☒ **CSMA/CD**

☐ CSMA/CA

☐ TOKEN PASSING

☐ None of the given

Question No: 11 (Marks: 1) - Please choose one

_----- have advantages arisen from the size and ease of computation.

☐ CRC

☐ Parity

☒ **Checksums (p-19)**

☐ None of given

Question No: 12 (Marks: 1) - Please choose one

_The term ----- is used to denote the definition of a packet used with a specific type of network.

☐ Packet

☒ **Frame (p-16)**

☐ Data

☐ None of the given

Question No: 13 (Marks: 1) - Please choose one

_----- has no way to determine the cause of the problem.

☒ **Ping (p-11)**

☐ Trace route

☐ ICMP

☐ Non of the given

Question No: 14 (Marks: 1) - Please choose one

_----- Program sends a message to a remote computer and reports whether the computer responds.

☐ **Ping (p-11)**

☐ Traceroute

☐ ICMP

☐ Non of the given

Question No: 15 (Marks: 1) - Please choose one

_In -----, network occupies the smaller area like a room a floor or a building

☐ **LAN (p-55)**

☐ WAN

☐ MAN

☐ None of the given

MIDTERM EXAMINATION

Spring 2010

CS610- Computer Network

Question No: 1 (Marks: 1) - Please choose one

In *Point-to-Point* topology there are two topologies.

a. Star and Tree (P-5)

- 497.** Star and Ring
- 498. Star and Bus
- 499.** None of the given

Question No: 2 (Marks: 1) - Please choose one

----- Program sends a message to a remote computer and reports whether the computer responds.

500. Ping (p-11)

501. Traceroute

502. ICMP

503. Non of the given

Question No: 3 (Marks: 1) - Please choose one

----- has no way to determine the cause of the problem.

504. ICMP

505. Non of the given

506. **Ping (p-11)**

507. Trace route

Question No: 4 (Marks: 1) - Please choose one

The term----- refers to the general concept of a small block of data

508. **Packet (p-16)**

509. Frame

510. Data

511. None of the given

Question No: 5 (Marks: 1) - Please choose one

----- scheme, which is designed to help detect transmissions errors, send one extra bit of information with each character

512. Parity (p-18)

513. Checksums

514. CRC

515. None of given

Question No: 6 (Marks: 1) - Please choose one

Local Talk is a LAN technology that employs -----

516. **Bus topology (p-30)**

517. Ring topology

518. None of the given

519. Star topology

Question No: 7 (Marks: 1) - Please choose one

Most LANs that employ ring topology use an access mechanism known as-----

520. CSMA/CD

521. CSMA/CA

522. **TOKEN PASSING (p-30)**

523. None of the given

Question No: 8 (Marks: 1) - Please choose one

Ethernet uses a ----- bit static addressing scheme in which each device is assigned a unique address by the manufacturer.

524. 64

525. **48 (p-98)**

526. 32

527. 8

Question No: 9 (Marks: 1) - Please choose one

Formally named _____ informally known as the thick wire Ethernet or Thick net.

a. 10 Base 2

528. 10 Base 5

529. None of the given

Question No: 10 (Marks: 1) - Please choose one

Formally named _____ informally known as the twisted pair Ethernet or TP Ethernet.

► 10 Base 2 (p-46)

530. 10 Base 5

531. 10 Base T

532. None of the given

Question No: 11 (Marks: 1) - Please choose one

The maximum size of an Ethernet segment is_____

533. 250 meters

534. 500 meters (p-27)

535. 700 meters

536. None of the given

Question No: 12 (Marks: 1) - Please choose one

A Bridge can _____

537. Filter a frame

538. Forward a frame

539. Extend a LAN

540. **Do all the above (p-50)**

Question No: 13 (Marks: 1) - Please choose one

_____ computes shortest paths in a graph by using weights on edges as a measure of distance.

541. Greedy algorithm

a. Distance vector algorithm

b. Dijkstra's algorithm (p-62)

c. Non of the given

Question No: 14 (Marks: 1) - Please choose one

_____is used for compressed audio and video where the data rate depends on the level of compression that can be achieved.

542. Constant Bit Rate (CBR) service

543. Variable Bit Rate (VBR) service (p-71)

544. Available Bit Rate (ABR) service

545. None of the given

Question No: 15 (Marks: 1) - Please choose one

Basic LAN technologies such as Ethernet, Token Ring, and FDDI use a _____.

► **Connectionless service paradigm (p-77)**

546. Connection-oriented service paradigm

547. Both Connectionless and Connection-oriented service paradigm

548. None of the given

Question No: 16 (Marks: 1) - Please choose one

The product of delay and throughput measures the _____ of data that can be present on the network.

549. Area

550. Volume

551. Length

552. **None of the given (p-80)**

Question No: 17 (Marks: 1) - Please choose one

A network with throughput T and delay D has a total of _____ bit in transit at any time.

553. T / D

554. $T \times D$ (p-80)

555. $T + D$

556. None of the given

Question No: 18 (Marks: 1) - Please choose one

One repeater _____, two repeaters _____ the maximum cable length limitation.

557. doubles, triple (p-49)

558. square roots, cube roots

559. and, triple

560. doubles, cancel

Question No: 19 (Marks: 1) - Please choose one

A network uses a ----- if all computers attach to a central point

561. Star Topology (p-25)

562. Ring Topology

563. Bus Topology

564. None of the given

Question No: 20 (Marks: 1) - Please choose one

Computers attached to an Ethernet use ----- in which a computer waits for the ether to be idle before transmitting a frame.

565. CSMA/CD (p-28)

566. CSMA/CA

567. TOKEN PASSING

568. None of the given

MIDTERM EXAMINATION

Fall 2008

CS610- Computer Network (Session - 2)

Ref No: 109816

Time: 60 min

Marks: 38

Question No: 1 (Marks: 1) - Please choose one

_____ has a jitter zero

- 569. None of the given
- 570. Virtual Private Network
- 571. Isochronous Network

572. Asynchronous Network (p-65)

Question No: 2 (Marks: 1) - Please choose one

Unlike Frame Relay and ATM, SMDS (Switched multi-megabit Data service) offers_____ .

► **Connectionless service paradigm (p-77)**

► Connection oriented service paradigm

► Both Connectionless and Connection-oriented service paradigm

► None of the given

Question No: 3 (Marks: 1) - Please choose one

ATM assigns each VC a _____ identifier that is divided two parts to produce a hierarchy.

► 21-bit

► 22-bit

► 23-bit

► **24-bit (p-67)**

Question No: 4 (Marks: 1) - Please choose one

Most WAN systems include a mechanism that can be used to eliminate the common case of duplication routing is called_____

- Hierarchal address
- **Default route (p-60)**
- Shortest path
- None of the given

Question No: 5 (Marks: 1) - Please choose one The next hop to which a packet is sent depends only on_____

- **Packet's destination (p-102)**
- Packet's original source
- Path the packet has taken
- Non of the given

Question No: 6 (Marks: 1) - Please choose one

An interface for twisted pair Ethernet must have an _____ connector , and must generate signals according to the _____ specification.

► **RJ-45, 10 Base T (p-46)**

- RJ-45, 10 Base 5
- BNC, 10 Base 2
- BNC, 10 Base T

Question No: 7 (Marks: 1) - Please choose one

When an application ----- data, it makes a copy of the data available to all other computers on the network.

► **Broadcasting (p-53)**

- Multicasting
- Unicasting
- None of the given

Question No: 8 (Marks: 1) - Please choose one

A ----- provide a mechanism that a customer can use to set a physical address.

- Static addressing scheme
- **Configurable addressing scheme (p-34)**
- Dynamic addressing scheme
- None of the given

Question No: 9 (Marks: 1) - Please choose one

FDDI can transmits data at a rate of -----

► **100 million bits per second (p-31)**

- 100 million bits per second
- 100 million bits per second

► None of the given

Question No: 10 (Marks: 1) - Please choose one

Computers attached to an ether use ----- in which a computer waits for the ether to be idle before transmitting a frame.

► CSMA/CD (not sure)

► CSMA/CA

► TOKEN PASSING

► None of the given

Question No: 11 (Marks: 1) - Please choose one

----- have advantages arisen from the size and ease of computation.

▶ CRC

▶ Parity

▶ **Checksums (p-19)**

▶ None of given

Question No: 12 (Marks: 1) - Please choose one

The term ----- is used to denote the definition of a packet used with a specific type of network.

▶ Packet

▶ **Frame (p-16)**

▶ Data

▶ None of the given

Question No: 13 (Marks: 1) - Please choose one

----- has no way to determine the cause of the problem.

▶ **Ping (p-11)**

▶ Trace route

▶ ICMP

▶ Non of the given

Question No: 14 (Marks: 1) - Please choose one

----- Program sends a message to a remote computer and reports whether the computer responds.

► **Ping (p-11)**

► Traceroute

► ICMP

► Non of the given

Question No: 15 (Marks: 1) - Please choose one

In -----, network occupies the smaller area like a room a floor or a building

► **LAN (p-55)**

► WAN

► MAN

► None of the given

Question No: 21 (Marks: 2)

What is the difference between the physical and logical topologies?

Every LAN has a topology, or the way that the devices on a network are arranged and how they communicate with each other.

PHYSICAL TOPOLOGY:

The way that the workstations are connected to the network through the actual cables that transmit data -- the physical structure of the network -- is called the physical topology. It depends on the wiring scheme.

LOGICAL TOPOLOGY:

The logical topology, in contrast, is the way that the signals act on the network media, or the way that the data passes through the network from one device to the next without regard to the physical interconnection of the devices. We can say that it is defined by the specific network technology.

Question No: 22 (Marks: 2)

Define Vector-Distance Algorithm.

Packet switches wait for next update message and they iterate through entries in message. If entry has shortest path to destination, insert source as next hop to destination

and record distance as distance from next hop to destination plus distance from this switch to next hop.

Question No: 23 (Marks: 3)

What is the concept of store and forward technology?

STORE AND FORWARD:

Data delivery from one computer to another is accomplished through store and forward technology. In this technology packet switch stores incoming packet and also forwards that packet to another switch or computer. For this purpose packet switch has internal memory into which it can hold packet if outgoing connection is busy. Packets for each connection held on queue.

Question No: 24 (Marks: 3)

How can Switched Virtual Network be established?

SWITCHED VIRTUAL CIRCUITS:

Most networks offer dynamic connections, which last for a relatively short time. To handle this, ATM can dynamically establish a switched virtual circuit (SVC), allow it last as long as necessary and then terminate it.

The terminology comes from the Telco's where switching system normally refers to all switching.

ESTABLISHING AN SVC:

The computer sends a connection request to the switch to which it is attached. Software in the switch finds a network path to the destination and sends along the connection request.

Each pair of switches in the path communicates to choose a VPI/VCI for their tables. Once the connection is established by the destination, a message is sent back to the originating computer to indicate the SVC is ready.

If any switch or the destination computer does not agree to setting up the VC, an error message is sent back and the SVC is not established

Question No: 25 (Marks: 5)

How can a bridge know whether to forward frames?

The bridge builds a list of MAC addresses on either side of the bridge. Therefore, it knows which packets should be forwarded to the other side and which should not. Most bridges are self-learning bridges. As soon as a frame arrives to a bridge, it extracts a source address from its header and automatically adds it in the list for that segment. In this way a bridge builds up address lists.

In the example of a packet that uses a MAC address not in its table it can err on the side of caution by forwarding the packet.

Question No: 26 (Marks: 5)

Compare connection oriented and connectionless Service.

Connection-Oriented vs. Connectionless Service

This characteristic specifies whether conversations take place in a more or less structured manner. When using a connection-oriented protocol, you incur the overhead of setting up a virtual circuit (a defined communications path) between the sender and

receiver, which is maintained until the sender and receiver have completed their entire conversation.

When the conversation is completed, you incur the overhead of tearing down the virtual circuit. Connection-oriented protocols provide guaranteed delivery of messages in the order in which they were sent.

Contrast this with Connectionless service, which does not require establishing a session and a virtual circuit. This can be found in the network layer or transport layer, depending on the protocol. You can think of a connectionless protocol as being akin to mailing a post card. You send it and hope that the receiver gets it. Common features of a connectionless service are:

- 573. Packets do not need to arrive in a specific order
- 574. Reassembly of any packet broken into fragments during transmission must be in proper order
- 575. No time is used in creating a session
- 576. No Acknowledgement is required.
- 577. The largest connectionless network in use today is the Internet

Question No: 21 (Marks: 2)

What is the difference between LAN and WAN?

Local area network (LAN)

LAN is small in size covers the area within a room, building or cities.

Wide area network (WAN)

WAN is large in size and covers the area cities, countries and continents.

Question No: 22 (Marks: 2)

Define the term Jitter.

The term Jitter is used for variance in transmission delays. Jitter is significance for voice, video and data. Jitter can occur when a packet is delayed because the network is busy.

Question No: 23 (Marks: 3)

Give a comparison of wiring Schemes.

Thick Ethernet wiring scheme.

This uses thick coax cable. AUI cable or transceiver or drop cable connects from NIC to transceiver. AUI cable carries digital signal from NIC to transceiver. The transceiver generates analog signal on coax cable. The wires in AUI carry digital signals power and other control signals. Thick Ethernet also requires terminators to avoid signal reflectance.

Thin Ethernet wiring scheme.

Thin Ethernet uses thin coax cable that is cheaper and easier to install than thick Ethernet coax. In thin ethernet wiring scheme transceiver electronics are built into NIC and NIC connect directly to network medium. Coax cable use connector on NIC. Coax runs directly to back of each connected computer by T-connector. The T-connector directly attaches to NIC.

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Question No: 25 (Marks: 5)

Describe permanent virtual circuits (PVC).

ATM can provide the virtual circuits that look like traditional leased digital circuits. The permanent virtual circuits PVC works as long as the customer pays the periodic fee for its use. The forwarding table enter configured the terms used by Telco Provisioning requires two steps. To determine a complete path.

To choose appropriate VPI/VCI for each step in the path and configures each adjacent pair of switches.

Question No: 26 (Marks: 5)

What are default routes, draw the table.

Routing table entries can collapse with a default route. If destination doesn't have in explicit routing table entry and then it use a default route. It is shown in the below table.

Destination Next

hop

Destination Next

hop

Destination Next

hop

Destination Next

hop

1-2-13,124,2

578. 1,342,423,24-

579. 2,33-*4,3

4 3,4

Node 1 Node 2 Node 3 Node 4

Question No: 19 (Marks: 3)

Why 10-base T wiring scheme is called star shaped bus wiring scheme. The main feature of twisted pair Ethernet is that it forms a classic star topology however functions like a bus. 10Base-T Ethernet is often called a star shaped bus.

Question No: 20 (Marks: 5)

What are the main factors due to which mechanism of error detections schemes differ? Question No: 21 (Marks: 10)

How much data can be present on an original Ethernet segment at one time? To find out,

compute the delay-throughput product. The original Ethernet operate at 10 mega bits per second, and a segment was limited to 500 meters. Assume the signals propagate down the cable at 66 percent of the speed of light.\

One Ethernet cable is sometimes called a segment. This segment is limited to 500 meters in length. The minimum separation between connections is 3 meters.

Question No: 20 (Marks: 5)

_Which type of information is obtained from network sniffer and in which mode Network sniffer operates?

Solution:

A network analyzer also called network monitor or a network sniffer is used to examine the performance of or debug a network. It can report statistics such as capacity utilization, distribution of frame size, collision rate or token circulation time

Most installations still use DIX Ethernet encoding in which there is no LLC/SNAP header in the frame. A network analyzer can tell from the values in the type field (small values are lengths, which mean an LLC/SNAP header is located in the first octets of the data area; large values are types, which mean no LLC/SNAP header is included).

The operation of network analyzer is a computer with a network interface that receives all frames, which is called promiscuous mode. So many desktop computers have interface that can be configured for promiscuous mode. When combined with software computer can examine any frame on LAN. In this way the communication across LAN is guaranteed to be private. Computer receives and displays frames on the LAN. Network analyzer can be configured to filter and process frames. It can count frames of specific type of size. It displays only frames from or to specific computers.

Question No: 21 (Marks: 10)

Thick Ethernet, Thin Ethernet and Twisted pair Ethernet.

Solution:-

Thick Ethernet:

Thick Ethernet, officially known as 10 Base-5, is the oldest form of Ethernet. One form of cabling supported by Ethernet is low-loss 50 Ohm coaxial cable as shown in the

figure below. This type of cable is 0.5" diameter (usually supplied with a yellow outer PVC coating) and rather inflexible. It has become known in the communications industry as "Thick Ethernet". The official name for this cable is 10 Baseband5 (10B5), indicating that it is specified for baseband communications at 10 Mbps over distances up to 500m.

Thin Ethernet:

Thin Ethernet, officially called 10 Base-2, is a less expensive version of 10 Base-5 (Thick Ethernet) technologies. It uses a lighter and thinner coaxial cable and dispenses with the external transceivers used with 10 Base-5.

10 Base-2 uses an RG-58A/U coaxial cable and is wired in a bus topology. Each device on the network is connected to the bus through a BNC "T" adapter, and each end of the bus must have a 50 Ohm terminator attached. Each node on the bus must be a minimum of 0.5 meters (1.5 feet) apart, and the overall length of the bus must be less than 185 meters (606 feet).

Twisted Pair Ethernet:

Twisted Pair Ethernet (10baseT), sometime also called "UTP" from "Unshielded Twisted Pair", is based on using a cable similar to phone-wiring. The cable is connected via RJ-45 connectors to the network card installed in the PC.

Question No: 21 (Marks: 2)

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1	-	2	-	1	3,1	2	4,2
*	1,3	4	2,4	2	3,2	4	-
		*	2,3	3	-	*	4,3
				4	3,4		
Node 1		Node 2		Node 3		Node 4	

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_____ is used for compressed audio and video where the data rate depends on the level of compression that can be achieved.

Constant Bit Rate (CBR) service

Variable Bit Rate (VBR) service

Available Bit Rate (ABR) service

None of the given

L_19, p71

VARIABLE BIT RATE (VBR)

It is used for compressed audio and video where the data rate depends on the level of compression that can be achieved.

A Bridge forwards or filters a frame by comparing the information in its address table to the frame's_____

Layer 2 source address

Source node's physical address

Layer 2 destination address

Layer 3 destination address

L_13, p50

A bridge keeps a list for each segment that consist of physical addresses of the computer attached to that segment. In this way a bridge knows on which segment a destination computer is attached.

Most bridges are self learning bridges. As soon as a frame arrives to a bridge, it extracts a source address from its header and automatically adds it in the list for that segment. In this way a bridge builds up address lists.

_____uses distance vector approach to define routing

BGP

OSPF

--

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RIP

None of the given

L_17, p64

COMPERASION

DISTANCE-VECTOR ROUTING

It is very simple to implement

Packet switch updates its own routing table first.

It is used in RIP.

Computers attached to an ethernet use _____in which a computer waits for the other to be idle before transmitting a frame.

CSMA/CD

CSMA/CA

TOKEN PASSING

none of the given

L_8, p28

After collision is detected computers stop transmitting. So Ethernet uses CSMA/CD to coordinate transmission. RECOVERY FROM COLLISION

Computers that detects a collision sends special signal to force all other interfaces to detect collision. A computer then waits for other to be idle before transmission.

Hub

Switch

Router

None of the given

L_12, p43

HUBS:

They are used for extension of connection with multiplexing concept.

A bridge uses _____to determine which computer is connected to which

segment, and uses the_____ to determine whether to forward a copy of frame.

--

every day

n address, destination address.

Destination address, source address.

source address, source address.

L_13, p50

FRAME FILTERING:

If both the source and destination are on the same segment, it does not forward the frame to the other segment. A frame is forwarded to the other segment, if it is destined to that segment.

A bridge keeps a list for each segment that consists of physical addresses of the computer attached to that segment. In this way a bridge knows on which segment a destination computer is attached.

**S
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s.**

A _____ provide a mechanism that a customer can use to set a physical address.

Static addressing scheme

Configurable addressing scheme

Dynamic addressing scheme

None of the given

L_9, p34

CONFIGURABLE

In this category, the address can be set by the end user either manually

In Direct point to point communication adding the Nth computer requires _____ new connections.

N2

N-1

**D
es
ti
na
ti
o**

ven

L_6, p23

Another disadvantage is that adding a new computer to the network requires N-1 new connections

A form of addressing; in which a set of computers is assigned one address.

Broadcast

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Handwritten text: "L_9, p34"

Multicast

Packet Switch

CRC

Bandwidth

connected in a closed loop.

Star Topology

Ring Topology

Bus Topology

None of the given

L_7, p25

RING TOPOLOGY

In this topology of network the computers are connected to each other in closed loop.

The term _____ is used to denote the definition of a packet used with a specific type of network.

L_4, p16

FRAME:

A frame or hardware frame denotes a packet of a specific format on a specific hardware technology.

A network uses a _____ if all computers attach to a central point

Star Topology

Ring Topology

Bus Topology

None of the given

Star Topology

1_ In this topology, all computers are attached to a central point, which
7, is sometimes called the Hub.

P
25



ical

address to a station when the station first boots.

A L_9, p34

DYNAMIC:

In this category the interface automatically assigns physical address each time it is powered up.

In *Point-to-Point* topology there are two topologies.

Tree and Ring

Star and Ring

Star and Tree

None of the given

L_1, p5

In *Point-to-Point* topology there are two topologies.

580. Star topology

581. Tree topology

Local Talk is a LAN technology that employs _____

Star topology

Bus topology

Ri at uses bus topology.

ng
to
P
ol
og
y
N
o
ne
of
th
e
gi
ve
n

In _____ network occupies larger areas like cities & countries.

LAN

WAN

MAN

L
_8
,
P
30

A
p
pl
e
in
ve
nt
e
d
th
e
L
A
N
te
c
h
n
ol
o
g
y
th

QUESTION

None of the given

L_1, p4

In WAN, network occupies larger areas like cities & countries.

sponds.

Ping

Traceroute

ICMP

Non of the given

L_2, p11

Most NICs contain _____ circuitry that allows the NIC to operate independent of the CPU.

L_11, p40

NIC AND CPU PROCESSING:

NIC contains sufficient hardware to process data independent of system CPU. In which some NICs contain separate microprocessor. In addition to this it also include analog circuitry interface to system bus, buffering and processing.

Missing eot indicates sending computer crashed in frame format.

True

False

L_4, p16

FRAMING IN PRACTICE

one extra bit of information with each

character Parity

—
—
—
—
—
—
sc
h
e
m
e,
w
hi
c
h
is
d
es
ig
n
e
d
to
h
el
p
d
et
ec
t
tr
a
ns
m
is
si
o
ns
er
ro
rs
,
se
n
d

he error there are different schemes in which parity checking is also common used. In parity checking, parity refers to the number of bits set to 1 in the data item. A parity bit is an extra bit transmitted with data item chose to give the resulting bit even or odd parity.

CRC can detect more errors than a simple shecksum.

True

False

L_5, P20

To enable a network system to detect more error without increasing the amount of information in each packet another most successful approach is mad which is called CRC.

The checksum of 0000 and 0000 is _____

1111

0000

1110

0111

L 5, p 19

In this procedure data is treated as sequence integers and their arithmetic sum is computed and the carry bits are added to the final sum.

0000

+0000

0000

A _____
B
ri
d Filter a frame
g Forward a frame
e
ca Extend a LAN
n



--

Handwritten text: *Handwritten text*

are device also used to connect two LAN segments to extend a LAN.

FRAME FILTERING

The most important task a bridge performs is frame filtering. If both the source and destination are on the same segment, it does not forward the frame to the other segment. A frame is forwarded to the other segment, if it is destined to that segment.

Formally named _____ informally known as the twisted pair Ethernet or TP Ethernet.

10 Base 2

10 Base 5

10 Base T

None of the given

L_12, p46

10Base- T version of twisted pair Ethernet operated at 10Mbps. 10Base-T twisted pair Ethernet operates at 100Mbps.

The Gigabit Ethernet hardware operates at a rate of _____

582. Mbps

583. Mbps

1000 Mbps None

of the given

L_7, p27

ETHERNET SPEEDS:

The Ethernet speed was originally 3Mbps, and the current standard is 10Mbps the fast Ethernet operates at 100Mbps. There are also gigabits Ethernet available now.

The original Ethernet hardware operated at a rate of _____

584. mbps

585. mbps



--

Handwritten text: *Handwritten text*

t speed was originally 3Mbps, and the current standard is 10Mbps the fast Ethernet operates at 100Mbps. There are also gigabits Ethernet available now.

The term_____ refers to the general concept of a small block of data

Packet

Frame

Data

L_4, p16

Packet is a generic term that refers to small block of data.

_____was especially concerned about the lack of high powered computers.

ARPA

IEEE

EIA

None of the given

L_2, p9

The efforts of ARPA was to enable all its research groups have access to latest computers.

Which error detection method involves polynomials?

Simple parity check

Two-dimensional parity check

CRC

C
he
ck
su
m

L_5, p20

To understand the concept of CRC considers data in a message as co-efficient of a polynomial. Their co-efficient set is divided by a known polynomial.



--

every day

adcast topology there are further two categories

586. SATELLITE\ RADIO

587. RING TOPOLOGY

The term _____ refers to the general concept of a small block of data

L_4, p16

Packet is a generic term that refers to small block of data.

The original Ethernet hardware operated at a rate of _____

588. mbps

589. m
bps
100
0
mb
ps
No
ne

L_7, p27

ETHERNET SPEEDS:

The Ethernet speed was originally 3Mbps, and the current standard is 10Mbps the fast Ethernet operates at 100Mbps. There are also gigabits Ethernet available now.

_____ computes shortest paths in a graph by using weights on edges as a measure of distance.

Distance vector algorithm

G
re
ed
y
al
go
rit
h
m

Dijkstra's algorithm

None of the given

L_17, p62

Dijkstra's algorithm can accommodate weights on edges in graph. The shortest path is then the path with lowest total weight.





QUESTION

_____ sense a signal on one cable and then transmits an amplified copy on the other cable.

Repeater

Bridge

Hub

11

ase the signal strength. It amplifies the weakening signal received from one segment and then retransmits onto another segment.

CRC can detect more errors than a simple checksum

True

False

L_5, p20

CYCLIC REDUNDANCY CHECK (CRC):

To enable a network system to detect more error without increasing the amount of information in each packet another most successful approach is made which is called CRC

A network technology that uses Collision Avoidance

Bridge

Broadcast

WAN

Wireless LAN

L_8, p29

CSMA/CA

Wireless uses collision avoid ness rather than collision detection.

A network with throughput T and delay D has a total of _____ bit in transit

at any time.



e multiplied by effective throughput.

A network with throughput T and delay D has a total of $T \times D$ bits in transit at any time. (e-books)

T
/
D

T
x
D

_____ is used for audio and video, since these have predefined maximum data rates

T
+
D

Constant Bit Rate (CBR) service

N
o
ne
of
th
e
gi
ve
n

Variable Bit Rate (VBR) service

Available Bit Rate (ABR) service

None of the given

L_19, p71

CONSTANT BIT RATE (CBR):

L
_2
2,
p
80

It is used for audio and video, since these have predefined maximum data rates.

It
is
c
o
m
p
ut
e
d
as
d
el
ay
ti
m

Most NICs contain _____ circuitry that allows the NIC to operate independent of the CPU.

L_11, p40

NIC AND CPU PROCESSING:

NIC contains sufficient hardware to process data independent of system CPU. In which some NICs contain separate microprocessor. In addition to this it also include analog circuitry interface to system bus, buffering and processing.

hop to which a packet is sent depends only on_____

T
h
e
n
ex
t

L_15, p57

NEXT HOP FORWARDING:

Vuaskari Team



many more

- 590. If the destination is local computer, packet switch delivers computer port.
- 591. If the destination is attached another switch, this packet switch forwards to next hop through connection to another switch.
- 592. The choice of another switch is based on destination address in packet.

M use an access mechanism known as_____

o
s
t

L_8, p30

L
A
N

Many LAN technologies that are ring topology use token passing for synchronized access to the ring

s

If the ASCII character G is sent and the character D is received, what type of error is this?

t

h

Single-bit

a

Multiple-bit

t

Burst

e

Recoverable

m

p

l

L 6, p22

o

When a small set of bits changes near a specific location due to lighting or electric motor starting nearby etc. then these types of errors are called burst errors.

y

r

i

A_____ relies on the hardware manufacturer to assign a unique physical address to each network interface.

n

g

Static addressing scheme

t

o

Dynamic addressing scheme

p

Configurable addressing scheme

o

l

None of the given

o

g

L_9, p34

y

S
T
A
T
I
C:

In this category the hardware manufacturer assigns permanent physical address to each network interface and manufacturer must ensure that every interface has a unique address.

variable length packet is variable and does not require a complicated interrupt scheme to detect completion of transmission.

True

False

L_20, p72

CELLS VS PACKETS:

The length of time required to send a variable length packet is variable and requires complicated interrupt scheme to detect completion of transmission.

LAN that use ATM technology have a -----

Star topology

Bus topology

Ring topology

None of the given

L_8, p30

Many LAN technologies that are ring topology use token passing for synchronized access to the ring.

Formally named _____ informally known as the twisted pair Ethernet or TP Ethernet.

10 Base 2

10 Base 5

10 Base T

none of the given

e-T

L This is another standard of wiring scheme. It is commonly called 10 Base-T,
_1 twisted pair or TP Ethernet.
2,

p
43 ATM was designed for _____ and use a _____

10
B
as L_17, p65



QUESTION

ATM (ASYNCHRONOUS TRANSFER MODE):

It was designed as a single technology for voice, video and data and has low jitter (variance in delivery time) and high capacity. It uses fixed size, small cells, 48 octet's data and 5 octets header. It can also connect multiple ATM switches into a network.

In which type of topology computers are connected to a central hub?

ANSWER

connected to a central hub.

Computers attached to an ether use ----- in which a computer waits for the ether to be idle before transmitting a frame.

CSMA/CD

CSMA/CA

TOKEN PASSING

None of the given

L_8, p28

After collision is detected computers stop transmitting. So Ethernet uses CSMA/CD to coordinate transmission.

RECOVERY FROM COLLISION:

Computer that detects a collision sends special signal to force all other interfaces to detect collision. Computer then waits for other to be idle before transmission.

OSPF is based on_____

Distance vector routing

Link state routing

Path vector routing

Distance vector routing and Link state routing



many bus

L_17, p64

LINK STATE ALGORITHM

It is used in OSPF

The network that uses a _____, usually consist of a single long cable to which computer attach.

TOPOLOGY:

In a bus topology all computers are attached to a single long cable and any computer can send data to any other computer.

The maximum size of an Ethernet segment is_____

593. meters

594. meters

595. mete

rs None of

the given

L_7, p27

EXAMPLE BUS NETWORK: ETHERNET

One Ethernet cable is sometimes called a segment. This segment is limited to 500 meters in length. The minimum separation between connections is 3 meters.

A network uses a _____if all computers attach to a central point

Star Topology

Ring Topology

Bus Topology

None of the given

l_7, p25

Star Topology

In this topology, all computers are attached to a central point, which is sometimes called the Hub.

closed

loop.

Star Topology

Ring Topology

Bus Topology

None of the given

L_7, p25

RING TOPOLOGY

In this topology of network the computers are connected to each other in closed loop.

The Fast Ethernet hardware operates at a rate of _____

596. Mbps

597. Mb

ps 1000

Mbps

None of
the given

L_12, p46

10Base-T version of twisted pair Ethernet operated at 10Mbps. 10Base-T twisted pair Ethernet operates at 100Mbps.

A popular LAN Technology that uses a shared Bus Topology is called.

WAN

CSMA

Ethernet

EXAMPLE BUS NETWORK; ETHERNET:

Ethernet is a widely used LAN technology. -----

-----The Ethernet uses bus topology.

L_8, p30

LOCAL TALK

Apple invented the LAN technology that uses bus topology.



--

Handwritten text: *Handwritten text*

hnology that employs_____

Bus topology

Ring topology

Star topology

None of the given

L_8, p30

Apple invented the LAN technology that uses bus topology.

In _____ network occupies larger areas like cities & countries.

LAN

WAN

MAN

None of the given

L_1, p4

In WAN, network occupies larger areas like cities & countries.

**L
o
c
a
l
T
a
l
k
i
s
a
L
A
N
t
e
c** Most LANs that employ ring topology use an access mechanism known as_____

CSMA/CD

CSMA/CA

TOKEN PASSING

None of the given

chnologies that are ring topology use token passing for
synchronized access to the ring.

8
,
p
30
M
a
n
y

Ethernet uses a _____ bit static addressing scheme in which each device is
assigned a unique address by the manufacturer.

64

48

L
A
N

32

8

t
e

598. *Ethernet uses a 48-bit static addressing scheme*

www.mutah.edu.jo

www.mutah.edu.jo/e-courses/ch9.ppt

Computers attached to an ethernet use _____ in which a computer waits for the other to be idle before transmitting a frame.

CSMA/CD

ion is detected computers stop transmitting. So Ethernet uses CSMA/CD to coordinate transmission. **RECOVERY FROM COLLISION**

Computers that detects a collision sends special signal to force all other interfaces to detect collision. A computer then waits for other to be idle before transmission.

A popular LAN Technology that uses a shared Bus Topology is called.

WAN

CSMA

Ethernet

Multicast

L_7, p27

EXAMPLE BUS NETWORK; ETHERNET:

Ethernet is a widely used LAN technology. -----

-----The Ethernet uses bus topology.

L_8, p30

LOCAL TALK

Apple invented the LAN technology that uses bus topology.

A network technology that uses Collision Avoidance

Bridge

Broadcast

WAN

Wireless LAN

L
_8

, p29



QUESTION

CSMA/CA

Wireless uses collision avoidance rather than collision detection.

FDDI can transmit data at a rate of _____

100 million bits per second

ANSWER

FDDI: Fiber distributed data interconnect (FDDI) is another ring technology. Its most important features are

It uses fiber optics between stations and transmits data at 100Mbps.

Most LANs that employ ring topology use an access mechanism known as _____

L_8, p30

Many LAN technologies that are ring topology use token passing for synchronized access to the ring

A network that permits many sources to take turns accessing a shared communication resource is providing a form of _____

Packet Division Multiplexing

Time Division Multiplexing

Frequency Division Multiplexing

Wave Division Multiplexing

L_4, p15

In TDM each packet leaves the source and is switched on the shared communication channel through a multiplexer.

Local Talk is a LAN technology that employs _____

Star topology

Bus topology

Ring topology



QUESTION

None of the given

L_8, p30

Apple invented the LAN technology that uses bus topology.

**For point-to-point configuration, addressing is
needed True**

t connections. Data from transmitters is routed directly through hub switches to destination.

The Gigabit Ethernet hardware operates at a rate of _____

- 599. Mbps
- 600. Mbps
- 1000 Mbps** None
- of the given

L_7, p27

ETHERNET SPEEDS:

The Ethernet speed was originally 3Mbps, and the current standard is 10Mbps the fast Ethernet operates at 100Mbps. There are also gigabits Ethernet available now.

When an application_____ data, it makes a copy of the data available to all other computers on the network?

Broadcasting

- Multicasting
- Unicasting
- None of the given

L_9, p34

Some applications want to broadcast messages to all stations on the LAN. For this purpose shared communication channel can make broadcast efficient in such a way that message is delivered to all stations.

A special broadcast address is used to identify broadcast message, which are captured by all stations



ical

address.

A Static addressing scheme

Configurable addressing scheme

Dynamic addressing scheme

None of the given

L_9, p34

CONFIGURABLE

In Direct point to point communication adding the Nth computer requires_____ new connections.

N^2

N-1

$(N^2 - N)/2$

None of the given

L_6, p23

Another disadvantage is that adding a new computer to the network requires N-1 new connections

Unlike Frame Relay and ATM, SMDS (Switched multi-megabit Data service) offers_____.

Connectionless service paradigm

Connectionless and Connection-oriented service paradigm

Both Connectionless and Connection-oriented service paradigm

None of the given

L_17, p65

SMDS:

Switched Multi megabit Data Service (SMDS) is also a Telco service. It is a connection less service. Any SMDS station can send information to any station on the same SMDS cloud. It is typically ranges from 1.5Mbps to 1000Mbps.

_____ have advantages arisen from the size and ease of computation.



very easy

Parity

None of given

L_5, P20

T
o

e
n
a
b
l
e

a

n
e
t
w
o
r
k

s
y
s
t
e
m

t
o

d
e
t
e
c
t

m
o
r
e

e

rror without increasing the amount of information in each packet another most successful approach is mad which is called CRC.

The third field of the header consists of _____ bit Ethernet frame type.

48

32

16

8

L_10, p36

Figure 10.1 Ethernet frame format

Header's 3rd field is frame type 2byts, 16bits

The term_____ refers to the general concept of a small block of data

Packet

Frame

Data

None of the given

L4, p14

Network systems divide data in small blocks or junks called packets, which they send individually.

An electronic device that connects LAN segments

Ethernet

10Baset

B
ri
d
g
e

10Base5

L_13, p50

BRIDGES:



QUESTION

A bridge is a hardware device also used to connect two LAN segments to extend a LAN.

A Bridge can _____

Filter a frame

Forward a frame

ANSWER

Extended LAN

It is a device also used to connect two LAN segments to extend a LAN.

It listens to all traffic and recognizes frame format. It also forwards only correct complete frames and discards the collided and error frames.

FRAME FILTERING

The most important task a bridge performs is frame filtering.

Download the file

In _____, network occupies the smaller area like a room a floor or a building.

LAN

WAN

MAN

None of the given

L_1, p3, p50

L_1, p4

In LAN network occupies the smaller area like a room a floor or a building.

Answer

FDDI can transmits data at a rate of _____

100 million bits per second

100 million bits per second

100 million bits per second

None of the given

Answer

L_8, p31

FDDI: Fiber distributed data interconnect (FDDI) is another ring technology. Its most important features are

It uses fiber optics between stations and transmits data at 100Mbps.



QUESTION

_____ has no way to determine the cause of the problem.

ICMP

Ping

Trace route

None of the given

ANSWER

ses from host computer it can't tell the reason of problem.

The number of connections needed for N computer in direct point to point communication is equal to:

$(N^2 - N) / 2$

601.
$$\frac{N(N-1)}{2}$$

None of the given

L_6, p23

For 'n' computers connections = $(n^2 - n) / 2$.

Hardware that calculates a CRC uses two simple components.

AND unit and XOR unit

Shift register and XOR unit

Shift register and AND unit

None of the given

L_5, p20

HARDWARE COMPONENTS USED IN CRC:

CRC uses just two hardware components:

602. Shift register

603. Exclusive OR (XOR unit)

The _____ is used to denote the definition of a packet used with a specific type of network.

Packet

— **Frame**



QUESTION

Data

None of the given

L_4, p16

A frame or hardware frame denotes a packet of a specific format on a specific hardware technology.

605. Mbps

1000 Mbps None

of the given

L_7, p27

ETHERNET SPEEDS:

The Ethernet speed was originally 3Mbps, and the current standard is 10Mbps the fast Ethernet operates at 100Mbps. There are also gigabits Ethernet available now.

_____ computes shortest paths in a graph by using weights on edges as a measure of distance.

Greedy algorithm

Distance vector algorithm

Dijkstra's algorithm

None of the given

L_17, p62

Dijkstra's algorithm can accommodate weights on edges in graph. The shortest path is then the path with lowest total weight.

Most LANs that employ ring topology use an access mechanism known as _____

CSMA/CD

CSMA/CA

604. TOKEN PASSING

None of the given

The Gigabit Ethernet standard was proposed at a rate of

—
—
—
—
—
—
—
—
—



L_8, p30

Many LAN technologies that are ring topology use token passing for synchronized access to the ring.

An _____ method, the network hardware designers specify how type information is included in the frame and the value use to identify various frame types.

L_10, p35

In this type the identifying value is included with frame describes types of included data.

_____ has advantages arisen from the size and ease of computation.

CRC

Parity

Checksums

None of given

L_5, P20

To enable a network system to detect more error without increasing the amount of information in each packet another most successful approach is mad which is called CRC.

ATM is designed to work on_____.

Twisted Pair

Coaxial

Radio Frequency

Fiber

L_20, p72

ATM is designed to work on fiber (but can be used with twisted pair). A typical port on an ATM switch operates at OC-3 speed (155Mbps) or higher.



ata

Packet

Frame

Data

L_4, p16

Packet is a generic term that refers to small block of data.

IEEE LLC/SNAP header is _____, which is used to specify the type of data.

606. octets

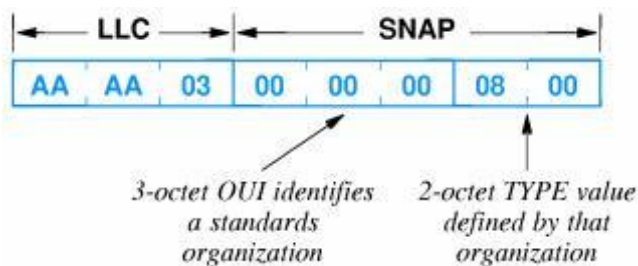
607. bytes

608. bits

None of the given

L_10, p37

Figure 10.3



The checksum of 0000 and 0000 is _____

11 L 5, p 19

11

In this procedure data is treated as sequence integers and their arithmetic sum is computed and the carry bits are added to the final sum.

00

00

0000

11

10

+0000

01

11



QUESTION

0000

The third field of the header consists of _____ bit Ethernet frame type.

48

32

16

ANSWER

FDDI can transmits data at a rate of _____

100 million bits per second

100 million bits per second

100 million bits per second

None of the given

L_8, p31

FDDI: Fiber distributed data interconnect (FDDI) is another ring technology. It most important features are

It uses fiber optics between stations and transmits data at 100Mbps.

The term_____ refers to the general concept of a small block of data

L_4, p16

Packet is a generic term that refers to small block of data.

A _____ provide a mechanism that a customer can use to set a physical address.

Static addressing scheme

Configurable addressing scheme

Dynamic addressing scheme

None of the given



QUESTION

CONFIGURABLE

An interface for thin Ethernet must have a _____ connector, and must generate signals according to the _____ specification.

RJ-45, 10 Base T

RJ
-
45
,
10
B
as
e
5
B
N
C,
10
B
as
e
2
B
N
C,
10
B
as
e
T

L
_1
2,
p
44
Fi
g
ur
e
12
.2

Which of the following is a connecting device?

Bridge

Repeater

Hub

All the given



*RJ-45 connector
for 10Base-T*

L_13, p50



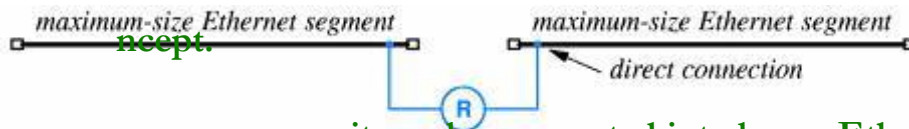
Figure 13.2

A bridge is a hardware device also used to connect two LAN segments to extend a LAN.

P49

Repeaters are used when copper wire communication is carries out.
Figure 13.2

One repeater doubles, two repeaters triple the maximum cable length limitation.



..... it can be connected into larger Ethernet.

A popular LAN Technology that uses a shared Bus Topology is called.

WAN

CSMA

Ethernet

Multicast

L_7, p27

EXAMPLE BUS NETWORK; ETHERNET:

Ethernet is a widely used LAN technology. -----

-----The Ethernet uses bus topology.

L_8, p30

LOCAL TALK

Apple invented the LAN technology that uses bus topology.

FDDI can transmits data at a rate of _____

100 million bits per second

100 million bits per second

100 million bits per second

None of the given

L_8, p31

FDDI: Fiber distributed data interconnect (FDDI) is another ring technology. Its most important features are

It **es fiber optics between stations and transmits data at 100Mbps.**
us



QUESTION

A_____ relies on the hardware manufacturer to assign a unique physical address to each network interface.

Static addressing scheme

Dynamic addressing scheme

Configurable addressing scheme

None of the given hardware manufacturer assigns permanent physical address to each network interface and manufacturer must ensure that every interface has a unique address.

_____ was especially concerned about the lack of high powered computers.

ARPA

L_9

IEEE

, p

EIA

34 None of the given

S
T
A

L_2, p9

T
I
C: The efforts of ARPA was to enable all its research groups have access to latest computers.

I
n _____ scheme, which is designed to help detect transmissions errors, send one extra bit of information with each character

t
h
i
s

Parity

c

Checksums

a

CRC

t
e

none of given

g
o

L_5, p18

r
y

Parity checking

t
h
e To detect the error there are different schemes in which parity checking is also common used. In parity checking, parity refers to the number of bits set to 1 in the data item. A parity bit is an extra bit transmitted with data item chose to give the resulting bit even or odd parity.



ernet.

10 Base 2

10 Base 5

10 Base T

None of the given

L_12, p46

10Base- T version of twisted pair Ethernet operated at 10Mbps. 10Base- T twisted pair Ethernet operates at 100Mbps.

Missing eot indicates sending computer crashed in frame format.

True

False

L_4, p16

FRAMING IN PRACTICE

The Fast Ethernet hardware operates at a rate of _____

609. Mbps

610. Mb

ps 1000

Mbps

None of
the given

L_12, p46

10Base- T version of twisted pair Ethernet operated at 10Mbps. 10Base- T twisted pair Ethernet operates at 100Mbps.

y concerned about the lack of high powered computers.

—
—
—
—
—
—
—
w
as
es
p
ec
ia
ll

ARPA

IEEE

EIA

None of the given

L_2, p9



many many

The efforts of ARPA was to enable all its research groups have access to latest computers.

When an application_____ data, it makes a copy of the data available to all other computers on the network.

Broadcasting

M
u
l
t
i
c
a
s
t
i
n
g

U
n
i
c
a
s
t
i
n
g

N
o
n
e
o
f
t
h
e
g
i
v
e
n

L
_9
,
p
34

S
o
m
e
a
p
p
l
i
c
a
t
i
o
n
s
w

ant to broadcast messages to all stations on the LAN. For this purpose shared communication channel can make broadcast efficient in such a way that message is delivered to all stations.

A special broadcast address is used to identify broadcast message, which are captured by all stations

One repeater _____ two repeaters _____ the maximum cable length limitation.

Doubles, cancel

Doubles, triple

Square roots, cube roots

And, triple

L_13, p49

REPEATERS:

One repeater doubles two repeaters triple the maximum cable length limitation.

An interface for thin Ethernet must have an _____ connector, and must generate signals according to the _____ specification.

RJ-45, 10 Base T

RJ-45, 10 Base 5

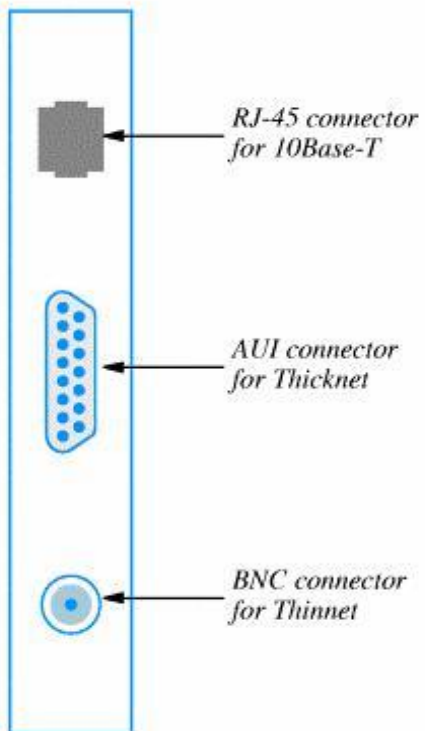
BNC, 10 Base 2

BNC, 10 Base T

L_12, p44



Figure 12.2



_____ device is used for multicasting.

Hub

Switch

Router

None of the given

L_12, p43

HUBS:

They are used for extension of connection with multiplexing concept.

In _____ network occupies larger areas like cities & countries.

LAN

WAN

MAN

None of the given

L_1, p4

In WAN, network occupies larger areas like cities & countries.



xial

Radio Frequency

Fiber

L_20, p72

ATM is designed to work on fiber (but can be used with twisted pair). A typical port on an ATM switch operates at OC-3 speed (155Mbps) or higher.

The Fast Ethernet hardware operates at a rate of _____

611. Mbps

612. Mb

ps 1000

Mbps

None of
the given

L_12, p46

10Base-T version of twisted pair Ethernet operated at 10Mbps. 10Base-T twisted pair Ethernet operates at 100Mbps.

A Bridge can _____

Filter a frame

Forward a frame

Extend a LAN

Do all the above

L_13, p50

A

is a hardware device also used to connect two LAN segments to extend a LAN.

b
r
i
d
g
e

It listens to all traffic and recognizes frame format. It also forwards only correct complete frames and discards the collided and error frames.

FRAME FILTERING



1111 0000

The most important task a bridge performs is frame filtering.

The checksum of 0000 and 0000 is _____

1111

0000

11 integers and their arithmetic sum is computed and the carry bits are added to
10 the final sum.

01 0000
11 +0000

L -----
5, 0000
p
19

I CRC can detect more errors than a simple checksum.

n
t
h True
is

P False
r
o
c
e

d L_5, P20
u
r
e
d

a To enable a network system to detect more error without increasing the
t amount of information in each packet another most successful approach
a is mad which is called CRC.
a

t Hardware that calculates a CRC uses two simple components.
a

is AND unit and XOR unit
tr

e Shift register and XOR unit
a

t Shift register and AND unit
e

d None of the given
a

s
s
e L_5, p20
q

u **HARDWARE COMPONENTS USED IN CRC:**

e CRC uses just two hardware components:
n

c 613. Shift register

e 614. Exclusive OR (XOR unit)
i





The network that uses a _____, usually consist of a single long cable to which computer attach.

Star topology

Bus topology

Ring topology

None of the given

L_7, p26

BUS TOPOLOGY:

In a bus topology all computers are attached to a single long cable and any computer can send data to any other computer.

A bridge uses _____ to determine which computer is connected to which segment, and uses the _____ to determine whether to forward a copy of frame.

Source address, destination address.

Destination address, destination address.

Destination address, source address.

source address, source address.

L_13, p50

FRAME FILTERING:

If both the source and destination are on the same segment, it does not forward the frame to the other segment. A frame is forwarded to the other segment, if it is destined to that segment.

A bridge keeps a list for each segment that consists of physical addresses of the computer attached to that segment. In this way a bridge knows on which segment a destination computer is attached.

Local Talk is a LAN technology that employs _____

Star topology

Bus topology

Ring topology

None of the given

L_8, p30

Apple invented the LAN technology that uses bus topology.

QUESTION

Which of the following is a connecting device?

Bridge

Repeater

Hub

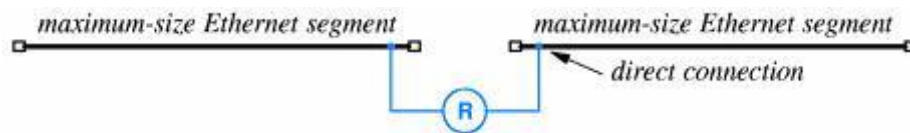
All the given

o connect two LAN segments to extend a LAN.

P49

Repeaters are used when copper wire communication is carries out.
Figure 13.2

One repeater doubles, two repeaters triple the maximum cable length limitation.



L_12, p42

They are used for extension of connection with multiplexing concept.

..... it can be connected into larger Ethernet.

Which error detection method involves polynomials?

Simple parity check

Two-dimensional parity check

CRC

Checksum

L_5, p20

To understand the concept of CRC considers data in a message as co-efficient of a polynomial. Their co-efficient set is divided by a known polynomial.

The maximum size of an Ethernet segment is_____

615. meters

616. meters

617. mNone of
e the given
t
eL_7, p27
r

s**EXAMPLE BUS NETWORK: ETHERNET**



QUESTION

One Ethernet cable is sometimes called a segment. This segment is limited to 500 meters in length. The minimum separation between connections is 3 meters.

The Gigabit Ethernet hardware operates at a rate of _____

618. p27 M

ETHERNET SPEEDS:

619. The Ethernet speed was originally 3Mbps, and the current standard is 10Mbps the fast Ethernet operates at 100Mbps. There are also gigabits Ethernet available now.

1
0 One repeater _____ two repeaters _____ the maximum cable length
0
0 limitation.

M
b Doubles, cancel
p Doubles, triple
s Square roots, cube roots

N And, triple

o
n
e L_13, p49

REPEATERS:

o
f One repeater doubles two repeaters triple the maximum cable length limitation.

t
h
e _____ have advantages arisen from the size and ease of computation.

g CRC

v Parity

e Checksums

n None of given

L
-7 L_5, P20
,



many more

To enable a network system to detect more error without increasing the amount of information in each packet another most successful approach is mad which is called CRC.

Unlike Frame Relay and ATM, SMDS (Switched multi-megabit Data service) offers_____.

Connectionless service paradigm

ice paradigm

None of the given

L_17, p65

SMDS:

Switched Multi megabit Data Service (SMDS) is also a Telco service. It is a connection less service. Any SMDS station can send information to any station on the same SMDS cloud. It is typically ranges from 1.5Mbps to 1000Mbps.

_____ Program sends a message to a remote computer and reports whether the computer responds.

Ping

Traceroute

ICMP

Non of the given

L_2, p11

_____ is used for audio and video, since these have predefined maximum data rates

Constant Bit Rate (CBR) service

Variable Bit Rate (VBR) service

Available Bit Rate (ABR) service

None of the given

L_19, p71

NT BIT RATE (CBR):

It is used for audio and video, since these have predefined maximum data rates.



QUESTION

The original Ethernet hardware operated at a rate of _____

620. mbps

621. m

bps

100

0

mb

pas originally 3Mbps, and the current standard is 10Mbps the fast Ethernet s operates at 100Mbps. There are also gigabits Ethernet available now.

N
oMost NICs contain _____ circuitry that allows the NIC to operate
n
e independent of the CPU.

L
_7
, L_11, p40

P
27
NIC AND CPU PRCESSING:

E
T
H
E
R
N
E
T
S
P
E
E
D
S:
NIC contains sufficient hardware to process data independent of system CPU. In which some NICs contain separate microprocessor. In addition to this it also include analog circuitry interface to system bus, buffering and processing.

The next hop to which a packet is sent depends only on_____

L_15, p57

NEXT HOP FORWARDING:

- h
e
E
t
h
e
r
n
e
t
s
p
e
e
d
w
622. If the destination is local computer, packet switch delivers computer port.
 623. If the destination is attached another switch, this packet switch forwards to next hop through connection to another switch.
 624. The choice of another switch is based on destination address in packet.

An electronic device that connects LAN segments

Ethernet

10Baset

B
ri
d

ge

10Base5



evice also used to connect two LAN segments to extend a LAN.

An _____ method, the network hardware designers specify how type information is included in the frame and the value use to identify various frame types.

Explicit frame type

Ideal frame type

Implicit frame type

None of the given

L_10, p35

In this type the identifying value is included with frame describes types of included data.

A network uses a _____arranges for computers to be connected in a closed loop.

Star Topology

Ring Topology

Bus Topology

None of the given

L_7, p25

RING TOPOLOGY

In this topology of network the computers are connected to each other in closed loop.

A Bridge forwards or filters a frame by comparing the information in its address table to the frame's_____

ess

La
ye
r
2
so
ur
ce
ad
dr

Source node's physical address

Layer 2 destination address

Layer 3 destination address

L_13, p50



QUESTION

A bridge keeps a list for each segment that consist of physical addresses of the computer attached to that segment. In this way a bridge knows on which segment a destination computer is attached.

Most bridges are self learning bridges. As soon as a frame arrives to a bridge, it extracts a source address from its header and automatically adds it in the list for that segment. In this way a bridge builds up address lists.

A Bridge can_____

ANSWER

L_13, p50

BRIDGES

A bridge is a hardware device also used to connect two LAN segments to extend a LAN.

FRAME FILTERING

The most important task a bridge performs is frame filtering. If both the source and destination are on the same segment, it does not forward the frame to the other segment. A frame is forwarded to the other segment, if it is destined to that segment.

_____ sense a signal on one cable and then transmits an amplified copy on the other cable.

Repeater

Bridge

Hub

None of the given

L_13, p49

A repeater is used to increase the signal strength. It amplifies the weakening signal received from one segment and then retransmits onto another segment.

IEEE LLC/SNAP header is _____, which is used to specify the type of data.

625. octets

626. bytes



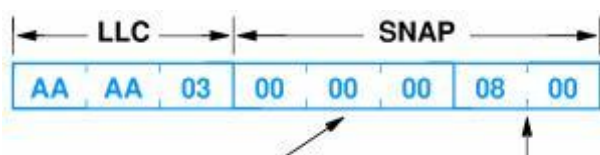
many many

8 bits

None of the given

L_10, p37

Figure 10.3



frame by comparing the information in its address table to the frame's_____

Layer 2 source address

Source node's physical address

Layer 2 destination address

Layer 3 destination address

A

B

L_13, p49

r

The most important task a bridge performs is frame filtering. A

i

d

frame is forwarded to the other segment, if it is destined to that segment.

g

e

Which of the following is a connecting device?

f

o

Bridge

r

w

Repeater

a

Hub

r

d

All the given

s

o

L_13, p50

r

A bridge is a hardware device also used to connect two LAN segments to extend a LAN.

f

i

P49

l

Repeaters are used when copper wire communication is carries out.

t

Figure 13.2

e

r

One repeater doubles, two repeaters triple the maximum cable length limitation.

s

a

L_12, p42



QUESTION

They are used for extension of connection with multiplexing concept.
..... it can be connected into larger Ethernet.

The Gigabit Ethernet hardware operates at a rate of_____

627. Mbps

628. Mbps

1 **PEEDS:**

0 The Ethernet speed was originally 3Mbps, and the current standard is 10Mbps
0 the fast Ethernet operates at 100Mbps. There are also gigabits Ethernet
0 available now.

M
b Computers attached to an ethernet use _____ in which a computer waits for
p the other to be idle before transmitting a frame.
s

N
o
n
e

CSMA/CD

CSMA/CA

TOKEN PASSING

o
f

none of the given

t
h
e

L_8, p28

After collision is detected computers stop transmitting. So Ethernet
uses CSMA/CD to coordinate transmission.

g
i
v
e
n

RECOVERY FROM COLLISION

Computers that detects a collision sends special signal to force all other
interfaces to detect collision. A computer then waits for other to be idle
before transmission.

L_7

CRC can detect more errors than a simple checksum

,
P
27

True

E
T
H
E
R
N
E
T
S

False

L_5, p20

CYCLIC REDUNDANCY CHECK (CRC):



To enable a network system to detect more error without increasing the amount of information in each packet another most successful approach is made which is called CRC

The next hop to which a packet is sent depends only on

Packet's destination

Packet's original source

Path the packet has taken

None of the given

L_15, p57

NEXT HOP FORWARDING

Packet switch must choose outgoing connection for forwarding the packet. There are two cases

If the destination is local computer, packet switch delivers computer port.
If the destination is attached another switch, this packet switch forwards to next hop through connection to another switch.

The choice of another switch is based on destination address in packet.

An interface for thin Ethernet must have an _____ connector, and must generate signals according to the _____ specification.

RJ-45, 10 Base T

RJ-45, 10 Base 5

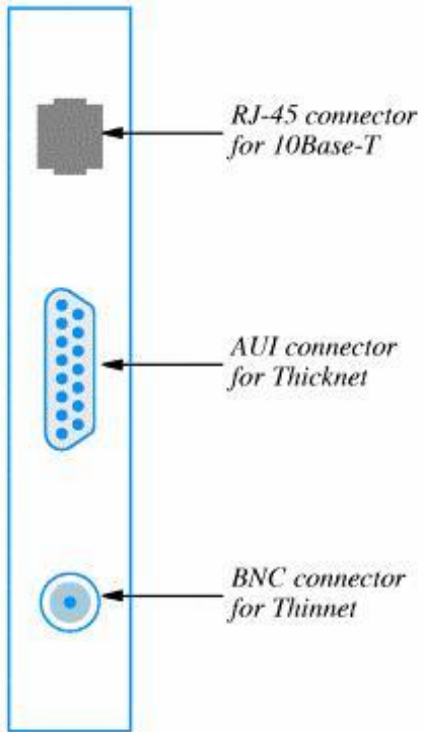
BNC, 10 Base 2

BNC, 10 Base T

L_12, p44

Figure 12.2





_____ was especially concerned about the lack of high powered computers.

ARPA

IEEE

EIA

None of the given

L_2, p9

The efforts of ARPA was to enable all its research groups have access to latest computers.

_____ computes shortest paths in a graph by using weights on edges as a measure of distance.

Greedy algorithm

Distance vector algorithm

Dijkstra's algorithm

Non of the given



many many

L_17, p62

Dijkstra's algorithm can accommodate weights on edges in graph. The shortest path is then the path with lowest total weight.

In broadcast topology there are two categories_____.

L
_1
,
p
5

In
br
o
a
d
ca
st
to
p
ol
o
g
y
th
er
e
ar
e
fu
rt
h
er
t
w
o
ca
te
g
or
ie
s

\RADIO

630. RING TOPOLOGY

_____uses distance vector approach to define routing

BGP

OSPF

RIP

None of the given

L_17, p64

DISTANCE-VECTOR ROUTING

It is very simple to implement

Packet switch updates its own routing table first

It is used in RIP

L_41, p138

ROUTING INFORMATION PROTOCOL (RIP)

It uses distance vector algorithm.

OSPF is based on_____

629. S

A Distance vector routing

T Link state routing

E

L

L Path vector routing

I

T Distance vector routing and Link state routing

E

L LINK STATE ALGORITHM

_1

7, It is used in OSPF

P

64



11/11/2020

_____ was especially concerned about the lack of high powered computers.

ARPA

IEEE

EIA

None of the given

ll its research groups have access to latest computers.

ATM assigns each VC a _____ identifier that is divided two parts to produce a hierarchy.

21-bit

22-bit

23-bit

24-bit

L_18, p67

The VC is identified by a 24 bit value formed from the VPI (8 bits) and VCI (16 bits)

_____ is used for typical data applications (where the data rate may be unknown and bursty) and allows use of whatever bandwidth is available at a given time.

Constant Bit Rate (CBR) service

Variable Bit Rate (VBR) service

Available Bit Rate (ABR) service

None of the given

L_19, p71

AVAILABLE BIT RATE (ABR)

It is used for typical data applications (where the data rate may be unknown and bursty) and allows use of whatever bandwidth is available at a given time.

_____ device is used for multicasting.

Hub



QUESTION

Switch

Router

None of the given

L_12, p43

HUBS:

They are used for extensions of the network.
The term _____ refers to the general concept of a small block of data

L_4, p16

Packet is a generic term that refers to small block of data.

The term _____ is used to denote the definition of a packet used with a specific type of network.

Packet

Frame

Data

None of the given

L_4, p16

A frame or hardware frame denotes a packet of a specific format on a specific hardware technology.

A typical port on an ATM switch operates at _____ or higher.

OC-2 speed (155Mbps)

OC-3 speed (100Mbps)

OC-3 speed (155Mbps)

OC-3 speed (155Gbps)

The

L ATM SPEED

_2

0,

P

72

A typical port on an ATM switch operates at OC-3 speed (155Mbps) or higher

The term_____ refers to the general concept of a small block of data



mall block of data.

P
ac
k
et

The length of time required to send a variable length packet is variable and does not require a complicated interrupt scheme to detect completion of transmission.

Fr
a
m
e

True

False

D
at
a

L_20, p72

CELLS VS PACKETS

L
_4

The length of time required to send a variable length packet is variable and requires complicated interrupt scheme to detect completion of transmission.

,
p1
6

The product of delay and throughput measures the _____ of data that can be present on the network.

P
ac
k
et
is
a
g
e
n
er
ic
te
r
m
th
at
re
fe
rs
to
s

Area

Volume

Length

None of the given

L_22, p80

It measures amount of data that can be present in the network.

The checksum of 0000 and 0000 is _____

1111

00
00

0111

11
10

L 5, p 19



Army

$$\begin{array}{r} 0000 \\ +0000 \\ \hline 0000 \end{array}$$

depends on the level of compression that can be achieved.

Constant Bit Rate (CBR) service

Variable Bit Rate (VBR) service

Available Bit Rate (ABR) service

none of the given

L_19, p71

VARIABLE BIT RATE (VBR):

It is used for compressed audio and video where the data rate depends on the level of compression that can be achieved.

The product of delay and throughput measures the _____ of data that can be present on the network.

Area

Volume

Length

None of the given

l_22, P80

DELAY THROUPUT PRODUCT

It is computed as delay time multiplied by effective throughput. It measures amount of data that can be present in the network.

One repeater _____ two repeaters _____ the maximum cable length limitation.

D ncel
o
u **Doubles, triple**
bl
es Square roots, cude roots
'
ca And, triple



two repeaters triple the maximum cable length limitation.

Which error detection method involves polynomials?

L
_1
3,
p
49

Simple parity check

Two-dimensional parity check

CRC

R
E
P
E
A
R
T
E
R
S:

Checksum

L_5, p20

To understand the concept of CRC considers data in a message as co-efficient of a polynomial. Their co-efficient set is divided by a known polynomial.

O
n
e

A network with throughput T and delay D has a total of _____ bit in transit at any time.

r
e
p
e
a
t
e
r

T / D

$T \times D$

$T + D$

None of the given

L_22, p80

It is computed as delay time multiplied by effective throughput.

d
o
u
b
l
e
s

A network with throughput T and delay D has a total of $T \times D$ bits in transit at any time. (e-books)

The product of delay and throughput measures the _____ of data that can be present on the network.

It measures amount of data that can be present in the network.

L
_2
2,
p
80

The maximum size of an Ethernet segment is_____

631. meters

632. meters

633. mete

rs None of
the given

L_7, p27

EXAMPLE BUS NETWORK: ETHERNET

O
n
e
E
t
h
e
r
n
e
t
c
a
b
l
e
i
s
s
o
m
e
t
i
m
e
s
c
a
l
l
e
d

a
s
e
g
m
e
n
t
.

This segment is limited to 500 meters in length. The minimum separation between connections is 3 meters.

Formally named _____ informally known as the twisted pair Ethernet or TP Ethernet.

10 Base 2

10 Base 5

10 Base T

None of the given

L_12, p46

10Base-T version of twisted pair Ethernet operated at 10Mbps. 10Base-T twisted pair Ethernet operates at 100Mbps.

Missing eot indicates sending computer crashed in frame format.

True

False

L_4, p16

FRAMING IN PRACTICE

A network technology that uses Collision Avoidance

Bridge

Broadcast

WAN

W
ir
el
es
s
L

AN

L_8, p29

CSMA/CA

QUESTION

Wireless uses collision avoidance rather than collision detection.

A typical port on an ATM switch operates at _____ or higher.

L_20, p72

1

A
ty
pi
ca
l
p
or
t
o
n
a
n
A
T
M
s
w
it
c
h
o
p
er
at
es
at
Q
C
-3
sp
ee
d
(1
55
M
b
ps
)
or
hi
g
h
er
.

A_____ provides mechanism that automatically assigns a physical address to a station when the station first boots.

L_9, p34

DYNAMIC:

In this category the interface automatically assigns physical address each time it is powered up.

By size there are_____classifications of networks.

L_1, p4

1) BY SIZE:

According to their size there are two classifications of networks.

- 634. Local Area Network. (LAN)
- 635. Wide Area Network (WAN)

In *Point-to-Point* topology there are two topologies.

Tree and Ring

Star and Ring

Star and Tree

None of the given

L_1, p5

In *Point-to-Point* topology there are two topologies.

topology

636.

S. 637.

Tree topology

t

a

r In _____, network occupies the smaller area like a room a floor or a building.



the smaller area like a room a floor or a building.

The product of delay and throughput measures the _____ of data that can be present on the network.

Area

Volume

Length

None of the give

L_22, p80

DELAY THROUGHPUT PRODUCT

It is computed as delay time multiplied by effective throughput. It measures amount of data that can be present in the network. In fast network with long delay times, sending computer can generate large amounts of data before destination receives first bit.

When an application_____data, it makes a copy of the data available to all other computers on the network.

Broadcasting

Multicasting

Unicasting

None of the given

L_9, p34

Some applications want to broadcast messages to all stations on the LAN. For this purpose shared communication channel can make broadcast efficient in such a way that message is delivered to all stations.

A 1 broadcast address is used to identify broadcast message, which are captured by all stations

s
P CRC can detect more errors than a simple checksum

e
c
i True

a



--

Handwritten text: *Handwritten text*

False

L_5, p20

CYCLIC REDUNDANCY CHECK (CRC):

To enable a network system to detect more error without increasing the amount of information in each packet another most successful approach is made which is called CRC

A _____ provide a mechanism that a customer can use to set a physical address.

Static addressing scheme

Configurable addressing scheme

Dynamic addressing scheme

None of the given

L_9, p34

CONFIGURABLE

_____ scheme, which is designed to help detect transmissions errors, send one extra bit of information with each character

Parity

Checksums

C
R
C

n
o
ne
of
gi
ve
n

Parity checking

To detect the error there are different schemes in which parity checking is also common used. In parity checking, parity refers to the number of bits set to 1 in the data item. A parity bit is an extra bit transmitted with data item chose to give the resulting bit even or odd parity.

_____ has no way to determine the cause of the problem.

L
_5
,
p1
8

ICMP

Ping

Trace route



QUESTION

None of the given

L_2, p11

If ping does not review any responses from host computer it can't tell the reason of problem.

A form of addressing; in which a set of computers is assigned one address.

FIGURABLE

B
ro
a
d
ca
st

A Bridge can _____

Filter a frame

Forward a frame

Extend a LAN

Do all the above

M
ul
tic
as
t

L_13, p50

A bridge is a hardware device also used to connect two LAN segments to extend a LAN.

It listens to all traffic and recognizes frame format. It also forwards only correct complete frames and discards the collided and error frames. FRAME FILTERING

Pa
ck
et
S
wi
tc
h

The most important task a bridge performs is frame filtering.

C
R
C

Formally named _____ informally known as the thick wire Ethernet or **Thick net.**

B
an
d
wi
dt
h

10 Base 2

10 Base 5

10 Base T

None of the given

L
_9
,
P
34

L_11, p41

C
O
N



THICK ETHERNET WIRING

THICK ETHERNET WIRING

It uses thick coax cable. AUI cable connects from NIC to transceiver. L_12, p42

10 Base-T

It replaces AUI cable with twisted pair cable and thick coax with hub.

computer in direct point to point communication is equal to:

$$(N^2 - N) / 2$$

$$638. \quad \left(\frac{N^2 - N}{2} \right)$$

None of the given

L_6, p23

For 'n' computers connections = $(n^2 - n) / 2$.

A bridge uses _____ to determine which computer is connected to which segment, and uses the _____ to determine whether to forward a copy of frame.

Source address, destination address.

Destination address, destination address.

Destination address, source address.

source address, source address.

L_13, p50

FRAME FILTERING:

If both the source and destination are on the same segment, it does not forward the frame to the other segment. A frame is forwarded to the other segment, if it is destined to that segment.

A bridge keeps a list for each segment that consists of physical addresses of the computer attached to that segment. In this way a bridge knows on which segment a destination computer is attached.

heme, which is designed to help detect transmissions errors, send

one extra bit of information with each character

Parity

Checksums

CRC

sc



QUESTION

None of given

L_5, p18

PARITY CHECKING:

PARITY BIT:

A parity bit is an extra bit transmitted with data item chose to give the resulting bit even or odd parity.

ed a unique address by the manufacturer.

64

48

32

8

639. *Ethernet uses a 48-bit static addressing scheme* www.mutah.edu.jo/e-courses/ch9.ppt

In which type of topology computers are connected to a central hub?

Tree

Mesh

Bus

Star

L_1, p5

In *star* topology each computer is connected to a central hub.

A_____ relies on the hardware manufacturer to assign a unique physical address to each network interface.

Static addressing scheme

Dynamic addressing scheme

Configurable addressing scheme

None of the given

L p34

_9

, STATIC:



QUESTION

In this category the hardware manufacturer assigns permanent physical address to each network interface and manufacturer must ensure that every interface has a unique address.

Basic LAN technologies such as Ethernet, Token Ring, and FDDI use a _____.

C Connectionless and Connection-oriented service paradigm
o None of the given

n
n
ec L_21, p77

ti CONNECTIONLESS:

o
nl An example of service paradigm is shown in the figure below.
es

s Figure 21.1
se

rv
ic
e
p
ar
a
di
g
m

Technology	Connection-Oriented	Connectionless	used for LAN	used for WAN
Ethernet		•	•	
Token Ring		•	•	
FDDI		•	•	
Frame Relay	•			•
SMDS		•		•
ATM	•		•	•
LocalTalk		•	•	

C
o
n
n
e
c
t
i
o
n-
or
ie
nt
ed
se
rv
ic
e
pa
ra
di
g
m

Local Talk is a LAN technology that employs_____

B
o
Bus topology

t
h
Ring topology

Star topology

N
o
ne
of
th
e
gi

L_8, p30

Apple invented the LAN technology that uses bus topology.



very easy

Twisted Pair

Radio Frequency

Fiber

(but can be used with twisted pair). A typical port on an ATM switch operates at OC-3 speed (155Mbps) or higher.

_____ Program sends a message to a remote computer and reports whether the computer responds.

Ping

Traceroute

ICMP

Non of the given

L_2, p11

A popular LAN Technology that uses a shared Bus Topology is called.

WAN

CSMA

Ethernet

Multicast

L_7, p27

EXAMPLE BUS NETWORK; ETHERNET:

Ethernet is a widely used LAN technology. -----

-----The Ethernet uses bus topology.

L_8, p30

LOCAL TALK

Apple invented the LAN technology that uses bus topology.

Formally named _____ informally known as the thick wire Ethernet or Thick net.

10 Base 2

10 Base 5



10 Base T

10 Base T

None of the given

L_11, p41

THICK ETHERNET WIRING

It uses thick coax cable. AUI cable connects from NIC to transceiver. L_12, p42

pair cable and thick coax with hub. The original Ethernet hardware operated at a rate of _____

640. mbps

641. m

bps

100

0

mb

ps

No

ne

L_7, p27

ETHERNET SPEEDS:

The Ethernet speed was originally 3Mbps, and the current standard is 10Mbps the fast Ethernet operates at 100Mbps. There are also gigabits Ethernet available now.

For point-to-point configuration, addressing is needed True

False

L_8, p31

Computer gets point -to-point connections. Data from transmitters is routed directly through hub switches to destination.

_____was especially concerned about the lack of high powered computers.

ARPA

IEEE

EIA

N
o
ne
of
th

e given
L_2, p9



11/11/2020

The efforts of ARPA was to enable all its research groups have access to latest computers.

Most NICs contain _____ circuitry that allows the NIC to operate independent of the CPU.

NIC contains sufficient hardware to process data independent of system CPU. In which some NICs contain separate microprocessor. In addition to this it also include analog circuitry interface to system bus, buffering and processing.

_____ have advantages arisen from the size and ease of computation.

CRC

Parity

Checksums

None of given

L_5, P20

To enable a network system to detect more error without increasing the amount of information in each packet another most successful approach is mad which is called CRC.

Unlike Frame Relay and ATM, SMDS (Switched multi-megabit Data service) offers_____.

Connectionless service paradigm

Connection oriented service paradigm

Both Connectionless and Connection-oriented service paradigm
None of the given

L_17, p65

SMDS:



Handwritten text: "Handwritten text"

Switched Multi megabit Data Service (SMDS) is also a Telco service. It is a connection less service. Any SMDS station can send information to any station on the same SMDS cloud. It is typically ranges from 1.5Mbps to 1000Mbps.

A network uses a _____ if all computers attach to a central point

Star Topology

R
in
g
T
o
p
ol
o
g
y
B
u
s
T
o
p
ol
o
g
y
N
o
n
e
o
f
t
h
e
g
i
v
e
n

l_
7,
p
25
St
ar
T
o
p
ol

ogy

In this topology, all computers are attached to a central point, which is sometimes called the Hub.

OSPF is based on_____

Distance vector routing

Link state routing

Path vector routing

Distance vector routing and Link state routing

L_17, p64

LINK STATE ALGORITHM

It is used in OSPF

LAN that use ATM technology have a -----

Star topology

Bus topology

Ring topology

None of the given

L_8, p30

Many LAN technologies that are ring topology use token passing for synchronized access to the ring.

An interface for thin Ethernet must have a _____ connector, and must generate signals according to the _____ specification.



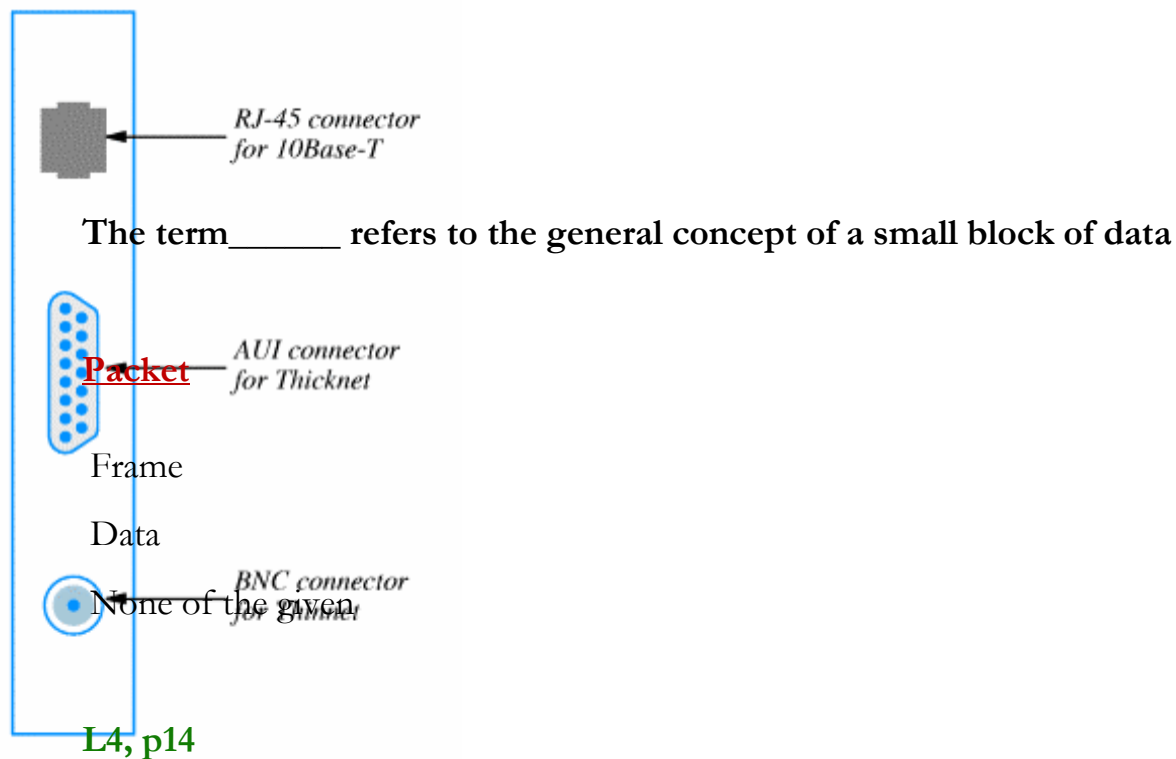
11/11/2020

RJ-45, 10 Base T

RJ-45, 10 Base 5

BNC, 10 Base 2

BNC, 10 Base T



Network systems divide data in small blocks or junks called packets, which they send individually.

Missing eot indicates sending computer crashed in frame format.



t more errors than a simple checksum.

True

False

L_5, P20

To enable a network system to detect more error without increasing the amount of information in each packet another most successful approach is mad which is called CRC.

A Bridge forwards or filters a frame by comparing the information in its address table to the frame's_____

Layer 2 source address

Source node's physical address

Layer 2 destination address

Layer 3 destination address

L_13, p49

The most important task a bridge performs is frame filtering. A frame is forwarded to the other segment, if it is destined to that segment.

When an application_____ data, it makes a copy of the data available to all other computers on the network.

Broadcasting

Multicasting

Unicasting

N
o
ne
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th
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gi
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n

L_9, p34

Some applications want to broadcast messages to all stations on the LAN.

For this purpose shared communication channel can make broadcast efficient in such a way that message is delivered to all stations.



QUESTION

A special broadcast address is used to identify broadcast message, which are captured by all stations

The product of delay and throughput measures the _____ of data that can be present on the network.

e are _____ classifications of networks.

L_1, p4

1) BY SIZE:

According to their size there are two classifications of networks.

642. Local Area Network. (LAN)

643. Wide Area Network (WAN)

A _____ provide a mechanism that a customer can use to set a physical address.

Static addressing scheme

Configurable addressing scheme

Dynamic addressing scheme

None of the given

L_9, p34

CONFIGURABLE

In this category, the address can be set by the end user either manually

Computers attached to an ether use ----- in which a computer waits for the ether to be idle before transmitting a frame.

CSMA/CD

CSMA/CA

TOKEN PASSING

No one of the given



L_8, p28

After collision is detected computers stop transmitting. So Ethernet uses CSMA/CD to coordinate transmission.

RECOVERY FROM COLLISION:

Computer that detects a collision sends special signal to force all other interfaces to detect collision. Computer then waits for other to be idle before transmission.

specific type of network.

L_4, p16

FRAME:

A frame or hardware frame denotes a packet of a specific format on a specific hardware technology.

The term_____ refers to the general concept of a small block of data

Packet

Frame

Data

L_4, p16

Packet is a generic term that refers to small block of data.

_____ has advantages arisen from the size and ease of computation.

CRC

Parity

Checksums

None of given

L_5, P20

To enable a network system to detect more error without increasing the amount of information in each packet another most successful approach is made which is called CRC.



QUESTION

A Bridge can_____

Filter a frame

Forward a frame

Extend a LAN

Do all the above

ANSWER

o used to connect two LAN segments to extend a LAN.

FRAME FILTERING

The most important task a bridge performs is frame filtering. If both the source and destination are on the same segment, it does not forward the frame to the other segment. A frame is forwarded to the other segment, if it is destined to that segment.

In _____ network occupies larger areas like cities & countries.

LAN

WAN

MAN

None of the given

L_1, p4

In WAN, network occupies larger areas like cities & countries.

The next hop to which a packet is sent depends only on_____

L_15, p57

NEXT HOP FORWARDING:

644. If the destination is local computer, packet switch delivers computer port.
645. If the destination is attached another switch, this packet switch forwards to next hop through connection to another switch.
646. The choice of another switch is based on destination address in packet.

The Gigabit Ethernet hardware operates at a rate of _____



QUESTION

647. Mbps
648. Mbps
1000 Mbps None
of the given

L_7, p27

ETHERNET SPEEDS:

rent standard is 10Mbps the fast Ethernet operates at 100Mbps. There are also gigabits Ethernet available now.

_____ is used for audio and video, since these have predefined maximum data rates

Constant Bit Rate (CBR) service

Variable Bit Rate (VBR) service

Available Bit Rate (ABR) service

None of the given

L_19, p71

CONSTANT BIT RATE (CBR):

It is used for audio and video, since these have predefined maximum data rates.

_____ is used for typical data applications (where the data rate may be unknown and bursty) and allows use of whatever bandwidth is available at a given time.

Constant Bit Rate (CBR) service

Variable Bit Rate (VBR) service

Available Bit Rate (ABR) service

None of the given

L_19, p71

AVAILABLE BIT RATE (ABR)

It is used for typical data applications (where the data rate may be unknown and bursty) and allows use of whatever bandwidth is available at a given time.

— _____ has no way to determine the cause of the problem.

—



Trace route

ICMP

Ping

Trace route

None of the given

L_2, p11

om host computer it can't tell the reason of problem.

A Bridge forwards or filters a frame by comparing the information in its address table to the frame's_____

Layer 2 source address

Source node's physical address

Layer 2 destination address

Layer 3 destination address

L_13, p50

A bridge keeps a list for each segment that consist of physical addresses of the computer attached to that segment. In this way a bridge knows on which segment a destination computer is attached.

Most bridges are self learning bridges. As soon as a frame arrives to a bridge, it extracts a source address from its header and automatically adds it in the list for that segment. In this way a bridge builds up address lists.

In broadcast topology there are two categories_____.

L_1, p5

In broadcast topology there are further two categories

649. SATELLITE\ RADIO

650. RING TOPOLOGY

When an application_____ data, it makes a copy of the data available to all other computers on the network?

Broadcasting

M	ting
ul	
tic	Unicasting
as	None of the given



11-11-2023

L_9, p34

Some applications want to broadcast messages to all stations on the LAN. For this purpose shared communication channel can make broadcast efficient in such a way that message is delivered to all stations.

A special broadcast address is used to identify broadcast message, which are captured by all stations

A
ty
pi
ca
l
p
or
t
o
n
a
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A
T
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0, p72

A typical port on an ATM switch operates at OC-3 speed (155Mbps) or higher.

A network that permits many sources to take turns accessing a shared communication resource is providing a form of _____

Packet Division Multiplexing

Time Division Multiplexing

Frequency Division Multiplexing

Wave Division Multiplexing

L_4, p15

In TDM each packet leaves the source and is switched on the shared communication channel through a multiplexer.

When an application_____data, it makes a copy of the data available to all other computers on the network.

Broadcasting

Multicasting

Unicasting

None of the given

L_9, p34

Some applications want to broadcast messages to all stations on the LAN. For this purpose shared communication channel can make broadcast efficient in such a way that message is delivered to all stations.

L

A special broadcast address is used to identify broadcast message, which are captured by all stations

s



ATM assigns each VC a _____ identifier that is divided two parts to produce a hierarchy.

21-bit

22-bit

23-bit

24-bit

L_18, p67

The VC is identified by a 24 bit value formed from the VPI (8 bits) and VCI (16 bits)

Basic LAN technologies such as Ethernet, Token Ring, and FDDI use a _____.

Connectionless service paradigm

Connection-oriented service paradigm

Both Connectionless and Connection-oriented service paradigm

None of the given

L_21, p77

CONNECTIONLESS:

An example of service paradigm is shown in the figure below.

Figure 21.1

Technology	Connection-Oriented	Connectionless	used for LAN	used for WAN
Ethernet		•	•	
Token Ring		•	•	
FDDI		•	•	

4
4
3
1
5
5
4
3
4
4
4



651. ATM assigns each VC a _____ identifier that is divided two parts to produce a hierarchy.

21-bit

22-bit

23-bit

24-bit (pg 67)

652. In -----, network occupies larger areas like cities & countries.

653. _____ is the scheme, which is designed to help in detecting transmission errors, sends one extra bit of information with each character.

Parity checking (Pg.
18) Checksums

CRC
LRC

654. The Fast Ethernet hardware operates at a rate of -----

655. When an application----- data, it makes a copy of the data available to all other computers on the network.

Broadcasting
Multicasting
Unicasting
Anycasting

656. Formally named _____ informally known as the thin wire Ethernet or Thin net.

657. NICs contain analogue circuitry and _____ that allows the NIC to operate independent of the CPU.

[Type text]

DMA (Direct Memory Access)

Multiplexer

Transceiver

Microprocessor

658. By size there are.....classifications of networks.

2 (pg 4)

3

4

5

659. In which type of topology computers are connected to a central device? Ring

Mesh

Bus

Star (pg 5)

660. ----- was especially most concerned about the lack of highly powered computers.

ARPA

IEEE EIA

Goverment

661. IEEE LLC/SNAP header is _____ which is used to specify the type of data.

8 octets (Pg. 112)

662. octets

663. bytes

664. bits

665. An interface for Thin Ethernet must have _____ connector and must generate signals according to the _____ specification.

RJ-45, 10 Base T (Pg. 126)
RJ-45, 10 Base 5

BNC, 10 Base 2
BNC, 10 Base T

666. _____ technology can support both LAN and WAN connections.

Frame relay a

Ethernet

SMDS

ATM (Pg. 66)

14. _____ is used for ordinary data.

Available Bit Rate (ABR) service

Uniform Bit Rate (UBR) service

Constant Bit Rate (CBR) service (pg 71)

[Type text]

Variable Bit Rate (VBR) service

667. A network which is based on packets uses a computer which may have _____, memory and _____ for switching.

Administration Rights, Algorithm

Source address, Destination
address Interface, Program

Input, Output

668. No error detection scheme is perfect because transmission errors can affect _____.

Data

Additional Information (Pg. 82)

Additional information & data

Transmission medium

669. In direct point to point communication adding the N^{th} computer requires _____ new connections.

$N(N - 1)$

N^2

$N - 1$

$(N^2 - N)/2$

670. The next hop to which a packet is sent depends only on _____.

Packet's destination

Packet's original source

Path the packet has taken

Routed Protocol (pg 93)

671. _____ in which delay among packets
can vary. Virtual Private Network

Isochronous Networks

Isochronous Networks

Asynchronous Networks (Pg. 65)

672. A typical port on an ATM switch operates at _____ or
higher. 2 OC-2 speed (155Mbps)

OC-3 speed (100Mbps)

OC-3 speed (155Mbps) (pg

72) OC-3 speed (155Gbps)

CS610 Computer Network

673. ATM assigns each VC a _____ identifier that is divided into two parts to
produce a hierarchy.

21-bit

22-bit

23-bit

24-bit (pg 67)

674. Hardware that calculates a CRC uses two simple components.

AND unit with XOR unit

Shift register with XOR unit (pg 20)

CS610 Computer Network

Shift register with AND unit

Shift register with shift XOR unit

03. ----- have advantages arisen from the size and ease of computation.

CRC

Parity (Pg.19)

Checksums

VRC

675. The Gigabit Ethernet hardware operates at a rate of -----

676. Mbps

677. Mbps

1000 Mbps

678. Gbps

679. When an application----- data, it makes a copy of the data available to all other computers on the network.

Broadcasting

Multicasting

Unicasting

Anycasting

680. A----- provides mechanism that automatically assigns a physical address to a station when the station first boots.

Static addressing scheme

Configurable addressing scheme

Dynamic addressing scheme (Pg.

34) Dynamic Host Configuration

681. NICs contain analogue circuitry and _____ that allows the NIC to operate independent of the CPU.

DMA(Direct Memory
Access) Multiplexer

682. Basic LAN technologies such as Ethernet, Token Ring, and FDDI use a _____. **Connectionless service paradigm**

Connection-oriented service paradigm

Both Connectionless and Connection-oriented service paradigm
Connection-oriented service paradigm without acknowledgment

683. ----- was especially most concerned about the lack of highly powered computers.

ARPA

IEEE EIA

Government

684. "Frame type" field of the header consists of _____ bits.

8

10

16

[Type text]

32 (Pg. 105)

685. An interface for Thin Ethernet must have _____ connector and must generate signals according to the _____ specification.

RJ-45, 10 Base T (Pg. 126)
RJ-45, 10 Base 5

BNC, 10 Base 2
BNC, 10 Base T

686. A bridge uses the _____ to determine which computers connect to which segment and uses the _____ to determine whether to forward a copy of frame.

Destination address, Source address
Destination address, MAC Address
Destination address, Destination address
Source address, Destination address

687. _____ senses a signal on a cable and then transmits an amplified copy of the signal.

Bridge Hub
Transceiver
Repeater

688. _____ is used for ordinary data.

689. _____ is used for delay sensitive applications requiring high data rate. Constant Bit Rate (CBR) service (Pg. 71)

Variable Bit Rate (VBR) service
Available Bit Rate (ABR) service
Uniform Bit Rate (UBR) service

690. The term _____ refers to the general concept of a small block of data in a network. Packet (Pg. 16)

691. A network with throughput T and delay D would have a total no of _____ bits in transit at any time.

T + D

T x D/Unit
Time T / D

T x D (Pg. 203)

692. A network that permits many sources to take turns accessing a shared communication resource is providing a form of _____.

Phase Division Multiplexing
Time Division Multiplexing
Frequency Division Multiplexing
Wave Division Multiplexing

[Type text]

693. No error detection scheme is perfect because transmission errors can affect _____ . Data

Additional Information (Pg. 82) Additional information & data Transmission medium

694. Formally named _____ informally known as the twisted pair Ethernet or TP Ethernet.

695. Base 2

696. Base 5

697. Base T (pg 43)

698. Base T

CS610 Computer Network

699. In -----, network occupies larger areas like cities & countries.

LAN

WAN (pg 4)

MAN

CAN

700. ----- have advantages arisen from the size and ease of computation.

CRC

Parity (Pg. 19)

Checksums

VRC

3. A network that uses a ----- usually consist of a single long cable to which computers attach.

Bus Topology (pg 26)

Tree Topology

Star Topology

Ring Topology

701. A----- provides mechanism that automatically assigns a physical address to a station when the station first boots.

Static addressing scheme

Configurable addressing scheme

Dynamic addressing scheme (pg 34) Dynamic Host Configuration

702. In which type of topology computers are connected to a central device?

703. If the sending computer crashes before transmitting a complete frame, the _____ will not arrive and the receiver will know that the frame is _____.

soh, complete

soh, incomplete

eot, incomplete (Pg. 16)

[Type text]

eot, complete

704. ----- was especially most concerned about the lack of highly powered computers.

ARPA

IEEE EIA

Goverment

705. Generally a Bridge is used to perform _____.

Extend a LAN, Filter and Forward a packet (Pg. 50)

Routing

Filter and Forward a frame

Filter and Forward a packet

706. Header of cell takes _____ of network capacity. 5% (pg 65)

10%

12%

15%

707. _____ senses a signal on a cable and then transmits an amplified copy of the signal.

Bridge Hub

Transceiver

Repeater

708. _____ is used for ordinary data.
Available Bit Rate (ABR) service Uniform Bit
Rate (UBR) service Constant Bit Rate (CBR)

709. A Bridge forwards or filters a frame by comparing the information in its address table to the frame's _____.

- Layer 2 source address
- Layer 3 source address
- Layer 2 destination address
- Layer 3 destination address

710. A network with throughput T and delay D would have a total no of _____ bits in transit at any time.

T + D

T x D/Unit
Time T / D

T x D (Pg. 203)

711. No error detection scheme is perfect because transmission errors can affect _____. Data

[Type text]

Additional Information (Pg. 82)

Additional information & data

Transmission medium

712. Most LANs that employ ring topology use an access mechanism known as_____.

CSMA/CD

CSMA/CA

Token Passing (Pg. 95)
Aloha

713. Local Talk is a LAN technology that employs _____.

Bus topology (Pg. 30)

Ring topology

Mesh topology

Star topology

714. Formally named _____ informally known as the twisted pair Ethernet or TP Ethernet.

715. Base 2

716. Base 5

717. Base T

718. Base T

719. Most WAN systems include a mechanism that can be used to eliminate the common case of duplication routing is called_____.

Dynamic route
Hierarchal address

720. _____ in which delay among packets can vary.

721. The product of delay and throughput measures the _____ of data that can be presented on the network.

Area
Volume(Pg.
80) Length

Size

Subjective

722. Suppose being an IT engineer in XYZ organization, you are given a task to avoid from duplication routes in routing table then how can you avoid from duplication of routes? 02 Answer: (Pg. 102 Lecture no.29)

723. Suppose a network engineer wants to transfer data that is less critical then which type of quality of service specification of ATM would be applied? 02

ANS: CONSTANT BIT RATE (CBR):

It is used for audio and video, since these have predefined maximum data rates. (Pg 71)

724. Suppose an organization wants voice communication in network by using ATM technology. Explain, how ATM eliminate the varying delays associated with variable size data packets? 03

Answer: To meet its goals, ATM uses small, fixed sized packets called cells. Each cell has 53 octets. VPI/VCI fields identify the cells destination. (pg 66)

725. Suppose a computer B on segment 1 of a LAN sends a frame to a computer F on segment 2 of the same LAN. If the bridge receives the frame from B, just after restarted then in this case will the bridge broadcast the frame or just send it to a particular computer? Justify your answer with reason. 03

Answer: Yes after restarted the bridge, the bridge can send this frame to the particular computer F on segment 2 because A frame is forwarded to the other segment, if it is destined to that segment.

726. While designing the ATM technology why the designers of the ATM have chosen cells over packets which are being used widely? Give five reasons to support your answer.05

Ans: 1. Cells are not variable length

a. Handling variable length packets leads to memory fragmentation

727. Variable length packets require hardware to accommodate the largest possible Packet.

728. The length of time required to send a variable length packet is variable and requires complicated interrupt scheme to detect completion of transmission.

729. QoS can't be guaranteed with variable length packets as easily as it can with fixed

length cells. (Pg. 72)

730. Routing is a key feature of the networking because it can perform more complex analysis to determine the optimal path for the packet and eventually reach the target machine. What is meant of source independence in routing and how it can be valuable for routing? 05

Answer: Next hop to destination does not depend on source of packet. This phenomenon is called 'Source Independence'.

It has several benefits.

731. It allows fast and efficient routing.

732. Packet switch need not have complete information about all destinations in spite it just has next hop so reduces total information and increases dynamic robustness.

733. The network can continue to function even if topology changes without notifying entire network. (Pg. 58)

21. Why the operation of NIC is isolated from the remaining computer system?02

ANS: NIC contains sufficient hardware to process data independent of system CPU. In which some NIC's contain separate microprocessor. In addition to this it also include analog circuitry interface to system bus, buffering and processing. (Pg. 40)

734. Suppose ABC Corporation has a small number of packet switches and computer systems in its computer network. If entire network is stable with single network link than which routing method will be most appropriate for this scenario? 2

Answer: - If entire network is stable with single network link then Manual routing method will be most appropriate for this scenario. (Pg. 62)

735. Suppose an organization wants voice communication in network by using ATM technology. Explain, how ATM eliminate the varying delays associated with variable size data packets? 03

Answer: To meet its goals, ATM uses small, fixed sized packets called cells. Each cell has 53 octets. VPI/VCI fields identify the cell's destination. (pg 66)

736. Suppose you want to add large amount of systems in your WAN to increase its capacity. Which type of switching devices will be used there? Write the name of each with its functionality

Answer: We can add capacity to WAN by adding more links and packet switches. Packet switches need not have computers attached. There are two types of switch according to the attached computers.

INTERIOR SWITCH:

The switch that has no attached computers is called an interior switch.

EXTERIOR SWITCH:

Prepared by: Dr. Tariq Hanif MIT-4th Sahiwal

[Type text]

Page 10

The switch that has computers attached with it is called exterior switch. Both interior and exterior switches forward packets and they also need routing tables. The routing table must have two things. (pg 59)

737. Suppose you are assigned to connect two site offices which are situated far away from each other. In your opinion, which one will be the appropriate solution for this, LAN extension or WAN? Justify your answer with four reasons 5

Answer: I choose wan if I assign to connect the to sites offices which are situated for a away from each other.

Reasons:

- a. Its cost will be increased because we will fiber optic which is very costly the sites are very far away.
- b. Its maintainers will be difficult we must use repeaters in that scenario.
- c. Its performance will be affected.
- d. Due to for of distance its signal will become weak.

738. A Connection-oriented service requires a session be established before start communication. When telephone call is made between two persons then which steps are involved to establish a connection? Write each step and tell how established connection is identified.05

Answer: The connection-oriented service paradigm for networking is similar to the manner in which telephones are used.

This is given as follows:

A caller dials a number of the destination. The telephone at the destination signals the arrival of a connection request. If the called person does not answer; the caller gives up after waiting for a timeout. If the called person does answer, then the connection is established. (Pg. 67)

21. Why the operation of NIC is isolated from the remaining computer system?02

Answer: NIC contains sufficient hardware to process data independent of system CPU. In which some NIC's contain separate microprocessor. In addition to this it also include analog circuitry interface to system bus, buffering and processing. (Pg. 40)

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Answer: If entire network is stable with single network link then Manual routing method will be most appropriate for this scenario. (Pg. 62)

[Type text]

Page 11

- 740. Suppose an organization wants voice communication in network by using ATM technology. Explain, how ATM eliminate the varying delays associated with variable size data packets? 03**

Answer: It was designed as a single technology for voice, video and data and has low jitter (variance in delivery time) and high capacity.

It uses fixed size, small cells, 48 octet's data and 5 octets header. It can also connect multiple ATM switches into a network. (pg 65)

- 741. Suppose you want to add large amount of systems in your WAN to increase its capacity. Which type of switching devices will be used there? Write the name of each with its functionality?**

Answer: We can add capacity to WAN by adding more links and packet switches. Packet switches need not have computers attached. There are two types of switch according to the attached computers.

INTERIOR SWITCH:

The switch that has no attached computers is called an interior switch.

EXTERIOR SWITCH:

The switch that has computers attached with it is called exterior switch. Both interior and exterior switches forward packets and they also need routing tables. The routing table must have two things. (pg 59)

- 742. Suppose you are assigned to connect two site offices which are situated far away from each other. In your opinion, which one will be the appropriate solution for this, LAN extension or WAN? Justify your answer with four reasons 5**

Answer: I choose wan if I assign to connect the to sites offices which are situated for a away from each other.

Reasons:Its cost will be increased because we will fiber optic which is very costly the sites are very far away.

- a. 2. Its maintainers will be difficult we must use repeaters in that scenario.
- b. It performance will be effected.
- c. Due to for of distance its signal will become weak.

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(Pg. 67)

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ENG-201 100 % CORRECT MCSQ

AND Many MoRe